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Development of a Wearable Sensor-based Ergonomic Monitoring System for Aviation Manufacturing Workers

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Abstract

Work-related Musculoskeletal Disorders (WMSDs) are injuries and disorders of soft tissue and the nervous system, resulting from excessive workload during working activities. Excessive workload is caused by an array of ergonomic risks, such as repetitive movements, improper postures, and excessive force. WMSDs are one of the most serious issues for both aviation manufacturing workers and industry, which can be prevented by closely monitoring ergonomic risks. However, considering the diversity and variety of ergonomic risks, it is challenging to monitor all risks simultaneously. Fatigue is an alternative indicator towards WMSDs, as workers with a significant accumulation of fatigue are more likely to suffer WMSDs. As a result, the framework of a real-time wearable sensor-based ergonomic monitoring system, named ErgoRelief, has been designed and developed in this thesis, in order to improve both aviation manufacturing workers and factory operations' awareness about workers' body conditions. There are four major components of the ErgoRelief framework, including real-time data collection, fatigue assessment, data logging, and visualisation. A pipeline for processing multimodal data has also been developed, that includes data pre-processing, feature extraction, data labelling, feature selection, and body fatigue classification. By implementing this methodology, multiple streams of data, which are sensitive to motion artefacts, have been processed and synchronised in order to detect fatigue during physical tasks.

To demonstrate the system's feasibility, two studies have been undertaken. The first study has been conducted to perform body fatigue classification during repetitive physical tasks, using body motion, heart rate, and external force exerted on the human body. The study has been undertaken for two purposes, which are evaluating the performance of different sensor systems, feature selection algorithms, and machine learning approaches on classifying fatigue, and optimising sensor deployment to prepare the system for real-time body fatigue assessment. An experiment has been conducted in a laboratory environment with thirty participants to simulate four repetitive physical tasks, and data have been collected while participants perform tasks. Results of body fatigue classification illustrated the highest accuracy of 97.17%. Furthermore, after optimising sensor deployment by minimising the number of sensors, results demonstrated an accuracy up to 86.17%, illustrating that the optimised sensor system can still achieve accurate body fatigue classification.

The first study has been conducted in a laboratory environment whilst participants undertook simple repetitive physical tasks. To achieve real-time fatigue assessment in an aviation-like factory environment, the second study has been conducted in a workshop to simulate more realistic aviation manufacturing tasks. Additionally, the hardware performance for the Bluetooth data streaming has

also been evaluated in this study, to ensure that the system is stable and durable for real-world deployment. This study was undertaken with thirty participants, with their fatigue being analysed while performing tasks. Results illustrated that the highest accuracies of 81.41% and 85.37% for the real-time body fatigue assessment and machine learning model training, respectively. Furthermore, the battery percentage decreased from 100% to approximately 65% after 5.5 hours data recording, and more than 99% of the packets can be received successfully through Bluetooth during the experiment.

Declaration by author

This thesis is composed of my original work, and contains no material previously published or written by another person except where due reference has been made in the text. I have clearly stated the contribution by others to jointly-authored works that I have included in my thesis.

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Publications included in this thesis

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Contributions by others to the thesis

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Statement of parts of the thesis submitted to qualify for the award of another degree

No works submitted towards another degree have been included in this thesis.

Research Involving Human or Animal Subjects

In this thesis, two studies have been undertaken with human subjects. Both studies have been approved by The University of Queensland Ethics Sub-Committee (approval number: 2019001586). The copy of the ethics approval letter has been included in Appendix 1 and 2 for the first and second studies, respectively.

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List of Abbreviations

ANN	Artificial Neural Network
ANOVA	Analysis of Variance
AT-BAN	Activity Tracking with Body Area Network
AUC	Area Under ROC Curve
BER	Balanced Error Rate
CFS	Correlation-based Feature Selection
ECG	Electrocardiogram
EEG	Electroencephalogram
EMG	Electromyography
FFT	Fast Fourier Transform
GSR	Galvanic Skin Response
IMU	Inertial Measurement Unit
IoT	Internet of Things
kNN	k-Nearest Neighbour
LDA	Linear Discriminant Analysis
MLR	Multiple Linear Regression
MSDs	Musculoskeletal Disorders
NIOSH	National Institute for Occupational Safety and Health
PCA	Principal Component Analysis
PPG	Photoplethysmogram
PSD	Power Spectral Density
P-SVM	Polynomial Kernel-based Support Vector Machine
REBA	Rapid Entire Body Assessment
RGB	Red, Green, and Blue
RMSE	Root-Mean-Square-Error
RULA	Rapid Upper Limb Assessment
RUS	Random Under-Sampling
SVM	Support Vector Machine
WMSDs	Work-related Musculoskeletal Disorders

Chapter 1

Introduction

1.1 Motivation

Musculoskeletal Disorders (MSDs) are injuries and disorders of the soft tissues (e.g. muscles, tendons, ligaments, joints, and cartilage) and the nervous system, which affects different parts of the body, including the arms, back, hands, wrists, fingers, legs, neck, and shoulders [1, 2]. Work-related Musculoskeletal Disorders (WMSDs) occur when a workload goes beyond the worker's physical capabilities. This can be caused by an array of ergonomic risk factors, such as excessive force, repetitive movements, awkward and/or static postures, quick motion, compression, vibration, and cold temperature [1]. Long-term exposure to such ergonomic risks can cause the accumulation of microdamage on different areas of the body, which contributes to a series of symptoms, including local or generalized pain, loss of sensation to touch, heat or pressure, and loss of muscle strength, endurance, and/or flexibility. [3, 4]. Industries that are particularly affected by WMSDs include manufacturing, health care and social assistance, and construction [4]. In particular, aviation manufacturing is an industry that has been classified as having high physical and cognitive demands, as it requires workers to be knowledgeable and experienced in order to deliver high-quality and precise products through highly intensive physical tasks [5]. WMSDs not only negatively affect workers' bodies, but also poses significant pressure on Occupational Health and Safety costs for the industry. In 2019/2020, approximately 480,000 workers were affected by WMSDs in Great Britain, which represents 1,420 per 100,000 workers and resulted in 8.9 million working days lost [6]. In Australia, 124,690 WMSD-related claims were made in 2019, with 50% of them classified as serious [4]. Furthermore, in the US during 1999, WMSD-related issues cost all industries between $45 and $54 billion per annum [7].

As WMSDs pose significant detrimental effects to both workers and the industry, it is essential to monitor the causal factors of such occurrences, in order to take proactive actions, such as adding short breaks, changing task sequences, and supplying ergonomically safe tools [1]. There are two major sets of parameters that can be monitored in order to detect WMSDs, which are ergonomic risk factors and health parameters, which includes pain and discomfort [8]. Current ergonomic assessment methods include self-report, observational, and direct measurement [9]. Self-report methods require

workers to complete questionnaires about their body conditions. However, the results can be biased as they can be easily influenced by memory, social desirability, and individual differences. Observational methods relate to professional observers viewing and evaluating the current setting to identify ergonomic risks. Whilst this is cost effective, more accurate compared to self-report methods, and does not disturb workers, the results are still subjective and based on expert's observations. Additionally, this method cannot be conducted for a long duration and can only cover a limited number of tasks. Direct measurement methods replace professional observations with sensor readings, such as from force sensors and accelerometers. This has been achieved through the development of the Internet of Things (IoT), where sensors have been developed that are smaller in size, have high computational power, and low energy consumption [10].

Current direct measurement methods rely on two types of devices, which include cameras and other sensors [9]. Red, green, and blue (RGB) cameras and depth cameras, such as the Microsoft Kinect, have been widely implemented to reconstruct body postures using skeletal tracking functionality. Using this tracking awkward postures can then be detected. In addition, ergonomic scores can also be calculated based on joint angles, using methods such as the Rapid Upper Limb Assessment (RULA). The higher the score is, the more likely the current posture/task will lead to WMSDs [11-16]. However, an aviation factory environment can pose significant challenges towards camera-based systems due to blocked fields of view, changing illumination, and moving objects. Additionally, deploying camera-based systems to a factory environment requires significant infrastructure spending, as it requires a considerable amount of hardware devices in order to perform computationally intensive processing on images and videos [17]. However, sensor-based systems can mitigate these problems by unobtrusively monitoring body-related information, where wearable sensors can be affixed to different body areas to measure body motion and physiological data, whilst non-wearable sensors can be deployed around the workplace to monitor environmental factors, such as temperature, humidity, and noise [9]. Therefore, we developed a wireless sensor network with wearable sensors to collect, analyse, and store data in real time. In the meantime, visualisation of the data through front-end smartphone applications is another avenue to explore, which aids in significantly improving awareness of body conditions [10, 18, 19].

Although monitoring ergonomic factors or health parameters is the most straightforward way of preventing WMSDs, it is impractical to monitor all ergonomic risks simultaneously due to the complexity of the factory environment and daily tasks that workers perform. Fatigue has been selected as an alternative standard to gauge the risk of suffering WMSDs, because the accumulation of fatigue

directly contributes to an increase in the possibility of suffering WMSDs [4, 20]. Fatigue has been defined as "An overwhelming sense of tiredness, lack of energy and a feeling of exhaustion, associated with impaired physical and/or cognitive functioning, which needs to be distinguished from symptoms of depression, which include a lack of self-esteem, sadness, and despair or hopelessness" [21]. From the current state-of-the-art, there is not a gold standard to quantify fatigue [22, 23]. Instead, many questionnaires have been developed to subjectively assess fatigue, such as the Brief Fatigue Scale, Fatigue Severity Scale, and Borg Rating of Perceived Exertion Scale [23, 24].

In summary, WMSDs pose a significantly adverse effect to both the aviation manufacturing industry and its workers. To minimize the risk of suffering WMSDs, this research aims to construct a real-time wearable sensor-based fatigue assessment system, in order to quantify fatigue using body motion, physiological data, and external force information. The accumulation of body fatigue can then be used as an indicator to alert workers before suffering WMSDs. For instance, in the case that the accumulation of body fatigue reaches a threshold, some proactive actions need to be performed, such as taking short breaks and changing task sequence.

1.2 Research Challenges

Although sensors have been widely used in the prevention of WMSDs and ergonomic analysis, there are still some areas that have rarely been explored [25-31]. Firstly, previous research has undertaken ergonomic assessment using either wearable or non-wearable sensors [28, 29, 32, 33]. Limited work has been undertaken that incorporates both wearable and non-wearable sensors together to perform ergonomic assessment. This is beneficial as data from both wearable sensors, such as body motion and physiological data, and non-wearable sensors, such as external force, can be combined together into one system to provide a more comprehensive set of data to monitor risks. Additionally, a comparison between features from wearable and non-wearable sensors for body fatigue classification has not been evaluated, which is essential, because the contribution from each type of sensor on detecting fatigue can be identified. In the condition that the contribution of one type of sensor outperforms another one significantly, the sensor system can be optimized by removing redundant sensors, as deploying both wearable and non-wearable sensors in a factory environment may incur high infrastructure costs, obtrusiveness, and privacy issues. Furthermore, there is a lack of systematic approaches to optimize sensor deployment by minimizing the number of sensors, in order to decrease obtrusiveness and increase comfort in a real factory application. The approach should take different

factors into consideration, such as fatigue classification performance, obtrusiveness, privacy issues, and data streaming quality.

Moreover, most studies investigate simple repetitive movements, such as lifting, lowering, reaching, working overhead, and carrying [25, 34-36]. Although daily factory work has been designed by combining these basic activities, it is still important to conduct ergonomic assessment on real factory tasks. It can be postulated that fatigue classification performance may deteriorate when assessing complex activities, as the classification model needs to detect the generality among basic activities. Moreover, previous research has been done in a strictly controlled laboratory environment to record error-free data [22, 26, 29, 32, 34, 37]. However, testing the system in a factory-like environment still requires investigation, as realistic conditions create external noise, such as signal interference, heavy foot traffic, and operating machines, all of which may have a negative effect on the data quality and on model performance. Lastly, most ergonomic monitoring systems have only been able to provide simple and straightforward feedback to workers without instructions, such as joint angles, body postures, and scores from ergonomic assessment tools [38-42]. Limited work has been done in constructing a real-time factory-wide ergonomic monitoring system, which takes both workers and factory operations into consideration, including real-time data collection, analysis, logging, and visualisation. From a worker's perspective, they should be able to receive not only body-related information, but also safety advice from factory operations. From the factory operations' point of view, they should be able to have a deeper understanding towards a group of workers in the same workplace, which can help in optimizing task design to prevent WMSDs. Finally, a bi-directional communication mechanism can also be established using smart devices, where workers are able to report any unexpected conditions to factory operations immediately, and factory operations can provide workers on-time instructions.

1.3 Aims and Objectives

Based on these research challenges, the following research question has been proposed:

What is an efficient and cost-effective system using wearable and non-wearable sensors that can detect fatigue with the goal of improving both workers' and factory operations' awareness of fatigue within an aviation manufacturing environment?

The aim of this project is to construct a real-time wearable sensor-based fatigue evaluation system for both aviation manufacturing workers and factory operations, including real-time data collection, analysis, logging, as well as visualisation, in order to improve knowledge and understanding of workers' body conditions towards preventing WMSDs. To achieve this aim, the objectives of this work are:

1. Designing a system framework for real-time fatigue assessment and feedback.
2. Designing a data processing procedure for multiple streams of artifact-sensitive data, including body motion, physiological data such as heart rate, and external force exerted on human body.
3. Evaluating the performance of different sensor systems, feature selection algorithms, and machine learning approaches for classifying fatigue during repetitive physical tasks.
4. Optimizing the sensor deployment by minimizing the number of sensors, in order to minimize obtrusiveness, avoid privacy issues, minimize computational cost, while maintaining fatigue assessment model performance.
5. Constructing a real-time wearable sensor-based fatigue assessment system in a factory-like environment that is capable of real-time data collection, analysis, and logging.
6. Investigating the machine learning model performance for classifying fatigue during real complex aviation-like tasks.
7. Evaluating the hardware system performance of wireless Bluetooth data streaming.

1.4 Novelties

In this thesis, a real-time wearable sensor-based fatigue assessment system has been constructed that addresses the aviation industry's need to design a cost-effective and scalable ergonomic monitoring system. In this way the system monitors workers' body condition (i.e. fatigue levels) to enable them to take proactive actions to prevent WMSDs before injury happens. This will in turn ensure workers' safety and decrease occupational health and safety cost. The research novelties have been listed below:

- **Development of a framework for a real-time ergonomic monitoring system for use in an aviation manufacturing environment using wearable and non-wearable sensors.**

The ErgoRelief framework has been proposed to achieve real-time ergonomic monitoring within an aviation manufacturing factory environment, using both wearable and non-wearable sensors. Compared with the ergonomic risk assessment system developed by Low et al. [43], the inclusion of both wearable and non-wearable sensors is both essential and a novelty because body motion,

physiological data, as well as external force all play an important role in fatigue assessment. The framework includes four components: real-time data collection, analysis, logging, and visualisation, as well as two stakeholders, the aviation factory workers and factory operations. In summary, the first novelty of this framework is that it includes not only wearable sensors, but also non-wearable sensors, in order to more comprehensively collect body-related information. Furthermore, the framework incorporated both aviation factory workers and factory operations, in order to provide workers with not only their body-related information, but also safety advice from ergonomic experts. Lastly, different types of data visualization dashboards have been designed, which enables factory operations to have a systematic overview about all workers' body conditions. In addition, workers can visualize their body condition through a smartphone application, as well as receive alarms through multiple ways, such as vibration and sound.

By implementing this framework, body-related information, such as fatigue, can be detected using multiple sensor measurements and machine learning algorithms. Concurrently, factory operations can visualise the body status of a group of workers, which can aid them to better allocate tasks. Furthermore, tasks can be optimized based on the history of workers' body conditions. Lastly, factory operations can provide workers with on-time safety notifications through smartphones. From the worker's aspect, they can visualise their body-related information, such as heart rate and fatigue levels. Meanwhile, they are also able to contact factory management immediately, to communicate any unexpected issues via smartphones. Factory operations have also been included during the data visualisation stage, in order to manage the factory efficiently and provide workers with professional advice.

Additionally, the framework has many advantages over current camera-based ergonomic monitoring systems. Such systems have difficulties in being implemented in an aviation manufacturing environment due to blocked field of view, changing illumination, and moving objects. It also requires significant infrastructure cost in order to include more hardware devices to process computationally intensive videos and images [17]. By implementing the ErgoRelief framework, issues suffered from camera-based systems can be mitigated by unobtrusively monitoring workers' body-related information through both wearable and non-wearable sensors. Furthermore, from an aviation industry's perspective, current mechanisms based on offline ergonomic monitoring systems are reactive. As such, ergonomic issues can only be observed *after* occurring, which have already resulted in health issues, workday loss, and/or increased cost to occupational health and safety. By applying this framework, monitoring workers' body conditions for instances of fatigue can be undertaken more

accurately and alerts can be provided to workers prior to injury. For example, workers will be alerted if they maintain a high body fatigue level for an extended period of time, which can help them in better allocating resting periods and adjusting task performance. In addition, from the factory operation's aspect, a systematic overview of all workers' body conditions can help them proactively optimise task sequence and design, which can therefore increase factory productivity and ensure workers' safety.

- **Development of a signal processing pipeline to process multiple streams of data from different systems for detecting body fatigue.**

In this research, a signal processing pipeline has been designed and implemented to process multiple streams of data in order to detect body fatigue, including steps for data pre-processing, feature extraction, data labelling, and body fatigue classification. Previous studies combined multiple sensors into one system with consistent timestamps [29, 44]. However, in this research, different sensors are operating separately, and data have been recorded using different sampling frequencies with different approaches, such as SD card data recording, and wired/wireless data streaming. Therefore, it is essential to properly synchronise data with inconsistent timestamps, in order to avoid time drift. Hence, a data pre-processing approach has been applied, including data synchronisation, segmentation, resampling, and filtering. Different sensor measurements being processed by this approach can be aligned to a consistent time basis, which can then be used for further analysis.

In addition, a novel data labelling approach for body fatigue classification has been implemented in this research. Previous studies represented body fatigue level for each repetition of activity using a subjective rating that was taken at the end of each repetition [22, 29, 35, 45]. However, we hypothesized that by obtaining subjective ratings at the beginning and end of each repetition, and thus calculating the change in score between them, is a better representation of fatigue. This is because the score taken before an activity occurs acts as a baseline to calculate change that has occurred over the course of an activity. Furthermore, rather than using a global threshold to classify fatigue for all participants, we labelled each repetition's data using a personalized threshold on a per task, per participant basis, since subjective feelings of fatigue differ between individuals.

Since different types of sensors have been included in this research, it is essential to comprehend the contribution of different sensors in classifying fatigue. Furthermore, based on different sensor systems, it is also important to select the optimal feature selection and machine learning approaches to achieve the highest accuracy for fatigue classification. Compared with previous studies [25, 44,

46, 47], which evaluated the performances of different feature selection and machine learning approaches for body fatigue classification, we also evaluated the performance of different sensors for classifying fatigue. This is essential, because the target of this research is to construct a real-time fatigue assessment system for real-world application. Therefore, the system design should take multiple factors into consideration, such as classification performance, obtrusiveness, privacy issue, and infrastructure cost. Hence, the result of the classification performance evaluation can be used for identifying the optimal sensor setup for body fatigue assessment, in order to optimize sensor deployment by minimizing the number of sensors.

After evaluating the performance of different sensor systems for detecting fatigue, a systematic approach for optimizing sensor deployment has been implemented, aiming at minimizing obtrusiveness, infrastructure cost, computational power, and privacy issues, as well as maintaining classification performance. The optimization procedure has been based on the fatigue classification result, correlation between different sensors, comfortability, and privacy protection, which addresses the research gap, since limited work has been done in previous studies in analysing the relationship between each individual sensor and reforming the sensor deployment.

Furthermore, this novelty addresses the industry's need for sensor system integration. In a real aviation factory environment, multiple types of sensors may operate independently, and it is impractical to use a pre-built system that integrates all sensors into one system. Therefore, by using this data processing pipeline, data from new types of sensors can be processed together with existing sensors, which significantly improves the system's extensibility.

- **Development of a real-time system using wearable sensors to detect fatigue during real and complex aviation-like tasks in an aviation factory-like environment.**

While some research has been completed to conduct ergonomic assessment in different industries, most of them analysed simple and repetitive physical tasks, such as stooping, lifting, lowering, and reaching [25, 26, 40]. Although factory tasks all consist of combining these basic tasks together, it is still important to evaluate the model's performance whilst undertaking complex tasks, as it may deteriorate due to task complexity and continuity. Additionally, most experiments have been conducted in a strictly controlled laboratory environment [22, 26, 29, 32, 34, 37]. However, evaluating the system's performance in a more realistic factory-like environment is necessary, as it may pose additional challenges to the system, such as operating machines, moving objects, and heavy traffic.

As a result, a real-time wearable sensor-based fatigue assessment system has been designed and implemented that undertakes real-time body fatigue detection during aviation-like tasks in an aviation factory-like environment. Body motion was recorded from wearable sensors and the performance of different feature selection and machine learning approaches on detecting fatigue has been evaluated.

Despite some real-time ergonomic assessment systems having been proposed in previous literature [28, 39, 41, 42, 48], the evaluation of the hardware performance has been rarely done, such as the latency, packet receiving rate, and battery consumption. These metrics are essential for a real-world application, as the system needs to not only be accurate, but also cheap and reliable. Therefore, we evaluated the hardware performance of our system to demonstrate that our system can provide reliable data streaming from multiple sensors for an extended period of time.

From an industry perspective, the real-time system accomplished the real-time data collection, analysis, and logging aspects of the ErgoRelief framework, which demonstrated the framework's feasibility. The real-time system can help factory operations to take proactive actions to prevent ergonomic issues and minimize workers' risks of suffering WMSDs, such as adjusting task sequence, allocating additional breaks, and optimising task design. In addition, the evaluation of the hardware performance provided industry with insight into the system's capability. In the future, based on the results of the hardware performance evaluation in this research, more sensors can be deployed and integrated into the system to monitor more workers simultaneously.

In summary, by accomplishing these novelties, a real-time wearable sensor-based fatigue assessment system has been constructed, which is able to detect body fatigue in real time, through measurements of body movements, physiological data, and external force exerted on the human body.

1.5 Thesis Structure

This thesis contains seven chapters, including the introduction, which have been organized as below:

Chapter 2 provides a review of the literature around both camera- and sensor-based ergonomic monitoring systems. A camera-based ergonomic monitoring system utilises both RGB and depth cameras to conduct an ergonomic assessment by tracking body posture. In contrast, sensor-based ergonomic monitoring systems utilise Inertial Measurement Units, physiological sensors, and non-wearable sensors to measure body motion, physiological data, and external force exerted on the

human body. The recorded data have then been used to identify improper postures, generate ergonomic scores, and quantify body fatigue.

Chapter 3 presents the ErgoRelief framework. Details of each stage of the framework have been displayed, including data collection, analysis, logging, and visualisation. Several front-end user interfaces have been designed for both aviation factory workers and factory operations about workers' body conditions with safety instructions.

Chapter 4 details the general methodology implemented in this thesis, including the sensor platform, data pre-processing procedures, data labelling approaches, and feature selection and machine learning approaches.

Chapter 5 discusses the first study with results and discussions, aiming at developing a machine learning approach for detecting fatigue during repetitive physical tasks. Thirty participants have been included in the experiment, with each of them completed four repetitive physical tasks that have been widely applied in daily factory work. Body motion, physiological data, such as heart rate, together with external force have been measured while participants performing tasks. The analysis has been conducted through both multiple linear regression and machine learning approaches on both per-task and overall basis. The feasibility of the ErgoRelief framework has been demonstrated first, followed by the optimal features for each machine learning approach. Lastly, a systematic approach to optimising sensor deployment has been illustrated based on the result to prepare the system for real-world application.

Chapter 6 details the second study with results and discussions. Rather than analysing simple repetitive movements in a strictly controlled laboratory environment, this study has been conducted in a workshop with aviation-like tasks, to simulate an aviation factory environment. Thirty participants have been included in the experiment, with each of them performing a 2-hour aviation manufacturing task. Body motion has been recorded during the experiment and a real-time fatigue assessment system has been constructed. The performance for the hardware system has also been evaluated relying on battery lifetime and packet receiving rate, to demonstrate that the system is capable for the real-world deployment. Lastly, the performances of different feature selection and machine learning approaches on the fatigue assessment using body motion have also been appraised to select the optimal combination for real-time fatigue assessment.

Chapter 7 concludes the thesis, which summarises major the findings and results. The limitations have then been elaborated, including experiment groups, experiment environment, and task design. Finally, the future work has been discussed, which details a few future tasks which are required to be finished to further improve the system.

Overall, these chapters cover the progress of designing a real-time wearable sensor-based fatigue assessment system in an aviation-like environment, including framework design, offline body fatigue classification evaluation utilising different sensors, feature selection algorithms, and machine learning approaches, and real-time system design and implementation in an aviation factory-like environment.

Chapter 2

Literature Review

Based on the measurement approach, current ergonomic monitoring systems can be classified as either camera-based or sensor-based [9]. For instance, the Microsoft Kinect was commonly used as a camera-based ergonomic monitoring system, which incorporated both RGB and depth cameras. Its skeletal tracking function allows people to measure body postures and joint angles, which can act as the data source for the ergonomic assessment [11].

Wearable sensors have been widely implemented in current sensor-based ergonomic monitoring systems to measure behavioural and physiological signals, in order to prevent WMSDs [49]. Inertial Measurement Units (IMUs) are the most common sensor for measuring behavioural signals, including body part acceleration, angular velocity, and joint orientation. A range of wearable sensors can also be used to measure physiological signals, including Electrocardiogram (ECG), Electroencephalogram (EEG), Electromyography (EMG), Photoplethysmogram (PPG), Galvanic Skin Response (GSR), as well as skin temperature [49].

Although wearable sensors have been extensively utilised in ergonomic assessment, they are not able to measure external force that has been exerted on the human body. This is important as prolonged exposure to excessive external force is one of the most important factors that contributes to WMSDs [3]. For instance, non-wearable sensors, such as force plates and load cells have been used in ergonomic assessment to detect external force [3, 50-54]. Force plates measure the ground reaction force, whilst load cells record hand exertion force. In conjunction with camera-based motion capture systems, human body biomechanical models have been constructed, which allow accurate muscle and joint analysis [32, 55].

Current ergonomic monitoring systems tend to detect instances of awkward postures, generate ergonomic scores, and estimate physical demand [26, 41, 44]. For instance, ergonomic scores are generated from a series of ergonomic assessment tools, including the National Institute for Occupational Safety and Health (NIOSH) lifting equation, Rapid Upper Limb Assessment (RULA), Rapid Entire Body Assessment (REBA), as well as the Strain Index [56-61]. The NIOSH equation evaluates lower back pain of a lifting task using three criteria: biomechanics, psychophysics, and

work physiology. A recommended weight limit can be estimated based on a few parameters, such as the distance between the object and body and floor, object moving distance, body twisting angle, and activity duration and frequency. In contrast, RULA and REBA assess a task using body postures and external loads for the upper and full body respectively [62, 63]. RULA utilises a series of joint angles on different body parts, such as neck, arm, and trunk, along with load weights to generate an ergonomic score. In addition, apart from joint angles in RULA, REBA also incorporates joint angles on the lower body in ergonomic assessment, such as the leg. Lastly, the Strain Index rates a task by six variables, which include intensity of exertion, duration of exertion per cycle, effort per minute, wrist posture, speed of exertion, and duration of task per day [64]. By multiplying these variables, the Strain Index can be calculated and classified into three levels. For tasks with Strain Indexes lower than 3 and greater than 7, they have been classified as safe and hazardous respectively. For tasks with values between 3 and 7, they may have an increased risk of WMSDs.

From the current state of the art, sensor-based ergonomic monitoring systems can be divided into two aspects: offline and real-time systems. Offline systems analyse ergonomic issues after data recording, while real-time systems can provide real-time feedback to users. Wearable sensors, including IMUs and physiological sensors, as well as non-wearable sensors, such as the force plate and load cell, will also be introduced in the literature review.

2.1 WMSDs and Fatigue

WMSDs include a vast array of inflammatory and degenerative symptoms, which are related to tasks and working environments [65]. These symptoms include pain, nerve disorders, and tendon inflammation in different body areas, such as the lower back, neck, shoulder, arm, hands, hip, knees, ankles, and feet [65, 66]. Workers from certain industries have a much higher chance of suffering WMSDs, including nursing, transportation, mining, food processing, and heavy and light manufacturing [65]. There are multiple causal factors of WMSDs, which can be summarised as occupational and non-occupational [65]. Occupational factors include heavy physical work, awkward postures, high repetition, and vibration, whilst non-occupational factors contain age, gender, smoking, diet habits, exercise, and sleeping quality [65, 66]. WMSDs occur in the general population, and are one of the major work-related diseases in many countries [65].

As discussed in the Introduction, body fatigue has been selected as an alternative indicator of fatigue, in order to mitigate the negative effect of ergonomic risk diversity towards preventing WMSDs. Body

fatigue has been a significant concern for workers in many industries, which may result in a decrease in the capability to perform tasks and a decline in the quality of life [67, 68]. There are a few cognitive characteristics that can be used to assess fatigue, including alertness and attention, cognitive control, motivation, and stress [67]. Alertness and attention can be measured using EEG sensors and questionnaires, whilst the cognitive control represents the time spent on processing information, which can be commonly measured by the reaction time. In addition, since motivation is a subjective feeling, it can only be measured through questionnaires. Lastly, the stress can be assessed through a few parameters, such as heart rate, blood pressure, and body movements [67]. Both self-report and direct measurement methods have been developed to quantify fatigue [68, 69]. The self-report method relies on workers filling in questionnaires, in order to subjectively assess their fatigue, such as the Fatigue Impact Scale, Chalder Fatigue Scale, and Piper Fatigue Scale [23]. In contrast, the direct measurement method utilised sensors to measure physiological or external data, such as heart rate, strength, and force variability [67]. The relationship between these parameters and fatigue has been explored first, fatigue can then be quantified directly through sensor measurements using this relationship.

2.2 Internet of Things (IoT)

The direct measurement method of ergonomic risks and fatigue utilising sensors has largely been achieved through the Internet of Things (IoT) [10]. IoT integrates computer-based systems with the physical real world, which constructs networks of electricity, sensors, network connectivity, and physical objects [19]. IoT is one of the most important revolutions in the next decade and is attracting significant attention from many industries [10, 18]. There are five major components in an IoT network, including radio frequency identification, wireless sensor networks, middleware, cloud computing, and IoT application software [18]. The radio frequency identification is responsible for automatically identifying objects, whilst the wireless sensor networks are implemented in conjunction with radio frequency identification, in order to better track the status of things, such as temperature and movement. In addition, the middleware acts as a software layer between IoT developers and software developers, allowing software developers to easily access components of IoT. Furthermore, the cloud computing provides users a back-end solution to handle significant amounts of computation and data, and the IoT application software enables the human-device interaction, which helps users to have a deeper and more straightforward understanding of the environment [10, 18].

The concept of the IoT has been widely implemented in the area of ergonomic assessment through sensors and cameras [70, 71]. With the innovation in micro-electromechanical systems, both cameras and sensors have been developing toward smaller size and higher computational power. Sensors are able to be affixed to different body areas or deployed around the workplace to capture body motion, physiological, environmental, and location information, whilst cameras are able to be deployed in the workplace to monitor body movements [49, 70-72]. These sensors and cameras can be combined into a wireless sensor network, and different types of health parameters can be integrated to perform ergonomic assessment [72]. A real-time IoT-based ergonomic risk assessment system named ActSen was implemented by Low et al. to conduct real-time ergonomic assessment, including wearable sensors, a data transmission gateway, and a cloud server [43]. Multiple wearable sensors were included in the system, such as accelerometers and pressure sensors, which measure body motion and insole pressure respectively. The Raspberry Pi then acted as a gateway to transmit data from wearable sensors to the cloud server through Wi-Fi. Data was then processed in the cloud server to estimate the RULA score, which can be displayed in an interactive dashboard to enable data visualisation. Results illustrated that the task classification accuracy was 73% utilising the logistic regression model. In summary, the ActSen system was able to conduct real-time ergonomic assessment and data visualization. However, the system's hardware performance has not been evaluated, such as the quality of simultaneous data streaming from multiple sensors and battery lifetime. Furthermore, the system extensibility has not been evaluated, which is essential, because an IoT system should be able to be extended with more sensors. Additionally, a case study was conducted by Greco et al. [73] to evaluate the biomechanical overload for manufacturing workers. Both EMG sensors and IMUs were affixed to different body areas in order to capture muscle activities and body motion, with data being transmitted to a computer using the Raspberry Pi as a gateway through Wi-Fi. The data was then processed in the computer to monitor muscle activities and joint angles. The results of the case study illustrated that there is a clear potential to implement an integrated IoT system in assembly lines to evaluate the biomechanical overload. However, the systems obtrusiveness in this study was significantly high. Multiple probes of EMG sensors needed to be attached to the spine region, which may cause discomfort. To construct a real-world IoT-based ergonomic monitoring system, sensors should not only record necessary information, but also not interrupt or distract workers while performing tasks.

Another IoT sensor-based construction workload measurement system was constructed by Moon et al. [74] to estimate load weight utilising plantar pressure insoles, which measure the foot's plantar pressure. Data was streamed to a PC first through Bluetooth, and then transmitted to the cloud server

through Wi-Fi. The cloud server is responsible for logging and analysing the raw data, as well as sending the analysed data to the PC for data visualisation. An experiment was conducted with 20 participants, and results illustrated that the average error rates for estimating load weight with loads greater and less than 10kg were 6.15% and 25.32% respectively. The system developed unobtrusively measured workload through plantar pressure insoles. However, worker's workload depends on not only load weights, but also other factors, such as posture and body motion. Therefore, more parameters should be included in the system to systematically analyse workload. Furthermore, a study was conducted by Campero-Jurado et al. [75], which developed a smart helmet for Industrial IoT, targeting at identifying ergonomic risks, such as lighting, temperature, air quality, and body movements. Hence, the smart helmet was designed to incorporate a few types of sensors, including a force sensitive resistor, accelerometer, temperature, air quality sensor, and light sensor, which are responsible for shock detection, fall detection, temperature monitoring, air quality monitoring, and brightness monitoring. Moreover, a microcontroller was also integrated into the helmet, which processed and transmitted the raw data to the web server. A lamp was also connected to the microcontroller, which lit up automatically when the environment's brightness was below the threshold. In the web server, multiple machine learning models were constructed to classify a few ergonomic risks using the collected data, including harmful air quality, lack of illumination, and very high force exertion. The classification results along with a summary of raw data were displayed on the web interface for data visualisation. Results illustrated that the Convolutional Neural Network achieved the highest accuracy on classifying ergonomic risks with a value of 92.05%. The smart helmet in this study was able to analyse multiple parameters simultaneously and provide feedback in real time to workers. However, due to the limitation of the microprocessor in this study, only a limited amount of data can be processed, which may cause misinterpretation of the data and lead to economic loss and human injuries.

In addition, an IoT ecosystem named sustAGE was developed by Papoutsakis et al. [76] to detect physical strain and fatigue for aging workers during assembly tasks. The heart rate monitor was responsible for detecting fatigue, and a weight was assigned to each specific range of heart rate, such as less than 80, 80 ~ 110, 110 ~ 130, and greater than 130. Total weights during a certain period were then utilised to calculate the fatigue score. The higher the fatigue score, the more fatigue that the worker was suffering. In addition, multiple stereo cameras were installed in the workplace to capture four types of improper body postures using joint angles on different body areas, such as waist, knees, and arms. Results illustrated that the improper posture classification accuracy was 71%. However, implementing a camera-based system in a factory environment requires a significant amount of

infrastructure cost, as many hardware devices are needed to process computationally intensive images and videos. Furthermore, although heart rate is an important indicator towards fatigue, it cannot be used alone to determine the body fatigue. More parameters need to be included to precisely detect body fatigue. Furthermore, an automated personal protective equipment pair check system using smart IoT was implemented by Yang et al. [77], in order to alert construction workers wearing improper personal protective equipment while performing specific tasks. The system incorporates three levels, including sensing, application, and storage. In the sensing level, multiple sensors are embedded in the hand tools and personal protective equipment, such as photoresistors, optical sensors, force stretchable resistors, and touch sensors. Each sensor is embedded with a Wi-Fi module. A processor was also embedded into the helmet to receive sensor data using Wi-Fi. Once a signal was received by the receiver indicating that a certain hand tool was used, the processor immediately checked the rules and determined whether a warning needs to be sent to workers in order to alert them to wear the corresponding equipment. Moreover, in the application level, data from the processor can be visualised through a web browser on both PC and mobile devices. Lastly, in the storage level, data could be recorded in the cloud server through Wi-Fi, and users were able to access historical data online. A preliminary experiment was conducted by two participants to simulate construction tasks. Results illustrated that the time difference between the actual non-wearing actions and corresponding warning actions was 1.229s, illustrating that the system was able to provide on-time instructions. In summary, this system constructed a low-cost IoT system by integrating multiple wireless wearable sensors, which were able to provide workers real-time warnings about their wearing states. However, Wi-Fi has been used in the system for wireless data communication, which consumes a significant amount of energy. Therefore, different ways of wireless data communication should be further explored, in order to support sensors operating for an extended period of time.

Furthermore, an IoT-based risk monitoring system was developed by Tsang et al. [78] to manage occupational safety risks in the cold supply chain. The system includes three major components, which are the wireless sensor network, cloud database services, and a fuzzy logic approach. The wireless sensor network is responsible for collecting environmental parameters, such as temperature, humidity, and lighting intensity, and the collected data were logged to the cloud database for further analysis. In addition, the fuzzy logic approach assessed the occupational safety risks of different cold chain parties utilising the recorded data coupled with the health conditions. Lastly, a web interface was developed for each operator in order to display environmental conditions and occupational risk levels. To demonstrate the system's feasibility, a case study was conducted in a cold chain which particularly handles frozen and fresh food. Results illustrated that cold-related accidents and injuries

can be significantly reduced by implementing this system. However, the fuzzy logic approach implemented in the system significantly relied on professional knowledge from experts. Instead, the application of other machine learning approaches should also be explored to improve the system's adaptability. Another hybrid wearable sensor network system towards IoT was implemented by Wu et al. [79] to improve workers' safety in the outdoor workplace. The system includes three major components, which are an IoT wearable sensor network, IoT gateway, and IoT cloud. The IoT wearable sensor network contains multiple sensors targeted at measuring physiological and environment parameters, such as heart rate, body temperature, ambient temperature, relative humidity, carbon dioxide, and ultraviolet light. In addition, the IoT gateway connected the wearable sensor system to the cloud through Wi-Fi and Ethernet connections relying on the Raspberry Pi. Lastly, the IoT cloud is responsible for analysing and logging the data, as well as displaying data to workers using both mobile application and web interface. By implementing this system, workers' safety can be improved by closely monitoring their body and environmental conditions. Nonetheless, more wireless technologies should also be tested, in order to reduce data communication latency. Furthermore, the performance of different machine learning approaches should also be evaluated, in order to improve the system's accuracy and efficiency.

In summary, although some studies have been conducted that construct IoT-based ergonomic monitoring systems, there are still outstanding research challenges that remain. Firstly, previous studies have not investigated the energy consumption of each IoT device, which is necessary because minimizing energy consumption will significantly increase the data analysis continuity, which will improve the efficiency of ergonomic monitoring [80]. In addition, current studies conducted experiments with a limited number of devices. As such, the systems extensibility should be further tested, as the system should establish a data communication mechanism in order to communicate with a large number of devices [80]. Lastly, previous studies did not evaluate the system's performance during periods of poor internet and/or limited bandwidth. Considering the large amount of data being collected by IoT devices, it is essential to develop a mechanism to process data while it cannot be transmitted to the server due to contingencies [80].

2.3 Camera-based Ergonomic Monitoring System

Microsoft Kinect has been widely used in ergonomic monitoring systems relying on its skeletal tracking function, with its feasibility in posture measurement validated by Clart et al. [11, 81]. An experiment was conducted by 20 participants to simulate three different postures, including forward

reach, lateral reach, and single-leg eyes-closed standing balance. Posture data was collected from both the Microsoft Kinect and a multi-camera 3D motion analysis system, which acted as the benchmark reference. Results illustrated that the Microsoft Kinect has comparable performance with the motion analysis system, with a correlation greater than 0.90 for most measurements. Therefore, results suggested that the Microsoft Kinect can be used as a tool for posture tracking.

A real-time RULA assessment system was developed by Manghisi et al. [11] utilising the Microsoft Kinect, which was able to locate 25 joint positions. Joint angles can then be derived using relative positions between adjacent joints. An experiment was conducted by one participant to simulate 15 static postures, including standing, kneeling, and sitting, and the Microsoft Kinect was placed in front of the participant while performing these postures. In addition, the RULA score of each posture was assessed by a professional observer as the reference. Results illustrated that the RULA score generated by the real-time assessment system matched perfectly with reference scores, with a difference in score of less than 1. Another real-time ergonomic monitoring system using the Microsoft Kinect was constructed by Martin et al. [13]. An application was developed to calculate and display the recommended weight limit during a lifting task based on the frequency and duration of awkward postures, such as hands above shoulders, hands below knees, and arm extension. Results illustrated that the recommended weight limit calculated by the application was consistent with visual inspection. However, the error increased when the hands were close to the body or the participant was twisting away from the camera.

Furthermore, Plantard et al. [16] evaluated the performance of the Microsoft Kinect in estimating a RULA score in real workplace conditions. Twelve participants were included in the experiment. The experiment was conducted in both laboratory and real workplace environments. The reference RULA score was calculated by a motion capture system in the laboratory environment and by ergonomic experts in the real workplace environment respectively. In the laboratory environment, participants were asked to manipulate boxes with different levels of occlusions towards Kinect, for instance, manipulating without box, with box, and with box and a 45-degree sensor placement. In addition, the experiment in the real workplace environment was conducted in a car manufacturing factory with five tasks. Results illustrated that RULA scores in more than 70% of the conditions could be calculated correctly in both laboratory and real workplace environments. Moreover, the Root Mean Square Errors of the RULA score estimation were less than 0.7 and 1 for the laboratory and real workplace environment respectively. Another study was conducted by Xu et al. [15] to assess 3D shoulder kinematics using a Microsoft Kinect while using a computer. An experiment was conducted

by eleven participants to simulate computer tasks, such as text typing, comprehensive reading, and icon clicking. Each participant was asked to perform three 10-minute sessions. During each session, both the Microsoft Kinect and a motion tracking system were used to capture body movements, with data from the motion tracking system used as reference data. Results illustrated that the Kinect sensor had better performance in estimating angles of shoulder flexion/extension, rather than shoulder adduction/abduction and shoulder axial rotation. In addition, results also demonstrated that placing the Kinect sensor in front of the participant yielded better measurement accuracy than placing it at the back.

Furthermore, Bortolini et al. [82] proposed a motion analysis system utilising four Microsoft Kinects to conduct automatic ergonomic risk assessment during manual manufacturing tasks. Four Microsoft Kinects were placed around workers to capture body posture. In addition, workers also needed to provide the system a few inputs, such as body height, workplace layout, product information, and tool position. By combining these inputs, an ergonomic score could be generated using the European Assessment Worksheet. To demonstrate the system feasibility, a case study was conducted to analyse ergonomic risks in a manufacturing factory, and results illustrated that the trunk frontal bending contributed significantly to the ergonomic score. This motion analysis system has been further developed and validated by Bortolini et al. [83] in an assembly station of industrial water pumps. Rather than generating a single score, the updated system was able to generate additional ergonomic scores, such as REBA. Moreover, the system was also able to provide some quantitative summaries about manufacturing activities, including upper limb movements, worker walking path, hand distribution, and time consumption on each activity. Results illustrated that the system could provide a precise estimation of all quantitative summaries. By implementing this system, production managers were able to better understand and improve the manufacturing process in order to protect workers' health.

A Microsoft Kinect has also been used to detect movement disorders in order to assess Parkinson's disease [84]. The assessment was mainly based on one task, which is a finger-to-nose maneuver. People with Parkinson's disease were not able to move their hands in a straight line. The Microsoft Kinect was placed above the participant and was mainly used to track hand positions. Features were then extracted from the hand trajectory, including the average amplitude, maximum amplitude, and frequency, and a Support Vector Machine (SVM) classifier was implemented to classify Parkinson's disease patients and control subjects. An experiment was conducted by 17 participants, with six of

them suffering Parkinson's disease and the rest were control volunteers. Results demonstrated that the classification accuracies were 82% and 89.5% for the right and left hand respectively.

Yu et al. [85] developed a joint-level vision-based ergonomic assessment tool for construction workers, which extracted skeleton data from 2D videos using deep learning approaches. The skeleton tracking data was then used to estimate the REBA score, which can be used to assess the task risk level. Two experiments were conducted in a laboratory environment and real workplace respectively. In the laboratory environment, participants were asked to perform three tasks to simulate real construction tasks, including rebar fixing, bricklaying, and plastering. A set of IMU sensors were affixed to different areas around the whole body to collect the reference data. Results illustrated that the mean errors for the joint coordinates and joint angle estimation were 4.10 cm/joint and -0.70 degrees respectively. Moreover, the joint score estimation accuracy was up to 96%. For the experiment in the real workplace, six workers were included to demonstrate the feasibility of the ergonomic assessment tool, including a bricklayer, concreter, pipelayer, bar fixer, scaffolder, and formwork erector. Similarly, to the laboratory experiment, IMUs were attached to the body to record the reference data. Results illustrated that the tool is feasible to be used in construction sites. Moreover, Zhang et al. [86] constructed a system for ergonomic posture recognition using a single ordinary camera relying on 3D view-invariant features. Nine postures were included in the experiment to evaluate the system performance, which can be described by joint angles from arms, back, and legs. In this system, joint angles were derived from the video using a convolutional neural network. Multiple classifiers were then implemented to classify postures using joint angles, such as SVM and k-Nearest Neighbour (kNN). Results illustrated that the posture classification accuracies for the three body parts were 94.9%, 93.9%, and 94.6% respectively.

Another 3D human skeleton extraction system was developed by Liu et al. [87] to conduct on-site ergonomic analysis using a stereo video camera. The skeleton extraction includes three steps, which are 2D body joint tracking, 2D skeleton matching, and 3D skeleton reconstruction. Two cameras were included in the system and affixed to different positions of the workplace, with each of them conducting body joint tracking first using an optical flow algorithm. Figures from each camera with joints being detected were then matched using feature descriptors, such as a scale-invariant feature transform, in order to conduct 2D skeleton matching. Finally, by combining pairs of corresponding body joints, camera parameters, and relative positions between cameras, the 3D human skeleton could be reconstructed, which could then be used for ergonomic analysis. An experiment was conducted in a laboratory environment by one participant to simulate two common activities in construction tasks,

including squat lifting and walking. The participant had been asked to perform each activity repetitively with multiple cycles. In addition, a commercial motion capture system named Optotrak was also included in the experiment to collect reference data. Results illustrated that the mean tracking error was 3.8cm, demonstrating that the system can conduct accurate skeleton extraction.

An automatic physical fatigue assessment system was developed by Yu et al. [88] for construction workers using an RGB camera and smart insoles. The RGB camera is responsible for capturing 3D body postures using a single RGB camera-based 3D motion capture algorithm, whilst the smart insole measures the external force exerted on the human body. By combining these two parameters, joint forces could be estimated using inverse dynamics. After that, joint forces were then used as the input for a physical fatigue assessment model, which quantified the physical fatigue based on the intensity, duration, and repetitiveness of the external workload and the personal muscle strength capacity. Two experiments were conducted to test the accuracy of body posture estimation and physical fatigue assessment respectively. The first experiment was performed in a laboratory environment with participants performing three construction tasks, including material handling, plastering, and rebar handling. Data was collected from both the camera and an IMU system, which acted as the reference system. Results illustrated that the mean error of 3D joint location was 3.9cm, and the standard deviation was approximately 1.59cm. For the second experiment, the participant had been asked to perform a scaffolding task and a masonry task and their fatigue levels were derived in real time. Results illustrated that the joint capacity contributed significantly to the fatigue index. The lower the joint capacity, the higher the fatigue index is. Furthermore, the highest fatigue index occurred on the left hip and right knee during the scaffolding task, indicating that the participant should pay more attention to these two joints while performing this task. Moreover, the fatigue index for the left leg was five times higher than the right leg during the masonry task, illustrating that the participant should balance the body better when performing this task. Overall, results illustrated that the system was able to provide joint-level physical fatigue assessment automatically.

Subedi and Pradhananga [89] developed an approach to prevent back injuries for construction workers utilising the body posture recorded by a Microsoft Kinect. Each unique body action could then be classified using a random forest classifier. After that, the lower back moment was calculated using a random forest regression algorithm based on the inverse kinematics and inverse dynamics relying on the body postures and velocity. Finally, the theoretical maximum attainable level of safety was calculated based on the body postures and joint moments. An experiment was conducted by three participants to simulate two construction tasks, including object lifting and lowering. Results

illustrated that the accuracy for the body action classification was up to 92.6%, and the root mean square error of the back moment estimation was 10.644 Nm between the results from the reference data and the random forest regressor. Lastly, the theoretical maximum attainable level of safety was computed by combining the unique body actions exerting minimum lower back moment, which can help safety managers to better design and allocate tasks.

2.4 Offline Sensor-based Ergonomic Monitoring Systems

2.4.1 Wearable Sensor Systems

A digital assembly glove was developed by Kucukoglu et al. [90] in order to identify defective assembly processes during connector assembly tasks. Both vibration and force sensors were attached to each finger of the glove, and an artificial neural network was developed to detect defective connections using the vibration and external force data. An experiment was conducted by 17 participants to simulate an assembly task with both proper and defective connections, including holding, inserting, and releasing. The collected data was pre-processed first, which then acted as the input for artificial neural network training, validating, and testing. Results illustrated that the classification accuracy was 95%, illustrating that the system was applicable for detecting defective operations. Furthermore, Petz et al. [91] developed a wearable sensor shirt system to conduct real-time monitoring of body postures for Occupational Health and Safety. Nine IMUs were weaved to the shirt to capture upper body movement. Furthermore, a data transmission node was weaved on the hip area with Wi-Fi and Bluetooth capabilities, which was connected to each IMU and was responsible for data transmission and power supply. An experiment was conducted with a participant in a laboratory environment to simulate a series of movements, including rotating, taking a parcel from a desk, placing a parcel on a desk, taking a parcel from a shelf, and placing a parcel on a shelf. An SVM model was trained using the recorded data for activity classification. Results illustrated that the classification accuracy was 80.4%.

Nath et al. [25] conducted a study to monitor task duration and frequency in order to prevent WMSDs. The system contains two stages: activity recognition and ergonomic risk assessment. Two smart phones were strapped to the upper arm and waist to collect body acceleration, linear acceleration, and angular velocity through their internal IMUs. The data was then used to identify three categories of tasks using an SVM, including lift/lower/carry, push/pull, and other no-risk activities. Once an activity was detected, its corresponding duration and frequency were calculated, which were then used for ergonomic assessment. In this instance, the larger the duration and frequency during a certain

period, the riskier the activity is determined to be. An experiment was conducted by two workers and the results illustrated an activity classification accuracy of 99.6% using the arm-mounted smartphone, with the error for the duration estimation of less than 10.8%. In other work, Chen et al. [26], identified improper postures in construction workers, including working overhead, kneeling, back bending forward, squatting, neck bending, and reaching, using a motion capture system named Perception Neuron with 17 IMUs. A camera has also been included in the experiment for the purpose of reference. This setup captured full body motion and joint angles. Motion tensor decomposition and an SVM were used to classify improper postures and resulted in a classification accuracy of 82.79%.

Physiological parameters have also been widely included in ergonomic analysis systems to measure workload. A wristband-type wearable device was designed by Hwang and Lee [92] to measure construction workers' physical demand by recording heart rate, which was derived from the PPG signal. Percentage of Heart Rate Reserve (%HRR) was calculated based on the current, maximum, and resting heart rate, and was used to gauge physical demand. Results demonstrated that working patterns, task types, individual factors, as well as environmental factors all contributed to the accumulation of physical demand. Another framework was proposed by Li et al. [93] to evaluate muscle activity during repetitive manual material handling using EMG sensors. The performance of EMG sensors for calculating muscle forces was compared with a human body model constructed by video recording. Results illustrated that the EMG sensors can be used to visualise muscle activities. However, it only worked properly with superficial muscle bundles in low fat areas. In addition, Sadat-Mohammadi et al. [46] assessed physical demand using a wearable respiration sensor during construction tasks. An experiment was conducted by 15 participants to simulate construction tasks with three different levels of physical demand. Participants had also been asked to complete the NASA-TLX questionnaire in order to record their physical demand after each task. Respiratory data was collected and filtered with features being extracted. Multiple machine learning approaches were then applied on the feature set, including kNN, SVM, artificial neural networks, and random forest. Results illustrated that the random forest classifier achieved the highest accuracy, with a value of 93.4%.

Shakerian et al. [47] constructed a heat-stress risk assessment system, targeted at estimating workers' heat strain during construction tasks utilising physiological parameters, which were measured by a wristband biosensor, including skin temperature, a PPG signal, and a GSR signal. An experiment was conducted by 18 participants to simulate a roofing task, including hammering, attaching, and adjusting shingles on a tilted surface. Participants were asked to perform the task within three

different climatic conditions, and the physiological data were recorded while participants performed the task. The recorded data was then processed, which included artifact removal, data resampling, and feature extraction. Each sample was then labelled in two different ways in order to assess participants' heat stress and heart strain. The heart stress included low-, moderate-, and high-heart stress levels, which was determined by the humidity and ambient temperature, whilst the heat strain was categorized into safe, alarm, and danger zone based on participants' subjective feedback. Multiple supervised machine learning models were trained based on the feature sets, including random forest, artificial neural network, SVM, and kNN, and results illustrated that the accuracies for the heat stress and heat strain classifications were up to 89% and 92% respectively.

Aryal et al. [22] designed a physiological sensor-based fatigue monitoring system, which used a heart rate monitor, an EEG sensor, as well as four infrared temperature sensors to estimate fatigue. An experiment was conducted using 12 participants to simulate manual material handling tasks. While performing each task, participants were also required to fill in the Borg Scale Questionnaire [24] to indicate four fatigue levels. A series of machine learning approaches were also tested, including SVM, Complex Tree, and Boosted Trees. Results illustrated that the highest accuracy of 82.60% was achieved using the Boosted Trees. Tsao et al. [35] used non-invasive wearable sensors to measure fatigue during manual material handling tasks, including load lifting/lowering and turning. Respiration, PPG, as well as GSR signals were recorded for four 5-minute sessions for each participant. During each session, participants were asked to perform each task with a certain pace, in order to incur different feelings of fatigue. Participants also recorded their subjective rating of fatigue using the Borg Scale Questionnaire [24]. Features were extracted from the physiological measurements using a 1-minute sliding window and fatigue was estimated based on multiple linear regression models. Results indicated that the classification accuracy for all participants was up to 75%.

Jebelli et al. [44] also assessed the physical demand of construction workers using wristband-type biosensors, which measured PPG, GSR, as well as skin temperature signals. An experiment was conducted at two construction sites to collect data from 10 participants. Recorded activities were divided into low and high intensities, and included walking, moving light materials, and moving heavy objects. Each worker's physical demand was labelled by an energy expenditure prediction program that relied on video recordings. A series of supervised machine learning algorithms were tested, including kNN, multilayer perceptron, and decision trees. Results illustrated that Principal Component Analysis (PCA) with a Gaussian kernel SVM achieved the optimal performance, with an

accuracy of 90% for the binary physical demand classifications. The study also evaluated the effect of window size on model performance. Results illustrated that a 30-second window produced the optimal performance for recognising physical demand and that the PPG and skin temperature signals were more important than the GSR signal for detecting physical demand.

Lee et al. [27] used an ECG sensor (to derive heart rate) and accelerometer, in conjunction with meteorological data, to estimate physical demand of on-duty and off-duty construction workers. Data was collected continuously for 5 days. During this period, workers were asked to fill in an array of questionnaires to indicate their stress levels, sleep quality, diet history, and strength levels. An analysis of variance (ANOVA) test was performed between the recorded data and questionnaire answers. Results illustrated that workers with relatively low sleep quality were more likely to consume more energy and lose more productivity. Another ergonomic analysis system for construction workers was developed by Cheng et al. [28], aimed at identifying and locating unsafe postures. Heart rate, breathing rate, and acceleration, as well as worker's location and video clips were recorded, and an algorithm was developed to localise ergonomically risky actions. Three construction tasks were simulated in a controlled indoor environment for data collection. Results demonstrated that the classification accuracy of improper actions with reference to visual inspection was up to 92.4%.

Maman et al. [29] proposed a penalised regression model to predict physical fatigue in a workplace using IMUs and a heart rate monitor. IMUs were affixed to the ankle, wrist, hip, and torso to record body motion, whilst a heart rate monitor was strapped to the chest to measure heart rate. An experiment was conducted with eight participants. Each participant finished three manual material handling tasks in three days, with each of them lasting for 4 hours. Participants filled in their subjective rating of physical fatigue every 10 minutes and a threshold was designed to classify high/low physical fatigue. A penalised regression model was then created with physical fatigue as the output, and heart rate and body motion as inputs. Results illustrated that the sensitivity and specificity of the developed model was 100% and 79% respectively. Garimella et al. [94] also utilised a heart rate monitor to estimate body fatigue, which was recorded using the Borg Scale Questionnaire. An experiment was conducted by 3 participants to simulate a manual material handling task, including lifting and carrying sandbags from one point to another, with a distance of 10 meters. The experiment lasted for 7 days for each participant. During each day, participants were asked to perform 100 instances of load lifting and lowering. A few machine learning models were trained and validated

using 10-fold cross validation. Results illustrated that the Polynomial SVM achieved the highest accuracy, with a value of 76.92%.

2.4.2 Mechanical Sensing Systems

Lee et al. [30] used two force plates along with a camera to analyse ergonomic risks during minimally invasive surgery. Seven participants were included in the experiment to simulate a camera navigation and retraction task. Statistical analysis results illustrated that the increase in distance between the camera head and holder increased the ergonomic risks. Additionally, the left or right leg supported 70% to 80% of body weight disproportionally, which may lead to a series of ergonomic risks. Chen et al. [36] incorporated an insole shaped plantar pressure recording system to identify ergonomic risk factors, such as improper postures, force exertions, and excessive repetitions. An experiment was conducted by 8 participants to simulate both quasi-static and sequential postures. Features were then extracted, including pressure magnitude, pressure centre, as well as contact area. A model has then been developed based on a convolutional neural network and long short-term memory, in order to recognise postures and count the number of repetitions. Results illustrated a 100% accuracy for quasi-static posture recognition, as there is a clear difference between plantar pressure between different postures. However, the accuracy for sequential motion recognition decreased to 74%, as there are many faulty recognitions between adjacent postures, such as bending and carrying. Furthermore, the accuracies for the force recognition and repetition detection were up to 100% and 80% respectively.

A novel framework was proposed by Kong et al. [32] to measure mechanical energy expenditure of workers using a 3D biomechanical model, which was constructed using 2D video and smart insoles. These videos and insoles were responsible for estimating 3D poses and measuring foot pressure and plantar acceleration, respectively. An individualised model with joint displacements and joint forces was developed to estimate joint energy consumption. An experiment was conducted in a laboratory environment by a male participant to simulate tasks, such as walking, stooping, and carrying. Results illustrated that the energy expenditure varied significantly between joint positions and activities. For instance, the lumbar spine had the highest level of energy consumption during stooping, indicating that it is highly possible to accumulate fatigue and proactive actions, such as stooping slowly, need to be conducted. Overall, results demonstrated that the framework could approximate daily energy expenditure and provide workers with suggestions about preventing WMSDs.

Another study was performed by Plamondon et al. [95] to evaluate the effect of three factors towards manual material handling tasks, including expertise, lifting height, and weight lifted. In order to

evaluate expertise, 30 participants undertook the experiment, including 15 experienced and 15 novice workers. An experiment was then conducted with two different load weights and lifting heights. Data was also collected from a force plate and an optical tracking system. Participants were permitted to perform the lifting task using their most favoured postures. Results of an ANOVA analysis illustrated that expertise had a significant influence on the lifting tasks, as experts had less lumbar spine flexion, less trunk inclination, and more left knee flexion than novices. Overall, lifting height and load weight affected back loading and posture variables more than expertise, and the lifting task was riskier with a lower lifting position and heavier load weight.

A study was conducted by Knapik and Marras [31] to assess spine loading at different lumbar levels during a pushing/pulling task, which was achieved using a load cell and EMG sensors. An experiment was conducted with 20 participants to simulate pushing/pulling tasks with three handle heights and three handle force levels. An ANOVA was performed, and results indicated that activity, handle height, and handle force levels influenced the spine loading. Additionally, results recommended pulling rather than pushing, as pushing activities are more likely to pose risks to the spine, and that pushing and pulling loads less than 20% of the body weight are considered safe. Additionally, Chow and Dickerson [55] assessed the effect of gender, handle height, exertion type, and handle orientation on manual force strength and joint moment of pushing/pulling tasks. Twenty-four participants were included in the experiment, including 12 males and 12 females. A load cell, in conjunction with a VICON motion capture system, were used to reproduce full body motion. Different exertion types, handle heights, and handle orientation were tested in the experiment. Results of an ANOVA demonstrated that exertion type has the largest effect on manual force strength, followed by gender and handle height. Additionally, gender and handle orientation significantly influenced joint moments. Overall, results indicated that push exertions were preferred to pull exertion and a horizontal handle orientation produced greater resultant moments for all joints, without increasing manual force strength.

McDonald et al. [33] conducted a study in order to investigate the factors that influence isokinetic shoulder strength, as well as develop a model to estimate isokinetic shoulder flexion and extension strength from isometric strength. Fifteen females were included in the experiment and performed a set of shoulder movements with different movement planes, angular velocities, and grip type. Data was recorded from a dynamometer and two linear regression models were constructed with grip type, velocity, angle, and direction on peak isokinetic torque as inputs, and isokinetic flexion and extension

as outputs. Results illustrated that 59% and 67% of isokinetic extension and flexion can be explained by independent variables using each of the two linear regression models.

2.5 Real-Time Sensor-based Ergonomic Monitoring Systems

The number of repetitions in a certain period is one of the essential parameters for detecting ergonomic risks. Therefore, a study was done by Pernek et al. [38] to count exercise repetitions during resistance training using the IMUs situated inside smart phones. This has been achieved using dynamic time warping, which tries to match the incoming data with a reference pattern. In the meantime, a smartphone application has been designed to display an athlete training information, such as feedback on recognized repetitions, including whether they have been too slow, too fast, or correctly timed. The experiment was conducted by 10 participants with different exercises, workloads, and environments. Results illustrated that the F-scores for repetition recognition from different exercises and environments were up to 99.3%.

Another real-time IMU-based motion warning system was designed by Yan et al. [39] to identify ergonomically risky postures for construction workers. Two IMUs were attached to the helmet and back to measure body part acceleration, angular velocity, and orientation. The recorded data was transmitted to a smart phone through Bluetooth and processed. A set of thresholds have been developed between the maximum holding time and trunk inclination. Once the maximum holding time is over the threshold, a warning beep and vibration were generated by the smartphone to alert workers to adjust their improper postures. Additionally, data had also been logged to a MySQL database through Wi-Fi for further analysis. Results demonstrated that the system was able to effectively help workers adjust their postures.

Valero et al. [37] proposed an IMU-based activity tracking system, which strapped 7 IMUs to the arms, back, and shins to deduce joint angles and identify several postures. A preliminary experiment was conducted in a laboratory environment to simulate lifting movement. Three poses of lifting, including stopping, squatting with back bending, and squatting without back bending, have been classified as high, medium, and low risk levels, respectively. An interface has then been developed to display the current posture with its corresponding risk level. Results illustrated that the system was capable of identifying three different poses of lifting. Since each pose has been assigned its corresponding risk level, results demonstrated that the system could conduct accurate risk level classification. Furthermore, Valero et al. [40] also designed an Activity Tracking with Body Area

Network (AT-BAN) system. Eight IMUs were placed on the lower back, upper back, upper legs, arm, and lower legs to calculate joint angles and identify postures, such as back bending, squatting, and arm elevation. A corresponding posture score has then been generated by taking joint angles and posture durations into consideration. Lastly, a user interface was designed to display the history of identified motions with durations.

Another real-time ergonomic feedback system for manufacturing workers was developed by Vignais et al. [41]. A model was used to calculate a RULA score with input from 6 IMUs and 2 goniometers placed on the upper body. In addition, a visual feedback interface was designed to show workers a human body model with highlighted risky body parts. Two groups of participants were included in the experiment and performed a series of manufacturing tasks. One group had access to the feedback interface, while another group did not. Results illustrated that workers who had the visual feedback had a much lower RULA score than those without feedback, indicating that the system can help workers adjust their ergonomically risky postures. In addition, a study was conducted by Maior et al. [96] that measured mental workload during high-workload tasks and analysed the relationship between feedback and task performance. Objective mental workload feedback was recorded from a headband sensor monitoring brain activity using functional Near-Infrared Spectroscopy (fNIRS), which measures changes in hemoglobin concentration. In the meantime, self-reported feedback was also collected through questionnaires during each task. Results indicated that there is a strong correlation between fNIRS and task demand, illustrating that the workload can be assessed without relying on subjective feedback. Furthermore, results also illustrated that self-reported feedback hindered task performance, while objective real-time feedback did not pose any negative effect on the task performance.

Huang et al. [97] developed a wearable sensor-based system for assessing WMSDs in the workspace utilising the Xsens motion capture system, which affixed 17 IMUs to different body areas in order to capture the body posture. The RULA and REBA scores can then be generated using the joint angles. In addition, by combining the joint angles with body weight, body height, and load weight, the biomechanical load can be estimated through the Xsens-based ergonomic assessment system. Finally, two user interfaces were developed to display information to the workers about current RULA and REBA scores, and external force exerted on the human body. To demonstrate the system's feasibility, an experiment was conducted by 20 participants to simulate a series of physical tasks, including handling, painting, and welding. Two ergonomic experts rated the RULA and REBA scores visually to generate the reference data. Furthermore, 3DSSPP, which is one of the most famous biomechanical

analysis programs, has also been utilised to generate reference data for the lower-back compression force. Results illustrated that the classification accuracies for the RULA and REBA scores were greater than 88%, whilst the correlation for the lower-back compression force between the developed and reference systems was greater than 0.89.

Another novel wearable system was designed by Peppoloni et al. [48] that conducted real-time risk assessment during repetitive tasks. The parameters of the RULA and Strain Index have been measured by IMUs and EMG sensors worn on the left arm, where IMUs measured joint angles, angular velocities, and angular acceleration, whilst EMG sensors measured the level of fatigue. Two user interfaces were designed, with one of them showing a visual animation of the RULA score and Strain Index on each joint, and another displaying the current scores with the number of cycles. Results illustrated that the estimation accuracies of the RULA score and Strain Index were 94.79% and 44.79% respectively compared to the visual evaluation. In addition, a wearable sensor-based posture monitoring system was developed by Carbonaro et al. [98] for surgeons in order to prevent WMSDs. Three IMUs were affixed to the head, back, and hip in order to measure the flexion, lateral bending, and twisting angles of spine and neck during surgeries. The recorded data was then passed into the RULA scoring system to generate RULA scores. An experiment was conducted by a surgeon during a real laparoscopic surgery operation. During the experiment, a camera was also used to record reference data. Results demonstrated that the surgeon maintained uncomfortable static postures for an extended period of time, which significantly increased the risk of suffering WMSDs. Furthermore, neck was extended during the whole surgery because the surgeon needed to watch the monitor. Therefore, the operating room layout needs to be better designed to reduce this ergonomic issue.

Yi et al. [42] created an early-warning system to monitor construction workers' heat strain level in a hot and humid environment. Heart rate, environmental temperature, humidity, air velocity, solar radiation, along with the personal characteristics of workers, work duration and activities were recorded by a smart bracelet, smartphones, and environmental sensors. Heat strain level was classified into four levels (mostly safe, caution, warning, and severe) based on a set of thresholds from the heart rate data, and a three-layer Artificial Neural Network (ANN) model was constructed to estimate heat strain levels from all sensor measurements except heart rate. A real-time feedback loop was also developed in this study. Data from all sensors except the heart rate monitor was recorded and the heat strain level was derived from smartphones, and the results were both stored in the cloud for data visualisation by supervisory staff, and issued to workers with corresponding interventions, such as drinking fluids and to avoid working under direct sunlight. An experiment was conducted with thirty-

nine participants in the field. Results illustrated that more than 90% of the variation of the heat strain level can be explained by the ANN model.

Lind et al. [99] developed a smart workwear system, targeted at monitoring adverse postures while workers perform tasks, as well as providing real-time haptic feedback in order to adjust their postures. Two IMUs were attached to upper arms in order to record postures, and a vibration actuation unit was affixed to the dominant arm to provide haptic feedback. Both acceleration and angular velocity data were transmitted to a smartphone application first. Coupled with the wearer's gender, age, weight, and height, the ergonomic assessment can be performed through a real-time risk assessment algorithm. The haptic feedback can then be provided depending on the assessment result. An experiment was conducted by 16 participants to simulate a mail sorting task, and the haptic feedback would be provided if the upper arm elevation angles were beyond 30 degrees for more than 50% of the work time, 60 degrees for more than 10% of the work time, or 45 degrees for more than 15% of the work time. Results illustrated that the haptic feedback could significantly decrease the period of adverse postures, and workstation design significantly influenced the postures while performing the task.

2.6 Chapter Summary

In summary, camera-based ergonomic monitoring systems have mainly been designed and implemented in a laboratory environment. The Microsoft Kinect, which incorporated both RGB and depth cameras, has been widely utilised to capture body posture and estimate joint angles utilising its skeletal tracking function. These parameters can then act as the input for ergonomic assessment tools in order to generate ergonomic scores and prevent WMSDs. However, there are significant challenges when implementing camera-based ergonomic monitoring system in a real factory environment due to occlusion, illumination change, camera positions, and cluttered background.

Both wearable and non-wearable sensors were included in sensor-based ergonomic monitoring systems. Wearable sensors, including IMUs and physiological sensors, were implemented to identify awkward postures, estimate task duration and frequency, assess ergonomic scores, as well as to predict physical demand. Non-wearable sensors were mainly used together with camera-based motion capture systems for detailed ergonomic assessment of joints and muscles. However, conducting ergonomic assessment by combining both wearable and non-wearable sensors has rarely been done, which is necessary, as body motion, physiological conditions, and external forces all contribute to the

analysis of ergonomic risk. The result can be made more comprehensive by taking not only body-related information, but also external parameters into consideration.

Additionally, most sensor-based ergonomic monitoring systems are designed to analyse simple repetitive physical tasks, such as lifting, stooping, reaching, and working overhead. As real factory tasks are designed by combining such repetitive physical tasks, it is essential to evaluate the model performance on more realistic complex factory work, as the model needs to identify the generality among simple activities. Furthermore, most studies were performed in a strictly controlled laboratory environment to record error-free data. It can be postulated that a more realistic factory environment will pose additional challenges to the system, which will decrease the model's performance, such as signal interference, operating machines, and moving objects. It is important to design and test the system in a realistic factory environment, in order to conduct real-world ergonomic analysis.

Lastly, for real-time sensor-based ergonomic monitoring systems, limited work has been done on evaluating the system's hardware performance during data collection, analysis, logging, and visualisation, which is essential, as the system needs to not only be accurate, but also stable and durable. Hardware performance needs to be further evaluated prior to a real-world application, in terms of battery lifetime and packet receiving rate to ensure that the system can operate continuously for an extended period of time with reliable wireless data streaming.

Authorship Declaration

Chapter 3 incorporates sections from the following publication:

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	Liu	**Dobbins**	**D'Souza**	**Phuong**
Research Gap Identification	75%	10%	5%	10%
Experiment Design	80%	10%	5%	5%
Experiment Data Collection	100%	0%	0%	0%
Experiment Data Analysis	80%	15%	5%	0%
Paper Writing and Editing	70%	20%	5%	5%

Chapter 3

System Framework

As the first step in constructing a real-time fatigue detection system, this chapter details the design of the ErgoRelief framework, including key components, stakeholders, and data flow. Examples of data feedback dashboard design will also be displayed, targeting at providing real-time feedback to both aviation factory workers and operations.

3.1 Introduction

To achieve real-time ergonomic assessment, a framework (ErgoRelief) has been proposed in this thesis to accomplish real-time data collection, analysis, logging, as well as data visualisation. The aim of this framework is to detect fatigue during repetitive physical tasks, whilst providing real-time feedback to both manufacturing workers and factory operations. The context of this work has been grounded in an aviation factory environment.

3.2 System Requirements

Table 3.1 illustrates the system requirements of constructing a real-time factory-wide fatigue assessment system, including scalability, capacity, availability, reliability, recoverability, maintainability, security, manageability, and usability. The description of each requirement is discussed in the table below.

Table 3.1 System Requirements

System Requirement	Description
Scalability	The system should be scalable with the addition of more devices and users that allow multiple devices to operate simultaneously. The extension should be simple without influencing the existing setup.
Capacity	The system should be able to include multiple sensors, local PCs, and smart devices in an aviation factory environment, with each of them functioning properly.

Availability	During the data visualisation stage, when factory operations and/or workers want to visualise data during a certain period, smart devices should be able to immediately retrieve data from the cloud database.
Reliability	During the data collection stage, sensors should be able to stream data to local PCs continuously and stably without disconnection. In addition, the fatigue classification model operating on the cloud server should also be able to execute steadily. It needs to handle exceptions properly, such as invalid packets.
Recoverability	During the data collection and analysis stages, the sensor connection and fatigue classification model may stop randomly due to unexpected conditions. In this case, the system should establish a reconnection/restart mechanism, allowing sensors and detection models to restart smoothly without interrupting other functionalities.
Maintainability	In the case that the sensors and/or local PCs are down, there should always be a backup plan. For sensor issues, replacement sensors should be provided and added in the system, whilst for local PC issues, multiple streams of data should be able to be redirected to a replacement PC for continuous data streaming.
Security	When logging data from a local PC to a cloud server, the data should be encrypted to prevent data access by unauthorized entities. In addition, data access permission to the cloud database for factory operations and workers should be configured properly. Factory operations should be able to access all workers data, but the data should be deidentified. In contrast, workers should only be able to access their own data.
Manageability	When the system is operating, factory operations should be able to access the status of each sensor and local PC, which should indicate whether they are connected and operated normally.
Usability	Although the fatigue assessment system is complicated, all front-end user interfaces should be simple and straightforward. From the

	factory operation's perspective, they should be able to access all information through simple clicks, rather than understanding the detailed data format in the database. Furthermore, from the factory workers' aspect, the body-related information displayed on the smartphone application should be straightforward, comprehensive, and constructive.

3.3 The ErgoRelief Framework

Figure 3.1 demonstrates the system outline, which includes a real-time feedback loop. The system is composed of three segments, including data collection, analysis, and visualisation, along with two stakeholders – aviation manufacturing workers and factory operations. Details of each component will be explained in the following sections.

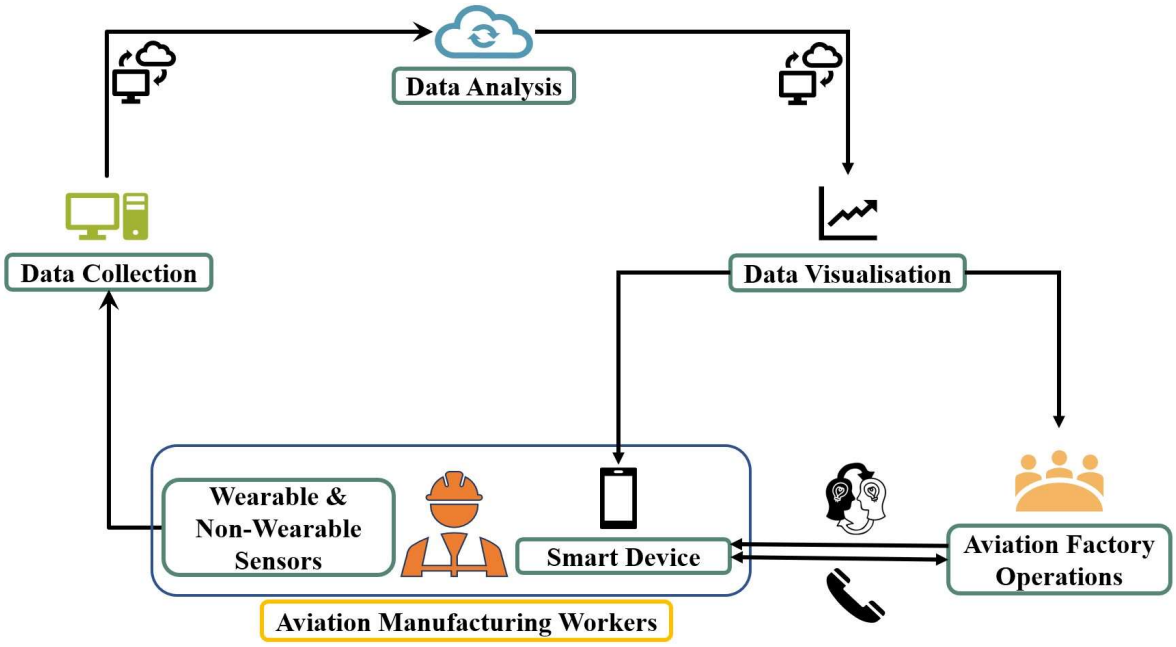


Figure 3.1 ErgoRelief Framework

3.3.1 Data Collection

The first component of the system is raw data collection. As illustrated in Figure 3.2, two types of sensors are included in ErgoRelief as the data sources to collect body-related information, which are wearable and non-wearable sensors. Wearable sensors are affixed to different body areas to collect body part motion and physiological data, such as heart rate, while non-wearable sensors, such as force

plates and load cells, are deployed in the workplace around workers to measure ground reaction force and hand exertion force. Therefore, both biometric and external factors related to body fatigue can be recorded simultaneously and unobtrusively.

Data from both wearable and non-wearable sensors are then transmitted to local devices, such as PCs, laptops, and smartphones, through either a wireless or wired connection. A packet is then generated, including body motion, heart rate, and external force measurements, which is then ready for uploading to the cloud server.

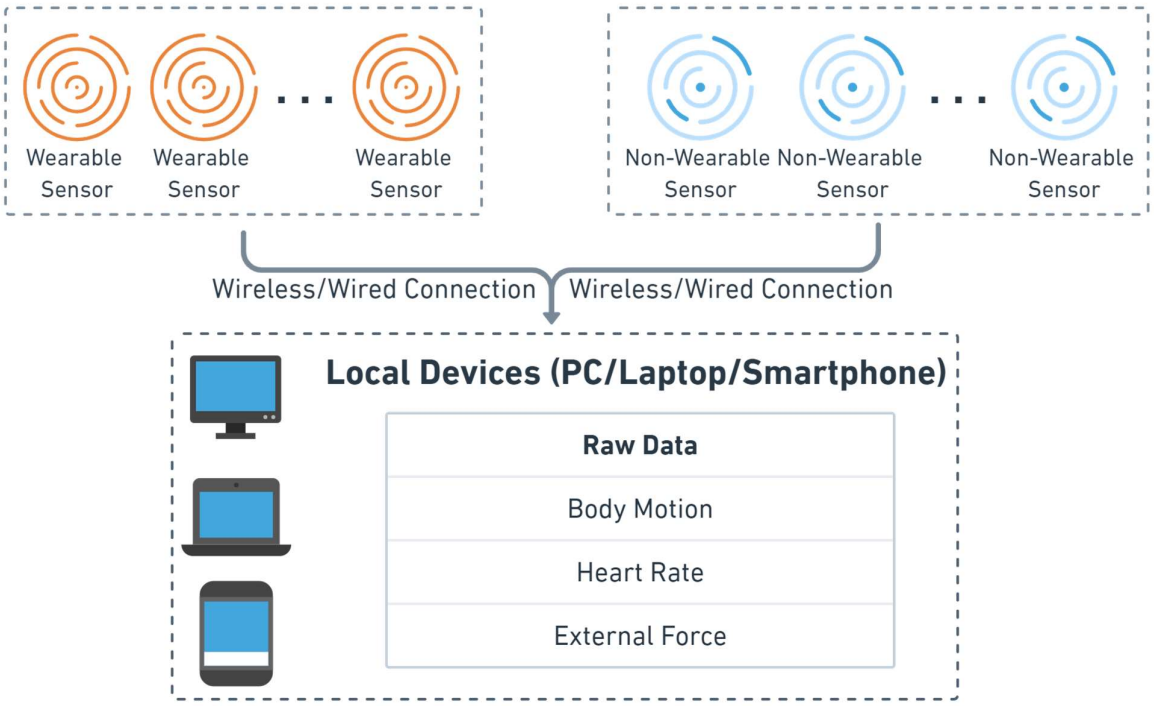


Figure 3.2 Data Collection Diagram

3.3.2 Data Analysis

Figure 3.3 demonstrates the data analysis procedure in the ErgoRelief framework. After being received in a local PC, laptop, or smartphone, raw data is transmitted to the cloud server through Wi-Fi. The cloud server is responsible for logging data to the cloud database and analysing the data, including detecting body fatigue. The server keeps obtaining packets from each local PC/laptop/smartphone, and the corresponding body fatigue can be derived using machine learning models through a series of signal processing procedures, including data synchronisation, cleaning, filtering, feature extraction, and body fatigue classification. Body fatigue is then logged to a cloud database, together with other body related information. The database can be updated continuously,

and multiple devices are able to request information for a certain period immediately when needed. In the event that there is limited or no internet, the data will be cached in the local device and transmitted to the cloud server once an internet connection has been re-established.

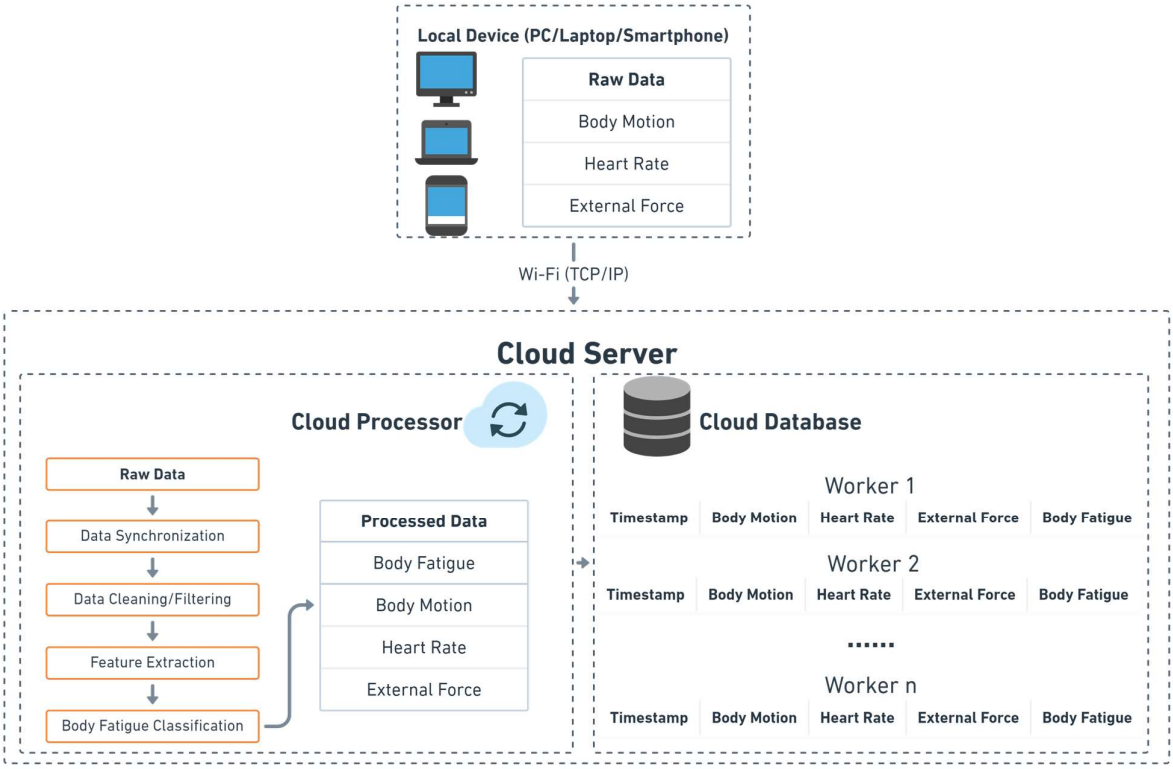


Figure 3.3 Data Analysis Diagram

3.3.3 Data Visualisation

The data visualisation aspect provides interactive feedback to workers to assist them in managing their body condition, in conjunction with providing a deeper understanding to aviation factory operations about occupational health and safety. From the operations' aspect, feedback should be professional, comprehensive, and accurate. In contrast, from the aviation manufacturing workers' perspective, the feedback should be instructive, concise, and meaningful [100].

Figure 3.4 demonstrates the interaction between aviation manufacturing workers and factory operations. Both aviation factory workers and factory operations need to log into the system first. They are then able to visualise individual data and data for a group of workers respectively. By visualising data for a group of workers, factory operations can provide regular safety advice to workers. In addition, in the case that a certain worker reports feeling unwell, factory operations can

visualise data for this worker, and provide specific safety advice or instructions to decrease fatigue and protect his/her safety.

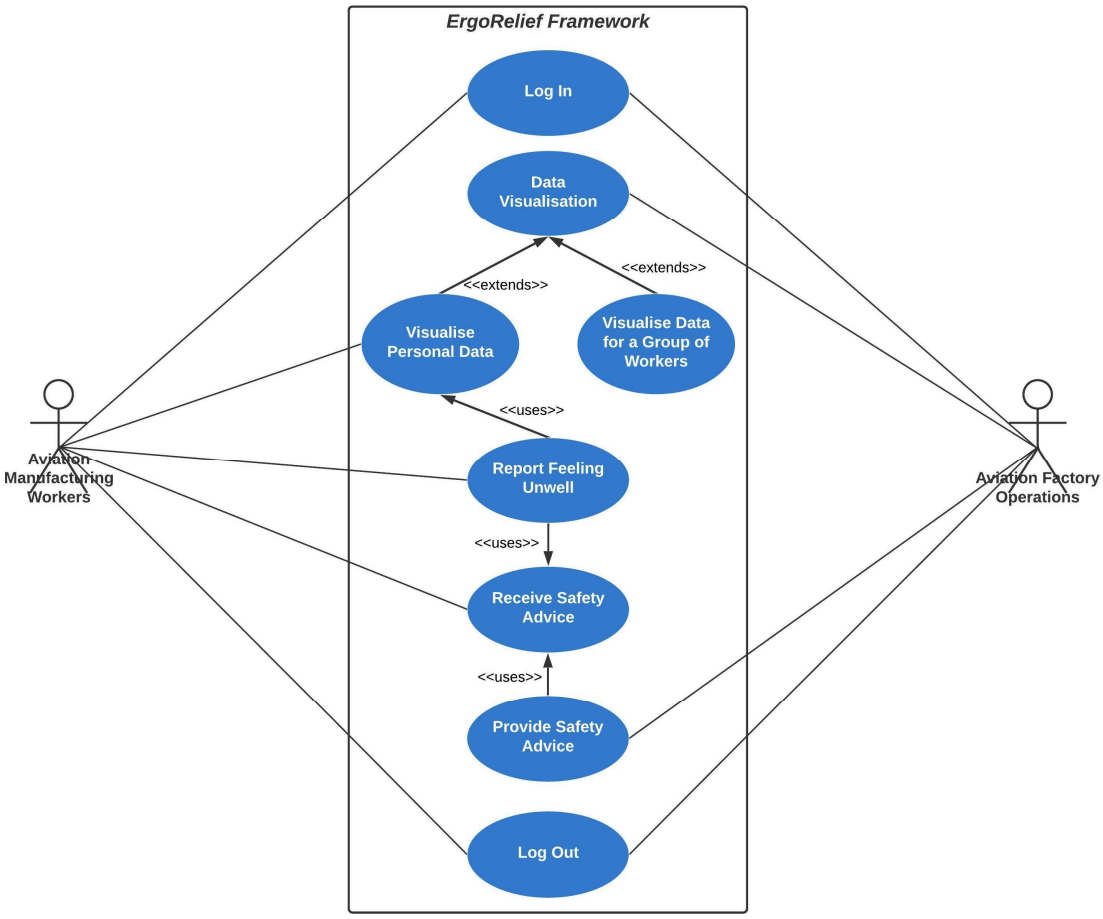


Figure 3.4 Interaction between Aviation Manufacturing Workers and Factory Operations

3.3.3.1 Aviation Factory Operations

Figure 3.5 and 3.6 display examples for feedback dashboard design for aviation factory operations. As illustrated in the data analysis section, processed data are logged to the cloud database continuously, and multiple devices are able to retrieve information from the database. Therefore, visualised data are being obtained from the database in real-time. Figure 3.5 illustrates the dashboard design for visualising data to a group of workers. The body fatigue level for each worker is displayed both as a duration and time percentage in the bar chart at the left side of the panel. Additionally, the line chart at the top right of the panel shows the change of body fatigue level for each worker in the time series. Lastly, the mean heart rate, of all workers, has also been shown at the bottom right of the panel.

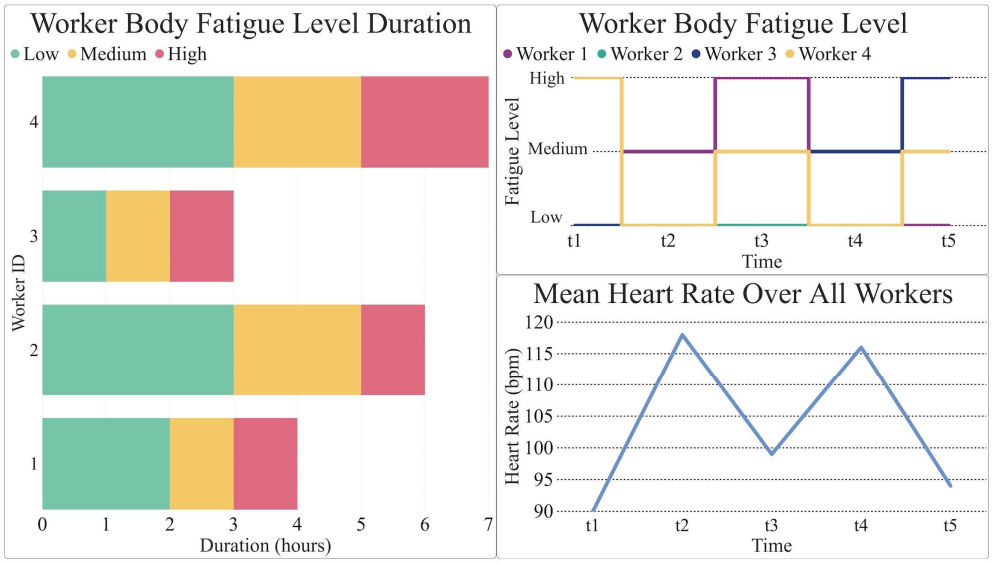


Figure 3.5 Example of Feedback Dashboard Design for a Group of Workers

Figure 3.6 illustrates the feedback dashboard design to an individual worker, which is also displayed to the factory operations. In this example, the worker reports feeling unwell, or he/she has been doing a task with a high body fatigue level for a long time. The absolute period of each body fatigue level during the shift is displayed as a bar chart at the bottom left of the dashboard.

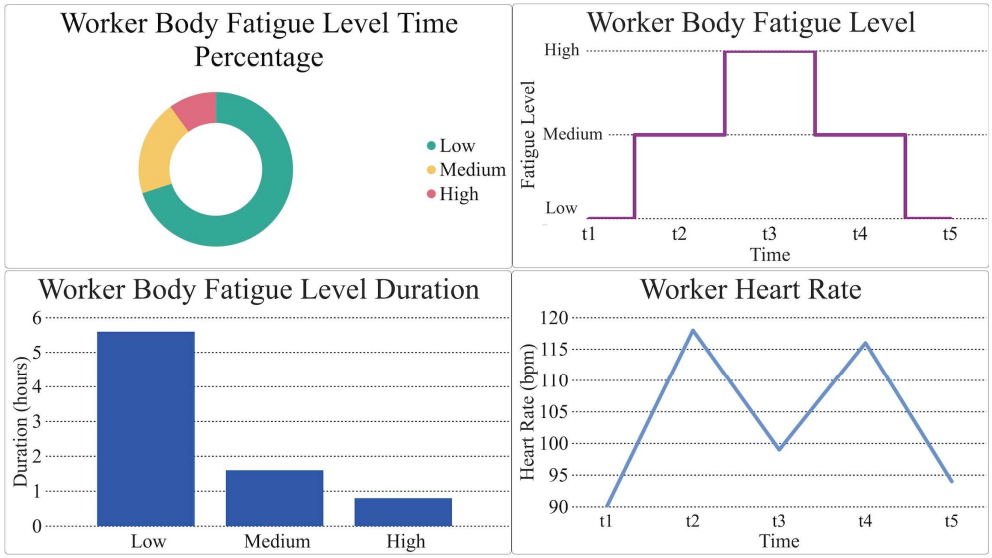


Figure 3.6 Example of Feedback Dashboard Design for an Individual Worker

3.3.3.2 Aviation Factory Workers

In an aviation factory environment, workers should also be able to monitor their body conditions closely. Therefore, a mobile application has been designed to provide feedback to workers. Similarly, to the feedback dashboards for aviation factory operations, the mobile application is able to obtain data from the cloud database directly through the network.

Figure 3.7 illustrates a sequence diagram for the smartphone feedback application. In the cloud database, each worker is assigned a unique ID and password. By entering their ID and password, the application obtains authorisation to access personal data in the database. The application then obtains body fatigue data from the database every 15 seconds, updates the fatigue information, and displays the real-time fatigue information to workers. In order to avoid significant battery energy consumption of the smartphone, the feedback application has been designed to account for shift work (and not continuous 24/7 use), which is approximately 8 hours. In addition, the application will operate in the background to communicate with the server, and the display will only be on when needed. The time interval between each data collection (15 seconds) can also be extended to meet on-site requirements of an aviation manufacturing environment.

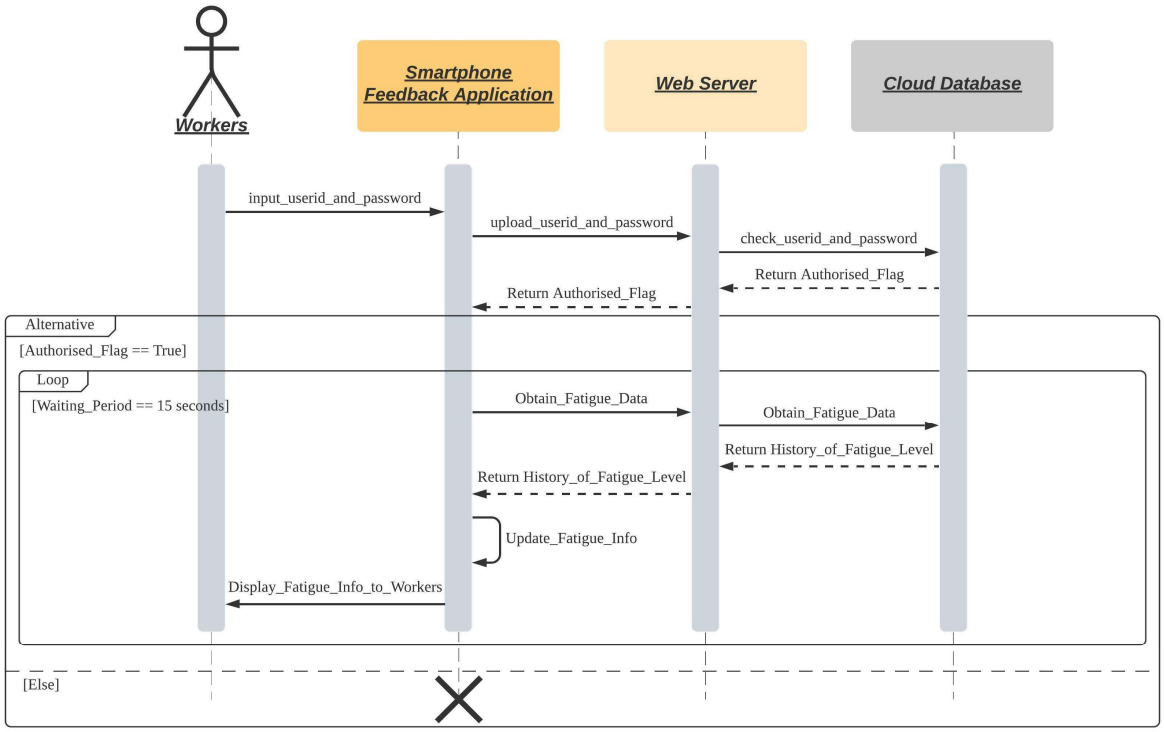


Figure 3.7 Smartphone Feedback Application Sequence Diagram

In the smartphone feedback application, it is essential to develop a valid and straightforward method to help workers understanding their energy and fatigue levels. As illustrated by Weigelt et al., a pictorial scale, such as a battery scale, performed significantly better than a purely verbal scale towards representing energy consumption on response latencies and user experience [101]. As such, as shown in Figure 3.8, a battery and face metaphor have been chosen to represent fatigue as these are easily understood interface metaphors that will be displayed in the application to represent energy and fatigue levels, in order to provide direct feedback to workers. For example, assuming workers are doing the same task in a similar environment, the higher the body fatigue level is, the more energy expenditure there will be for a certain period. Workers will be notified via a warning beep and vibration every time the body energy decreases by 20%.

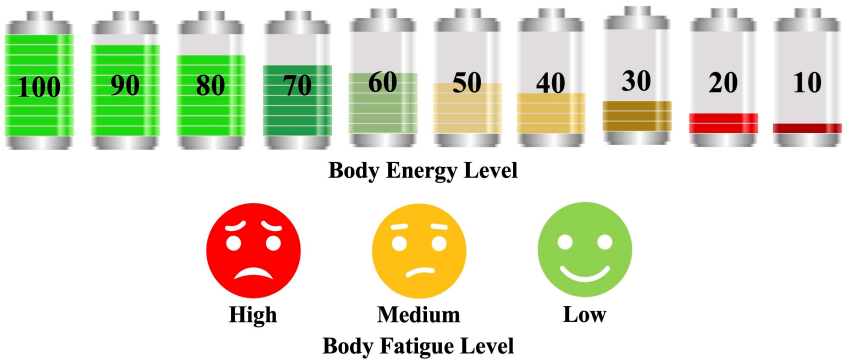


Figure 3.8 Example of Body Energy and Fatigue Level Representation

Figure 3.9 illustrates the design of the worker's feedback smartphone application, which represents energy and fatigue levels. The application has been split into three parts. The top part includes information about the current body energy level, task name, and safety advice. The middle part contains the duration for each body fatigue level in the current shift, coupled with the current body fatigue level. Lastly, the bottom part reveals the change in heart rate over the current shift. In addition, the utilisation of smart devices, such as smartphones, allows more types of feedback to be delivered to workers. For instance, if a worker maintains a high fatigue level for an extended period of time, he/she can be alarmed through signals such as a warning beep and vibration. Furthermore, the factory operations are able to provide on-time and professional advice and instructions to workers by visualising their body conditions through feedback dashboards. Workers are also able to report any unexpected conditions to supervisors and/or factory operations and/or management.

Figure 3.9 Design of the Worker's Smartphone Feedback Application

In this research, the prototype of the Worker's Smartphone Feedback Application has been developed, which is displayed in Figure 3.10, with each screen corresponding to the fatigue states of *Rest, Low,* and *High,* respectively. As illustrated in Figure 3.7, the application keeps obtaining body fatigue data from the database every 15 seconds. If the latest fatigue level is *Rest*, *Low*, or *High*, the face metaphor is displayed as green, yellow, and red, respectively. In addition, the duration of each fatigue level during the current shift is displayed on the left of the face metaphor. Lastly, the history of the body fatigue level for the previous one hour is also displayed as a line chart at the bottom of the application.

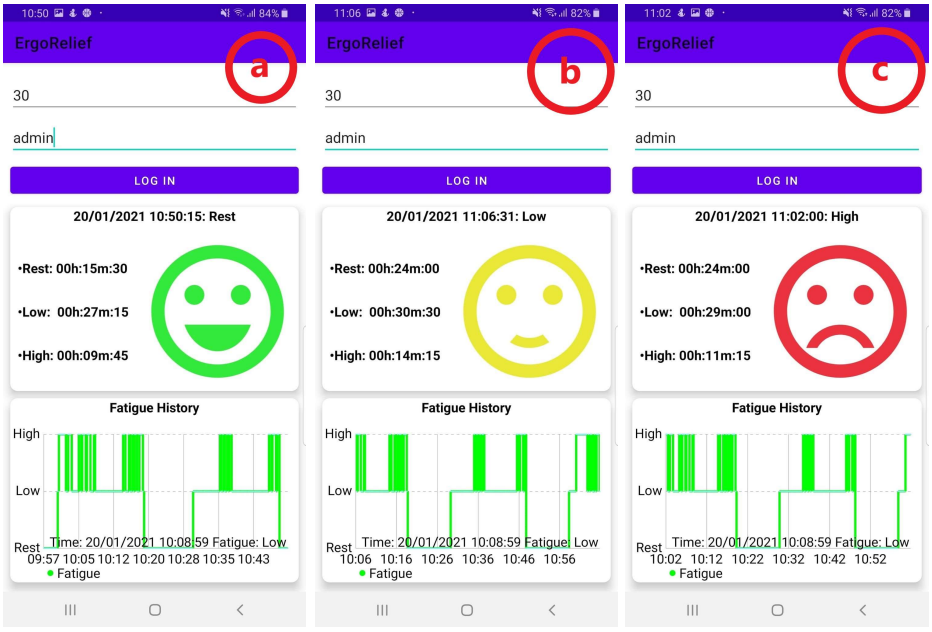


Figure 3.10 Prototype of the Worker's Smartphone Feedback Application a) Rest Fatigue Level b) Low Fatigue Level c) High Fatigue Level

3.4 Chapter Summary

In conclusion, the ErgoRelief framework has been designed to conduct real-time fatigue detection with a series of collaborators to improve awareness about workers' body conditions. In this framework, body motion, physiological data, as well as external force are measured by wearable and non-wearable sensors, and data are analysed, logged, and visualised by both factory management and workers. By implementing this framework, it can be anticipated that workers' safety and productivity can be significantly improved and supported by the continuous detection of fatigue, as workers are able to adjust their task performance and sequence based on professional safety advice in order to decrease fatigue.

Authorship Declaration

Chapter 4 incorporates sections from the following publication:

G. Liu, C. Dobbins. M. D'Souza, N. Phuong, "A Machine Learning Approach for Detecting Fatigue during Repetitive Physical Tasks," in *Personal and Ubiquitous Computing*, Feb. 2022 (Under Review).

	Liu	**Dobbins**	**D'Souza**	**Phuong**
Research Gap Identification	75%	10%	5%	10%
Experiment Design	80%	10%	5%	5%
Experiment Data Collection	100%	0%	0%	0%
Experiment Data Analysis	80%	15%	5%	0%
Paper Writing and Editing	70%	20%	5%	5%

Chapter 4

General Methodology

4.1 Introduction

In this research, two studies were undertaken. The first study evaluated the performance of different feature selection algorithms and machine learning approaches in detecting fatigue during repetitive physical tasks. Furthermore, the contribution of features from different sensors in detecting fatigue has also been evaluated. Additionally, the second study constructed a real-time fatigue assessment system using tasks that were constructed based on those found in real aviation manufacturing environments.

Both studies followed a similar data analysis protocol, including processing data from multiple devices, such as body motion, physiological data, along with external force, which are susceptible to motion artefacts. Therefore, a common data processing pipeline has been designed to process and analyse multimodal data, in order to align them to a common time basis and prepare them for machine learning model training and testing. The data processing pipeline includes data pre-processing (data synchronisation, segmentation, resampling, filtering), feature extraction, data labelling, feature selection, as well as fatigue classification. Details of the sensors and data processing pipeline are introduced in the following sections.

4.2 Sensor Platform

Figure 4.1 illustrates an overview of the data analysis pipeline, including the sensor platform. There are two aspects to the data collection platform, which includes wearable and non-wearable sensors. Wearable sensors are responsible for capturing body part acceleration, angular velocity, and heart rate, whilst non-wearable sensors record ground reaction force, torso moment, centre of pressure, centre of gravity, as well as hand exertion force. Since data with artefacts has been recorded from multiple types of sensors, it is essential to properly process the data in order to prepare them for further analysis. Previous studies recorded data using an integrated system, with each sensor having consistent timestamps. However, in this research, multiple sensors are operating independently, and data have been recorded using different sampling rates through multiple approaches, such as on-board SD card logging and wired/wireless data streaming. Multiple streams of data have inconsistent

timestamps, which need to be properly synchronised in order to avoid time drift. Therefore, a signal pre-processing pipeline has been developed for multimodal data fusion, which includes data synchronisation, segmentation, resampling, and filtering. By implementing this pipeline, different sensor measurements can be aligned to the same time basis, which can then be used for further fatigue assessment. Features will then be extracted and each sample in the feature set will be labelled by its corresponding fatigue level. After that, different feature selection algorithms will be applied to remove redundant features and generate an optimal feature set. Finally, a series of machine learning approaches will be trained and validated with the optimal feature set. Their performances will be evaluated systematically to select the optimal one for fatigue assessment.

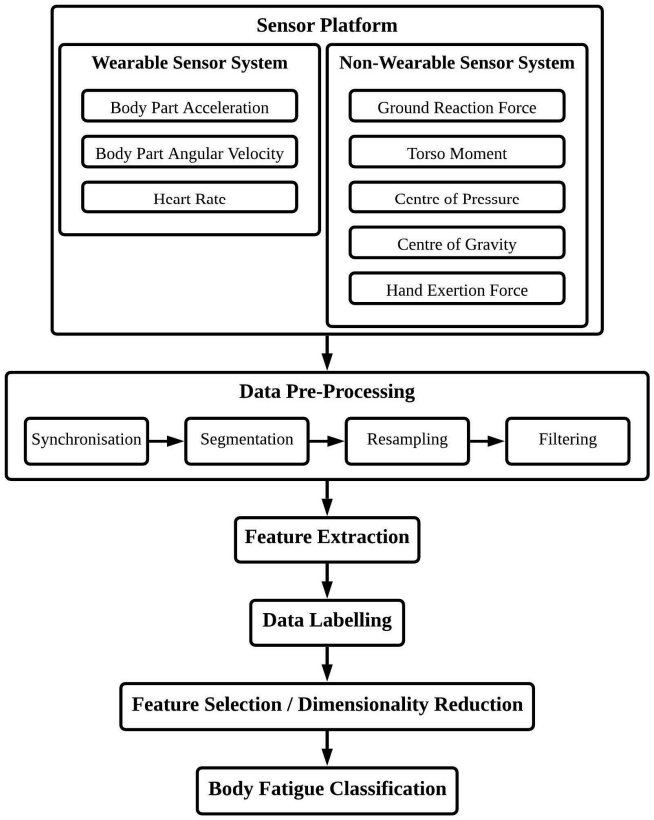


Figure 4.1 Overview of the Data Analysis Pipeline

In this research, body fatigue assessment will be based on the nature of different data measurements. Therefore, sensors should be able to provide raw data, as well as not interrupt and disturb workers while performing tasks. As such, for these reasons, the implementation of the wearable aspect of the sensor platform consists of up to six Shimmer3^{TM} Inertial Measurement Units (IMUs) (see Figure 4.2a) and a Shimmer3^{TM} PPG ear clip (see Figure 4.2b) [102]. Each IMU has been secured to different body areas to measure 3-axis acceleration and 3-axis angular velocity. Moreover, the Shimmer3^{TM}

PPG ear clip has been attached to the left ear lobe to measure Blood Volume Pulse, which has then been used to derive heart rate. Additionally, according to previous studies in the area of activity recognition and ergonomic analysis [103-105], 100Hz has been selected as the sampling frequency to ensure continuous and sufficient data collection. Due to a hardware limitation, the sampling frequency of the Shimmer3™ sensors have been configured as 100±0.21Hz.

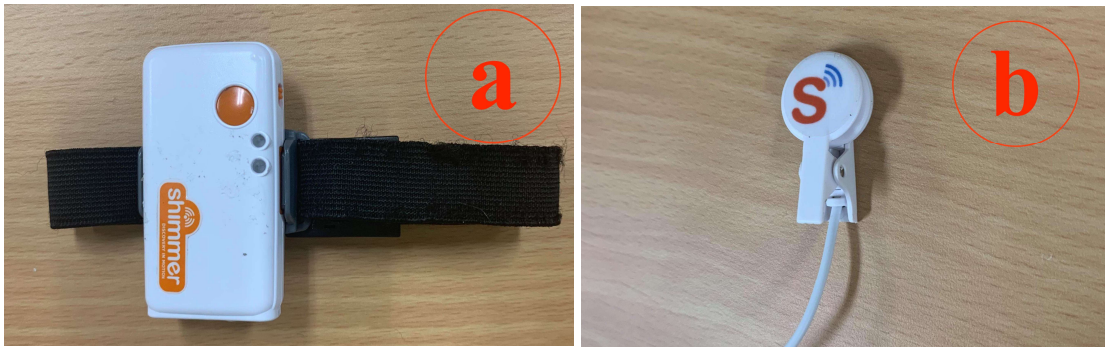


Figure 4.2 a) Shimmer3™ GSR+ Unit b) Shimmer3™ PPG ear clip

The non-wearable aspect of the sensor platform contains a Sparkfun™ Load Cell [106] (see Figure 4.3a) and a Bertec™ Force Plate [107] (see Figure 4.3b). The load cell's handle has been attached to a box, which measures hand exertion force against the box. The force plate measures 3-axis ground reaction force, 3-axis torso moment, 2-axis centre of pressure, and 2-axis centre of gravity. The sampling rate for the force plate has been configured to 100Hz. Due to hardware limitations, the maximum sampling rate for the load cell is 10Hz. Both the load cell and force plate have been connected to a PC through a wired connection and data has been logged directly to a local PC for further analysis.

Figure 4.3 a) Load Cell – 200kg, S-Type (TAS501) b) Force Plate FP4060-NC

All sensors were calibrated before conducting the experiment. Since each participant wore and used the sensors in a similar way, to ensure consistency between each participant, sensor re-calibration was unnecessary during the experiment.

4.3 Data Pre-Processing

Data pre-processing is necessary to prepare the data for further analysis. Figure 4.4 demonstrates a high-level overview of the data pre-processing procedure, including data synchronisation and segmentation, resampling, and filtering. Details for each step will be elaborated in the following sections. For development and testing purposes, data pre-processing has been undertaken using MATLAB R2019b [108].

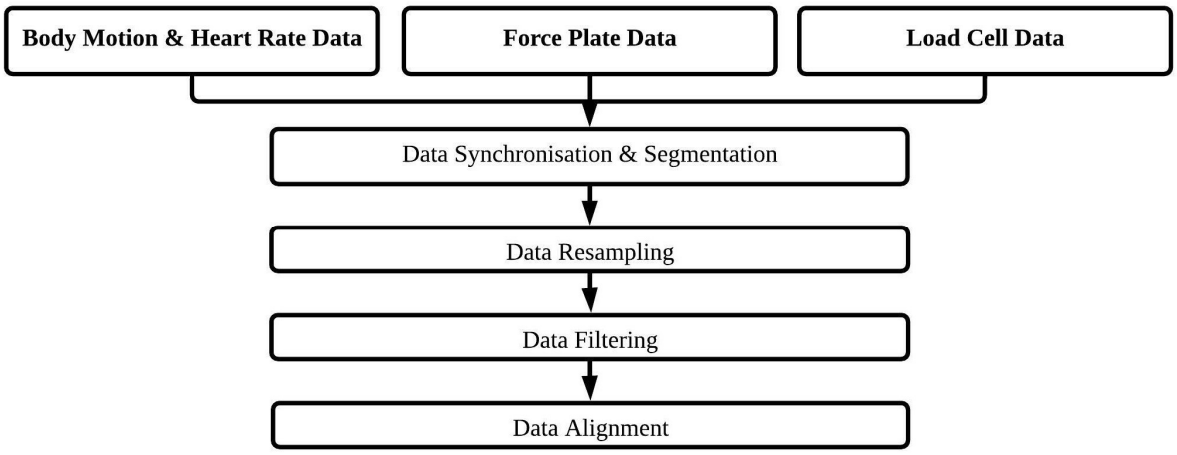


Figure 4.4 Data Pre-Processing Procedure

4.3.1 Data Synchronisation and Segmentation

Since multiple sensors have been utilised, which sample at different rates, it is essential to synchronise these different streams of data to a consistent timestamp in order to avoid time drift, which can pose detrimental issues for further analysis. Additionally, in both studies, each task has been segmented into multiple repetitions, with a resting period between each of them. Therefore, a synchronisation pattern has been designed and participants have been asked to perform it at the beginning and end of each repetition, which has then been used to align the data. The synchronisation pattern includes three steps:

1. Stand still and upright at origin for 10 seconds.
2. Perform the synchronisation signal by raising the body up slowly to stand on tiptoes and descend to the floor immediately.
3. Stand still and upright at origin for 10 seconds.

By performing the synchronisation pattern, an acceleration peak in gravity direction has been generated in the signals. Figure 4.5 shows the details for this stage.

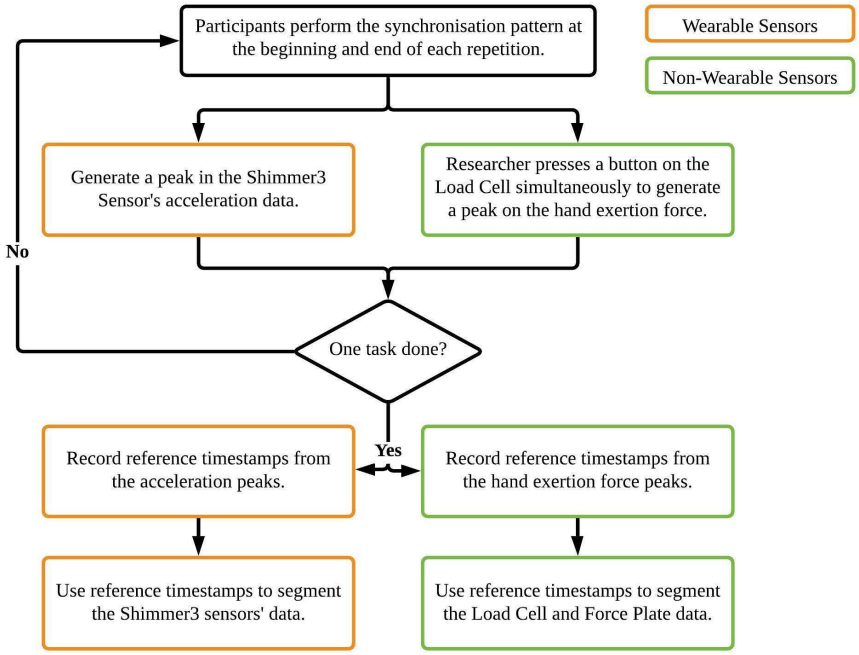


Figure 4.5 Data Synchronisation and Segmentation Procedure

As declared in the Shimmer User Manual [109], multiple Shimmer3^{TM} sensors will have a consistent timestamp when they have been configured together at the same time. Hence, the timestamps corresponding to acceleration peaks at the beginning and end of each repetition have been recorded as the reference timestamps to segment the data from the other Shimmer3^{TM} sensors. Figure 4.6 illustrates the key MATLAB function (*findpeaks*) for extracting reference timestamps using acceleration peaks. The body part acceleration in the gravity direction has been extracted first in line 3, and the minimum peak height has been configured as two times gravity in line 4. After that, *findpeaks* function has been used in line 7 to detect acceleration peaks, with minimum peak distance of 10 seconds, since the time difference between adjacent peaks should theoretically be greater than 5 minutes, and which should not be less than 10 seconds. Finally, the timestamps corresponding to the acceleration peaks have been extracted and combined as the reference timestamps for the wearable sensor data segmentation and synchronisation.

```
function [ref_timestamp] = shimmer_task_segmentation(shimmer)

    acc = shimmer.acc_wr(2, :);
    min_peak_height = 2;

    % Find acceleration peaks (Minimum Peak Height = 2g, Minimum Peak Distance = 10s)
    [~, locs] = findpeaks(acc, 'MinPeakHeight', min_peak_height, 'MinPeakDistance',
1000);

    % Obtain timestamps corresponding to each acceleration peak
    start_timestamp = shimmer.timestamp(locs(1:2:length(locs)));
    stop_timestamp = shimmer.timestamp(locs(2:2:length(locs)));

    ref_timestamp = [start_timestamp.' stop_timestamp.'];
end
```

Figure 4.6 Wearable Sensor Task Segmentation Code

Figure 4.7 displays an example of the acceleration on gravity direction of a Shimmer3™ sensor, with green and red circles indicating the repetition start and end points, respectively.

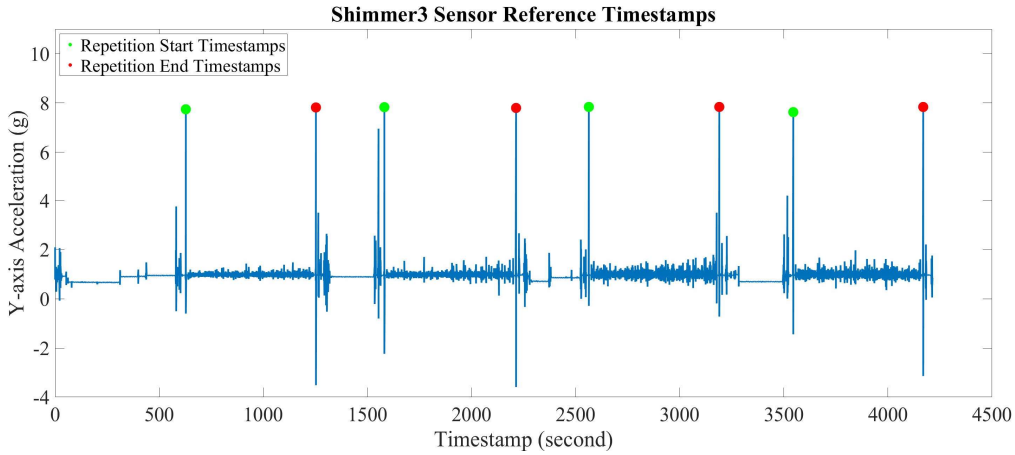


Figure 4.7 Example of Shimmer3™ Sensor Reference Timestamps

During the synchronisation pattern, at the point when participants perform the synchronisation signal, the researcher simultaneously presses down a button on the load cell, which generates a peak in hand exertion force. As both the load cell and force plate data are connected to the same PC, the reference timestamps from the hand exertion force peaks can be used to segment the force plate data. Similarly to the reference timestamp extraction for the wearable sensors, *findpeaks* has also been used as the key function to extract reference timestamps based on the hand exertion force peaks. The minimum peak height has been configured as 900kg, because the hand exertion force during the experiment should not be greater than 20kg, and the hand exertion force has 1000kg added every time when the researcher pressed the button. Figure 4.8 illustrates an example of the load cell reference timestamps. Green and red circles correspond to the hand exertion force peaks at the beginning and end of each

repetition, and their timestamps have been extracted for the non-wearable sensor system data segmentation.

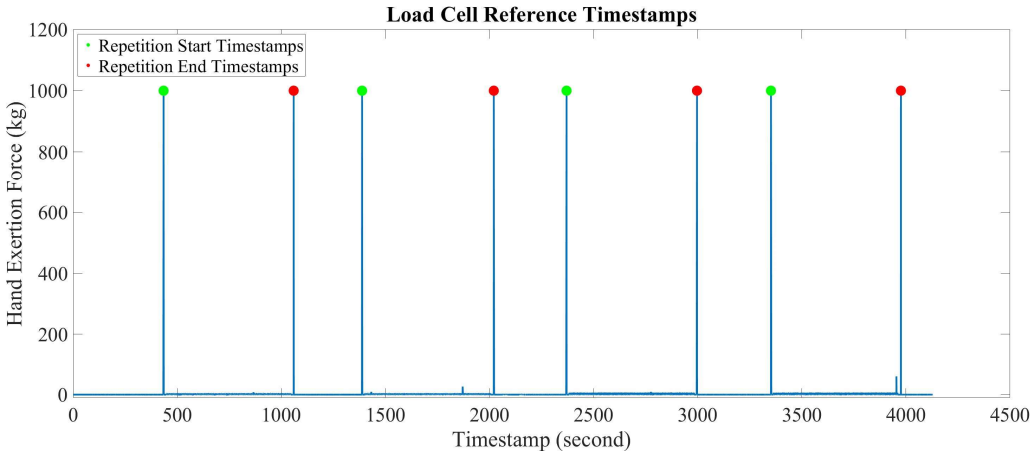


Figure 4.8 Example of Load Cell Reference Timestamps

4.3.2 Data Resampling and Filtering

After synchronising and segmenting the data to a consistent timestamp per repetition, it is essential to resample the data to the same sampling rate, in order to ensure that there are consistent number of samples for each sensor within each window. Since the sampling rate for the Shimmer3^{TM} sensors is 100.21Hz, the Shimmer3^{TM} sensors' data have been resampled to 100Hz [103-105]. Acceleration data from each IMU and data from the force plate have been filtered using a 2nd order zero-phase Butterworth low-pass filter, with a cutoff frequency of 3Hz (see Figures 4.9 and 4.11) [110-112]. Angular velocity data from each IMU has been filtered using a 2nd order zero-phase Butterworth high-pass filter, with a cutoff frequency of 0.1Hz (see Figure 4.10) [113]. Lastly, the PPG signal has been filtered using a 2nd order zero-phase Butterworth band-pass filter, with a cutoff frequency between 0.5 and 3.5Hz [114, 115]. Figure 4.9 to 4.11 demonstrate the examples of filtered acceleration, angular velocity, and ground reaction force data. For the acceleration and ground reaction force data, it can be discovered that the high frequency component greater than 3Hz has been removed, while the low frequency component less than 0.1Hz has been removed from the angular velocity data.

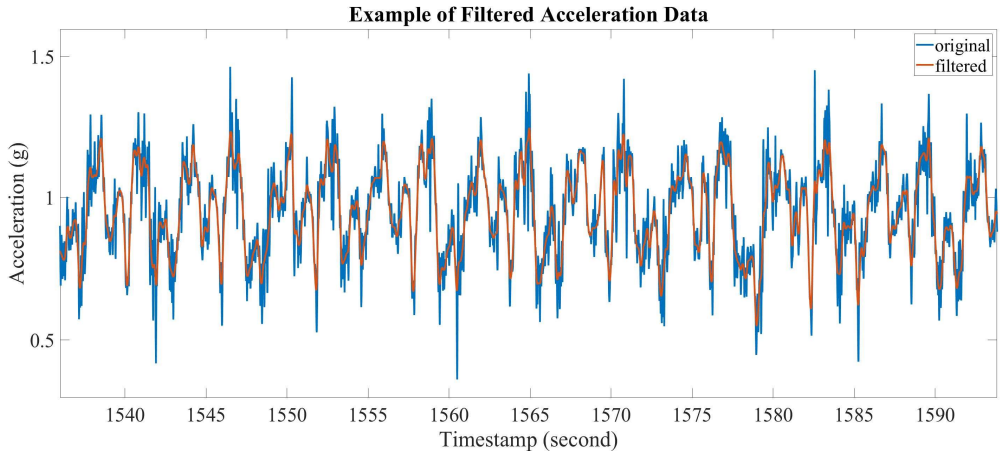


Figure 4.9 Example of Filtered Acceleration Data filtered using a 2nd order zero-phase Butterworth low-pass filter, with a cutoff frequency of 3Hz

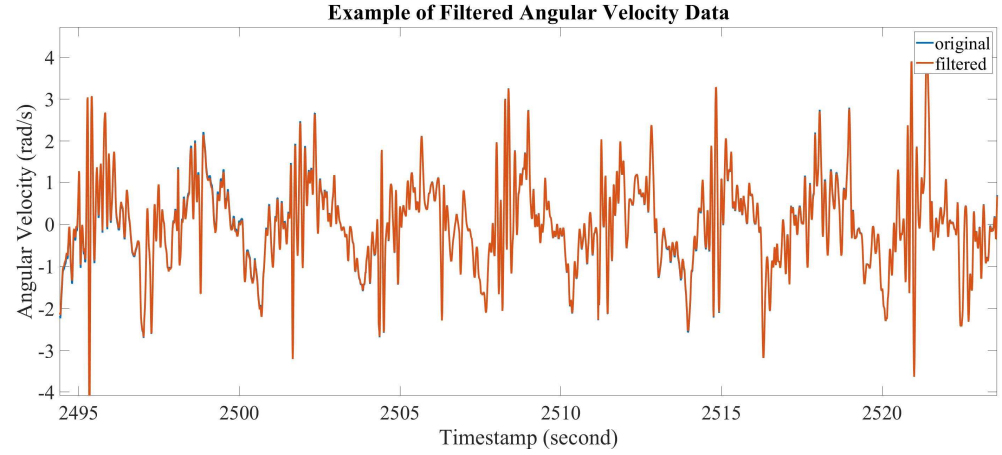


Figure 4.10 Example of Filtered Angular Velocity Data filtered using a 2nd order zero-phase Butterworth high-pass filter, with a cutoff frequency of 0.1Hz

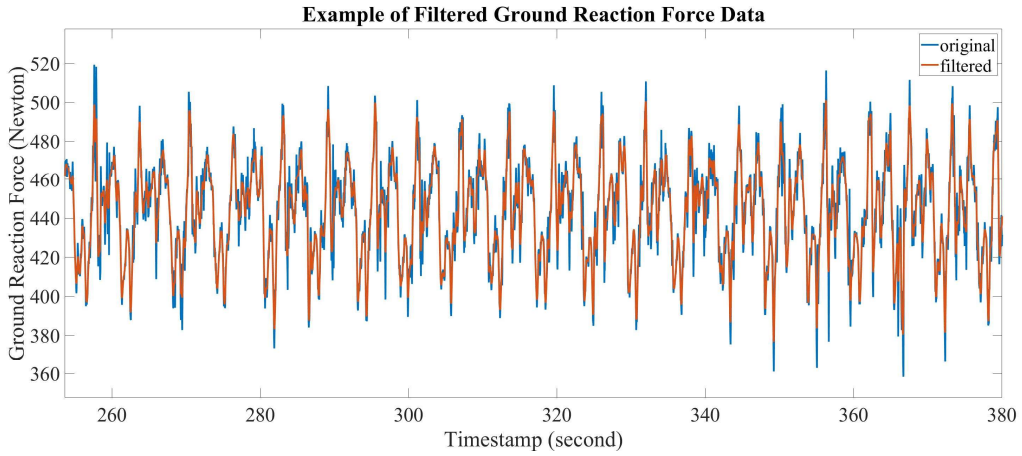


Figure 4.11 Example of Filtered Ground Reaction Force Data filtered using a 2nd order zero-phase Butterworth low-pass filter, with a cutoff frequency of 3Hz

Figure 4.12 illustrates the key functions for data resampling (*resample*) and filtering (*butter* and *filtfilt*). The sampling frequency has been configured as 100Hz at line 1, and the *resample* function has been used to resample the original data to 100Hz. After that, line 6 to 8 demonstrate three types of Butterworth filters created by the *butter* function. Finally, line 9 displays the creation of a zero-phase filter through the use of the *filtfilt* function, which filters the data without time distortion.

```
1. % Data resampling
2. sampling_frequency = 100;
3. [resampled_data, resampled_timestamp] = resample(original_data, original_timestamp,
   sampling_frequency, 1, 1, 'linear');
4.
5. % Data filtering
6. [B,A]=butter(2, fc/(fs/2),'low'); % Low pass filter
7. [B,A]=butter(2, fc/(fs/2),'high'); % High pass filter
8. [B,A]=butter(2, fc/(fs/2),'bandpass'); % Band pass filter
9. output = filtfilt(B, A, data); % zero-phase filter
```

Figure 4.12 Data Resampling and Filtering Code

Furthermore, outliers appear in the load cell data which cannot be easily removed by filtering alone. The outliers are caused by the wired data streaming, since noise can be randomly added to the digital signal due to the magnetic interference. As displayed in Figure 4.13, occasionally the hand exertion force measurement increases to over 20kg, which is invalid, because the maximum hand exertion force during the experiment should be less than 10kg. As such, these outliers have been removed using a sliding window algorithm, with a window size equalling 20 samples. Based on experimentation, for each window, if the sample's absolute value is greater than two times of the median absolute deviation, it has been treated as an outlier and replaced by a null value. Finally, all null values have been interpolated by their nearest valid samples. The MATLAB function named *rmoutliers* has been used to accomplish outlier removal, which is displayed in Figure 4.14.

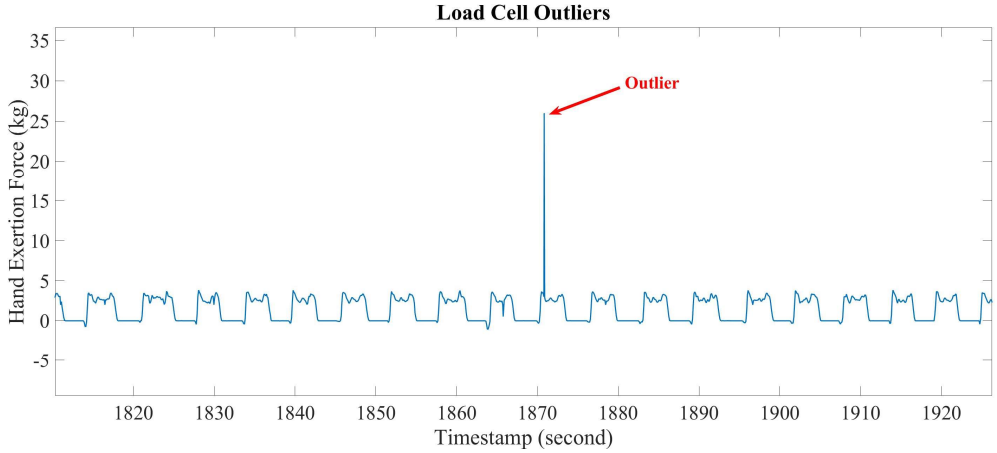


Figure 4.13 Example of Load Cell Outliers

```
1. % Detect outliers
2. [loadcell_force, outlier_idx] = rmoutliers(loadcell_force,'movmedian', 20,
   'ThresholdFactor', 2);
```

Figure 4.14 Hand Exertion Force Data Outlier Removal Code

As illustrated in Section 4.2, 100Hz has been selected as the sampling rate for the system to ensure continuous and sufficient data collection. Therefore, after removing outliers, the hand exertion force data have been resampled to 100Hz to make it consistent with other measurements and have then been filtered by a 2nd order zero-phase Butterworth low-pass filter, with a cutoff frequency of 3Hz [116, 117]. Figure 4.15 illustrates an example of filtered hand exertion force data. Similarly, to the acceleration and ground reaction force data, high frequency components greater than 3Hz have been removed from the signal.

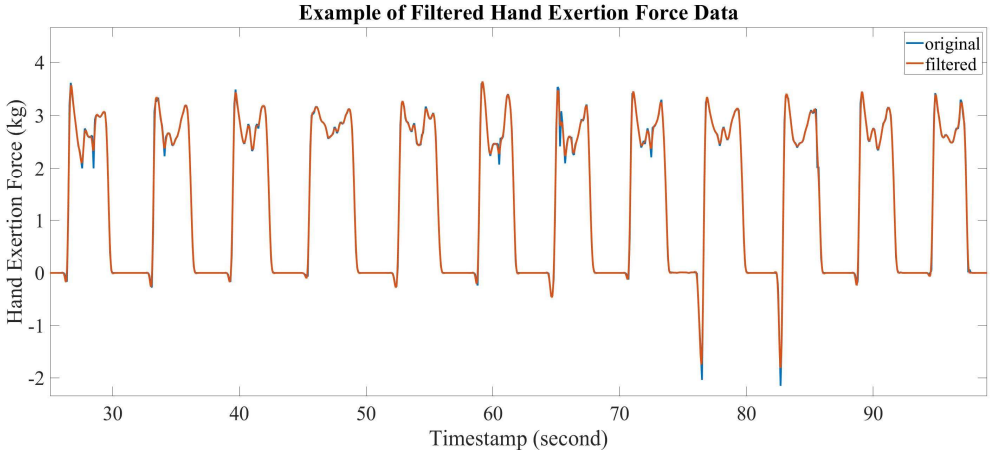


Figure 4.15 Example of Filtered Hand Exertion Force Data

Physiological data, such as heart rate, has also been included in the data analysis, which has been measured by a PPG ear clip. The PPG sensor measures the blood volume change inside the capillary vessels, as blood flows through the vessel at each heartbeat [118]. Therefore, the filtered PPG signal has been used to derive beats per minute (BPM) [119, 120]. This has been achieved by calculating the time difference between adjacent peaks (see Figure 4.16 and Equation 4.1). Figure 4.16 displays an example of the filtered PPG signal, with red circles indicating detected peaks. Each peak represents the time that a heartbeat occurs, and heart rate in beats per minute has been calculated by the time intervals between adjacent peaks using Equation 4.1.

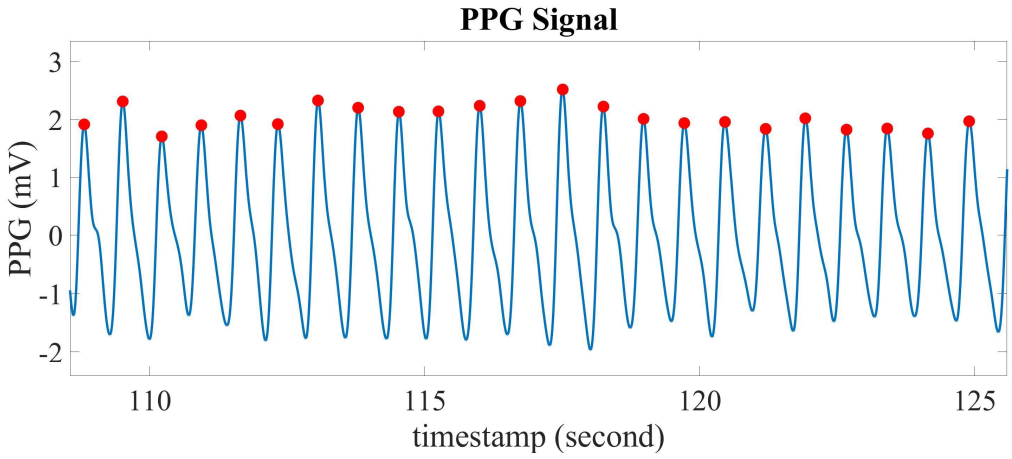


Figure 4.16 Example of Filtered PPG Signal with Detected Peaks

$$Heart\ Rate\ (beats\ per\ minute) = \frac{60}{\Delta t_{peak-peak}(seconds)} \tag{4.1}$$

Figure 4.17 illustrates the key MATLAB function for calculating the heart rate. Based on experimentation, the heart rate has been calculated through a sliding window algorithm, using an 8-second window with 50% overlapping area, which has been configured from line 3 to 5. The total number of windows has then been calculated in line 7. Furthermore, a for loop has been implemented to loop through data in each window in line 9. The PPG signal in each window has been normalised first in line 13, and the negative values in the PPG signal have been replaced by 0 in line 15. After that, the peaks in the PPG signal have been located using the *findpeaks* function in line 22. The minimum peak distance has been configured as 30, which represents 180 beats per minute, since the participants' heart rate should not exceed 180 beats per minute during the experiment. Finally, the mean heart rate in this window has been calculated through Equation 4.1 in line 25.

```
1.  heart_rate = [];
2.
3.  fs = 100; % Sampling rate of 100Hz
4.  window = 8 * fs; % Window size of 8 seconds
5.  step = 4 * fs; % Step size of 4 seconds
6.
7.  windowNb = floor((size(PPG, 2)-window)/step) + 1;  % Total number of windows
8.
9.  for i = 1:windowNb % Loop through each window
10.
11.     curSegment = (i-1)*step+1 : (i-1)*step+window;
12.     PPG_seg = PPG(:, curSegment); % Segment PPG signal in current window
13.     PPG_seg = normalize(PPG_seg, 2, 'zscore', 'std'); % Normalise PPG signal using z-
    score normalisation
14.     PPG_pos = PPG_seg;
15.     PPG_pos(PPG_pos < 0) = 0; % Remove negative values in a PPG signal
16.
17.     % Find peaks in the PPG signal. Each peak represents a heart beat
18.     % Minimum Peak Distance has been configured as 30. The reason for
19.     % choosing 30 is that
```

```
    % 30 samples/beat = 1/3 seconds/beat = 3 beats/second = 180 beats/min
    % In this study, the participants' heart rate would not go over 180bpm.
    [~, locs] = findpeaks(PPG_pos, 1:size(PPG_pos, 2), 'MinPeakDistance', 30);

    % Calculate mean heart rate in the current PPG segment.
    heart_rate_seg = 60 * fs/mean(diff(locs));
    heart_rate = [heart_rate, heart_rate_seg];

end
```

Figure 4.17 Heart Rate Calculation Code

4.4 Feature Extraction

After the data has been pre-processed, features have then been extracted from the data. In this research, both time and frequency-domain features have been extracted using a 30-second sliding window, with 50% overlapping area [121]. Figure 4.18 demonstrates an overview of the types of extracted features, with green and red boxes indicating frequency- and time-domain features, respectively.

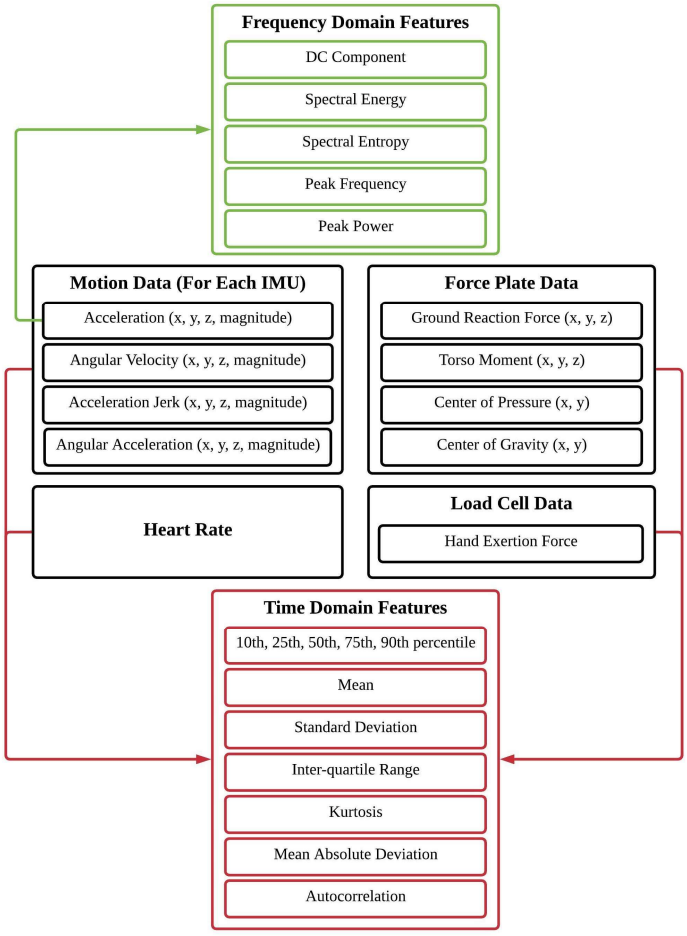


Figure 4.18 Feature Set

Time-domain features have been extracted from all sensor measurements. Motion data that have been recorded from each IMU include body part acceleration and angular velocity in each axis (x, y, z), as well as the magnitude of body part acceleration and angular velocity. Furthermore, acceleration jerk and angular acceleration have also been included in feature extraction, which are the first order time derivative of acceleration and angular velocity (see Equation 4.2 and 4.3) [25, 29, 122]. In Equation 4.2, $\vec{a}(t)$ represents the acceleration and $\vec{j}(t)$ is the first order time derivative of the acceleration, which is the acceleration jerk. In Equation 4.3, $\vec{\omega}(t)$ indicates the angular velocity and $\vec{\alpha}(t)$ is the first order time derivative of the angular velocity, which is named angular acceleration.

$$Acceleration\ Jerk\ \vec{j}(t) = \frac{d\vec{a}(t)}{dt} \tag{4.2}$$

$$Angular\ Acceleration\ \vec{\alpha}(t) = \frac{d\vec{\omega}(t)}{dt} \tag{4.3}$$

Heart rate and external force have also been included in feature extraction. External force contains the load cell and force plate data, with each axis of ground reaction force, torso moment, centre of pressure, centre of gravity, along with hand exertion force. Time-domain features contain 10^{th}, 25^{th}, 50^{th}, 75^{th}, 90^{th} percentiles, mean, standard deviation, inter-quartile range, kurtosis, mean absolute deviation, and autocorrelation.

Body part acceleration from each IMU, including x, y, z-axis, and magnitude, have been used for frequency-domain feature extraction. Frequency-domain features include the DC component, spectral energy, spectral entropy, peak frequency, and peak power. Figure 4.19 illustrates the procedure for frequency-domain feature extraction. Acceleration data have been processed using the Fast Fourier Transform (FFT), in order to generate the Power Spectral Density (PSD). The DC component has been extracted from the position where the frequency equals 0Hz. The DC component has then been removed from the acceleration data by subtracting from them their mean values. PSD has been recalculated with the FFT in order to extract spectral energy, spectral entropy, peak frequency, and peak power. Equation 4.4 displays the calculation for spectral energy, which is the sum of squares of all frequency components. Spectral entropy measures the spectral power distribution, which has been derived using Equation 4.5 by calculating the sum of normalized frequency components multiplied by their logarithm values, where $F(n)$ and $\hat{F}(n)$ are the magnitude and normalised value of n^{th} FFT frequency component respectively [123].

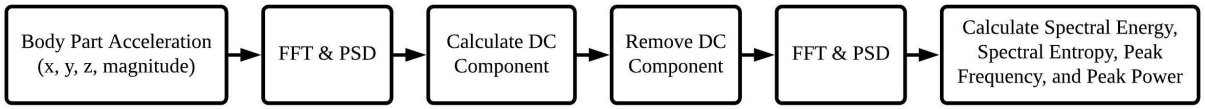


Figure 4.19 Frequency Domain Feature Extraction Procedure

$$S_E = \sum_{i=1}^{N} F(n)^2 \tag{4.4}$$

$$S_{EN} = -\sum_{i=1}^{N} \hat{F}(n) * \log\left(\hat{F}(n)\right) \tag{4.5}$$

$$\hat{F}(n) = \frac{F(n)}{\sum_{i=1}^{N} F(n)} \tag{4.6}$$

4.5 Data Labelling

After feature extraction, the feature was labelled before the data can be trained in the machine learning model. The data labelling approach has been undertaken on a per repetition, per participant basis utilizing a personalized threshold method using subjective labels generated from the Borg Scale Questionnaire [24], which has been utilised to record subjective ratings of fatigue at the beginning (pre) and end (post) of each repetition (see Appendix 3). The questionnaire includes 15 levels ranging from 6 to 20. A rating of 6 corresponds to the lowest level of fatigue, such as reading a book or watching television, while a rating of 20 indicates the highest level, such as a finishing kick in a race or other activities that cannot be maintained for a long time. Previous research labelled fatigue level for each repetition using a single value recorded at the end of each repetition [22, 29, 35, 45]. However, we hypothesised that the change in score for the levels of exertion between the end and beginning of each repetition can better represent fatigue. Therefore, we proposed a novel data labelling approach, where the fatigue level of each repetition has been labelled by a change in score. Figure 4.20 demonstrates the data labelling procedure. The corresponding subjective fatigue level rating for each repetition is represented by a 'change score', which has been calculated by subtracting the pre-level from the post-level (see Equation 4.7).

$$Subjective\ Fatigue\ Level = Post\ Level - Pre\ Level \tag{4.7}$$

Additionally, as different people have different perceptions of the level of exertion, rather than labelling fatigue through a fixed program/threshold for all persons, we calculated a personalised threshold based on their changes in fatigue level during each task. As such, samples belonging to the resting period between each repetition have been labelled as *Rest*. Each repetition's data have then been labelled as either *high* or *low* fatigue on a per task, per participant basis. Change scores for each repetition per task have been calculated first. Data from repetitions with a change score equalling 0 have been discounted, because there is no change in fatigue during these repetitions. A threshold has then been set to differentiate between high and low fatigue using the mean value of all change scores per task. For repetitions with a change score lower than the threshold, corresponding samples have been labelled as *Low* fatigue. On the other hand, data from repetitions with change scores greater than the threshold have been labelled as *High* fatigue. Lastly, data from repetitions where change scores equal the threshold have been discounted, as they are on the boundary and don't contain a sufficient change in fatigue to be included in the class.

Additionally, after the data has been labelled, the number of samples in each class is typically imbalanced. This can bias the body fatigue classification result, as the model is more likely to detect the majority classes rather than the minority class [124]. Therefore, it is important to balance the number of samples from each class. In this instance, Random Under-Sampling (RUS) has been used to randomly remove samples from the majority classes, in order to balance the numbers of samples with the minority class [125].

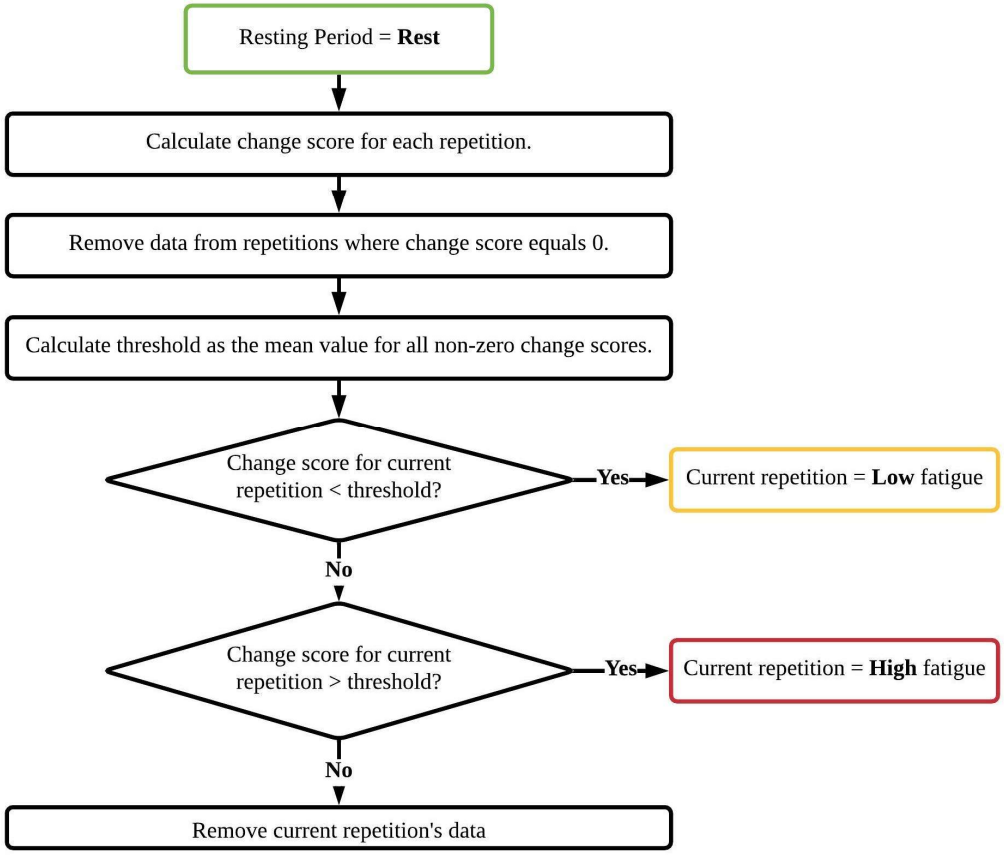


Figure 4.20 Data Labelling Procedure

4.6 Feature Selection

During the feature extraction stage, hundreds of features have been extracted from different streams of data. However, some of them may be highly correlated with others. These redundant features need to be removed from the feature set to remove noise from predictors, decrease computational cost, and improve model performance. ReliefF and PCA are feature selection methods that have been widely applied in the area of activity recognition and ergonomic analysis [25, 104, 126]. ReliefF uses a k-nearest neighbour approach to generate a weight for each feature, depending on how well the feature discriminates between classes from its neighbours, whilst PCA projects the original feature set to a lower dimension with a set of principal components, whilst maintaining as much of the data's variation as possible. In this research, three methods have been applied, which are:

1. ReliefF with 10 nearest neighbours (ReliefF-10) [127]
2. ReliefF with 30 nearest neighbours (ReliefF-30) [127]
3. PCA with 95% of variance being explained [128]

Both ReliefF-10 and ReliefF-30 are included in the feature selection approach to test the results when using the algorithm with a k value that is relatively low (k = 10) and high (k = 30), respectively. This allows us to compare the model's performance utilising different hyperparameters. Figure 4.21 illustrates an example of the ReliefF feature selection method. The calculated weight of each feature has been sorted in descending order and plotted. The elbow position has been selected as the threshold for retaining features, which is the point when the weights stop decreasing significantly. For those features with weights greater or equal to the weight at the elbow position, they have been retained in the optimal feature set, while other features have been trimmed [127].

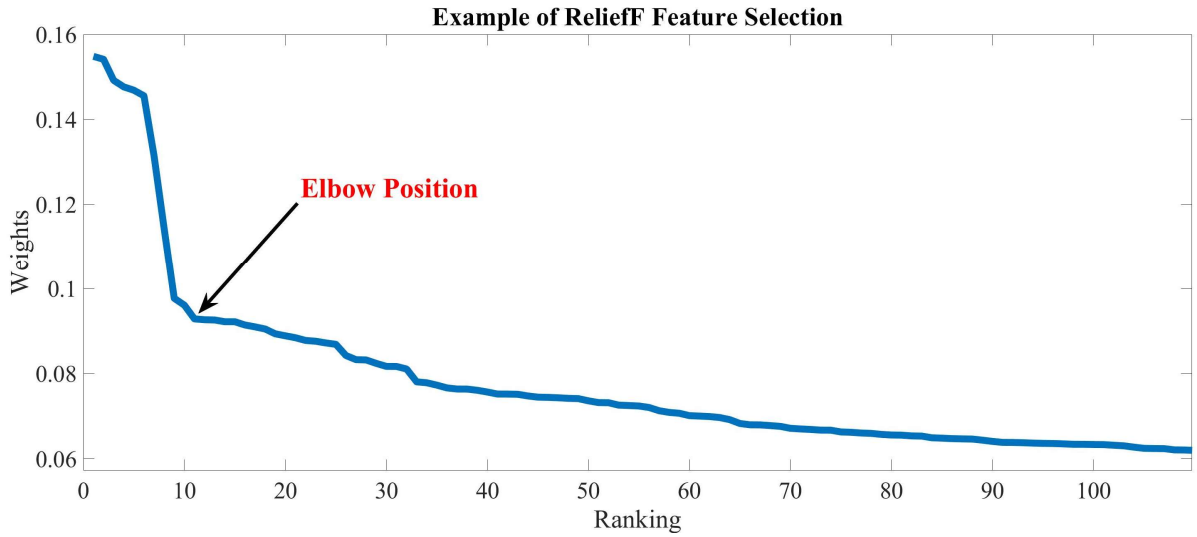


Figure 4.21 Example of ReliefF Feature Selection

Compared with the ReliefF feature selection algorithm which selects optimal features directly through weights, PCA projects features in the original feature space to a lower dimension. The new feature space contains a number of principle components, with the first few components containing most of the variation of the original feature set. Therefore, the first few components can be maintained, which keeps most of the variation in the data, as well as minimising the size of the feature set [128].

4.7 Fatigue Classification

Three supervised machine learning algorithms were implemented using RStudio v1.3.1093 [129] to classify fatigue, which can be distinguished into parametric and non-parametric methods. Parametric methods are targeted at constructing a strict mapping relationship between features and classes. In contrast, non-parametric methods do not make any assumptions of the relationship between features and classes, meaning that they are able to resolve more complex problems [130]. Linear Discriminant Analysis (LDA) [131], kNN [131], and Polynomial Kernel-based Support Vector Machine (P-SVM)

[126] have been included in this research, where LDA belongs to parametric methods and kNN and P-SVM are non-parametric methods. The reason for including these three machine learning approaches is to evaluate the classification performance of machine learning approaches with different complexities. In this research, LDA is one of the simplest machine learning algorithms, whilst the P-SVM is a significantly complicated approach. In addition, the complexity of kNN is between the LDA and P-SVM. Figure 4.22 illustrates the classification process that has been undertaken. Three feature selection methods and three machine learning algorithms have been applied to the feature set, with nine models being constructed in total. Each model has been trained and validated using repeated 10-fold cross validation, with 100 repetitions. Hyperparameters for each model have been determined using the grid search method, as this method can train the model using an array of possible hyperparameters configured by the researcher and select the optimal one for the body fatigue classification.

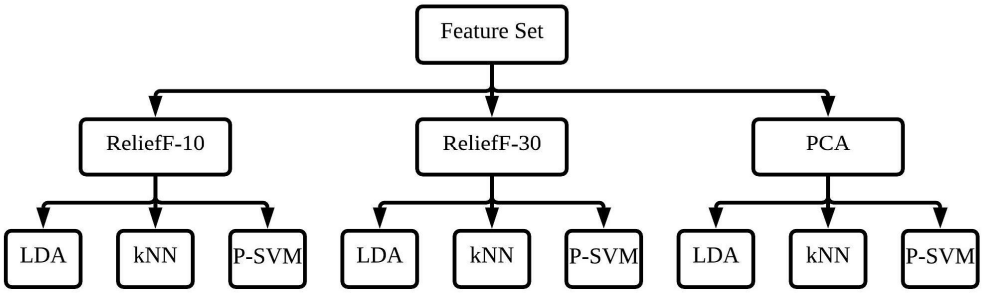


Figure 4.22 Machine Learning Process

After training the models, their performance has been evaluated and compared based on four measures, including:

1. Accuracy: The ratio of correct predictions over all predictions.
2. F1: A measurement of classification accuracy by balancing precision and recall, where precision is the number of true positives divided by the number of all positives, whilst recall is the number of true positives divided by the number of all samples that should be classified as positive.
3. Balanced Error Rate (BER): The average errors of each class.
4. Area Under ROC Curve (AUC): An aggregate measure of model performance across all possible classification thresholds.

As the F1 and AUC have been calculated on a per class basis, for multi-level body fatigue classification, the mean values for the F1 and AUC have been calculated from a series of "one-versus-all" results and implemented for the model performance evaluation.

4.8 Chapter Summary

In conclusion, a standard procedure for processing multimodal data has been presented in this chapter. Data have been pre-processed first, including data synchronisation, segmentation, resampling, and filtering. After that, features have been extracted, labelled, selected, and multiple machine learning models have been constructed to evaluate their performance and select the optimal model for fatigue classification.

Authorship Declaration

Chapter 5 incorporates sections from the following publication:

G. Liu, C. Dobbins, M. D'Souza, and N. Phuong, "Analysis of Biometric Sensor Data for Predicting Fatigue: A Framework Towards Reducing Work-Related Musculoskeletal Disorders in Aviation Manufacturing Workers," in the *2021 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2021, pp. 6928-6932.

	Liu	**Dobbins**	**D'Souza**	**Phuong**
Research Gap Identification	75%	10%	5%	10%
Experiment Design	80%	10%	5%	5%
Experiment Data Collection	100%	0%	0%	0%
Experiment Data Analysis	80%	15%	5%	0%
Paper Writing and Editing	70%	20%	5%	5%

G. Liu, C. Dobbins. M. D'Souza, N. Phuong, "A Machine Learning Approach for Detecting Fatigue during Repetitive Physical Tasks," in *Personal and Ubiquitous Computing*, Feb. 2022 (Under Review).

	Liu	**Dobbins**	**D'Souza**	**Phuong**
Research Gap Identification	75%	10%	5%	10%
Experiment Design	80%	10%	5%	5%
Experiment Data Collection	100%	0%	0%	0%
Experiment Data Analysis	80%	15%	5%	0%
Paper Writing and Editing	70%	20%	5%	5%

Chapter 5

Study One: A Machine Learning Approach for Detecting Fatigue during Repetitive Physical Tasks

5.1 Introduction

To construct a real-time fatigue assessment system, the first step is to demonstrate the feasibility of our proposed framework, and to select the optimal sensors, feature selection algorithm, and machine learning approach to conduct body fatigue classification. Therefore, the first study focuses on undertaking fatigue classification during repetitive physical tasks in a laboratory environment. The aims of this study were to:

1. Demonstrate the feasibility of the ErgoRelief framework using multiple linear regression analysis with body fatigue as output, and sensor measurements as input.
2. Evaluate the performances of different feature selection and machine learning approaches using binary body fatigue classification (high/low).
3. Explore the performances of different sensor systems on body fatigue classification, including non-wearable, wearable, and full (non-wearable and wearable) sensor systems.
4. Optimise the sensor deployment to prepare the system for a real-world application, targeting at minimising obtrusiveness, computational cost, avoiding privacy issues, as well as maintaining the classification performance.

5.2 Methodology

This study utilises a combination of wearable and non-wearable sensors to record data, including body motion, heart rate, and external force, whilst participants undertook tasks that were designed to simulate common activities found in routine factory work. According to the National Institute for Occupational Safety and Health [132], the most common physical tasks that factory workers encounter are lifting, lowering, pushing, pulling, carrying or moving of a load. Complex aviation manufacturing tasks, such as hand tool operation and assembling different structures, are based off combining these elementary tasks together [5].

In addition, the tasks were designed based on feedback provided by our industry partner. As this study aims to detect body fatigue during physical tasks, tasks that are considered to occur frequently/constantly (e.g. more than 35 times per day) have been included in the experiment task design. These tasks have been designed by combining a series of simple tasks, such as standing, reaching, and lifting. As such, simple tasks often occur in most complex tasks, such as standing, reaching, gripping/grasping, walking, pushing/pulling, lifting, and carrying. Hence, by combining common simple tasks together from real aviation manufacturing tasks, four repetitive physical tasks have been designed and undertaken in this study, including Two-handed Box Lifting, Two-handed Box Carrying, Two-handed Box Pushing/Pulling, and Two-handed Trolley Pushing/Pulling. Participant details, sensor placement, experiment procedure, and data analysis approach have been introduced in this section.

5.2.1 Participants

Thirty participants, including 13 males and 17 females, undertook study one, with an age between 20 and 54 years old (mean = 25.67, Standard Deviation = 6.66). Participants did not have a history of cardiovascular illness or any physical injuries prior to participating in the experiment. This experiment has been approved by The University of Queensland Ethics Sub-Committee (see Appendix 1).

5.2.2 Sensor Placement

As mentioned in Section 4.2, Shimmer3^{TM} IMUs and a Shimmer3^{TM} PPG ear clip have been used in this study in order to measure body part acceleration, angular velocity, and heart rate. In order to capture major body movements around the entire body, six Shimmer3^{TM} IMUs have been strapped to the Hip, Upper Spine, Left Arm, Right Arm, Right Wrist, and Right Shank [29, 39, 40, 133]. Additionally, the Shimmer3^{TM} PPG ear clip has been clipped to the left ear lobe and connected to the Shimmer3^{TM} sensor at the left arm, to measure blood volume pulse to derive heart rate (see Figure 5.1). Data have been directly recorded to the onboard SD card of each sensor while participants performed the tasks and were exported after each participant finished for further analysis.

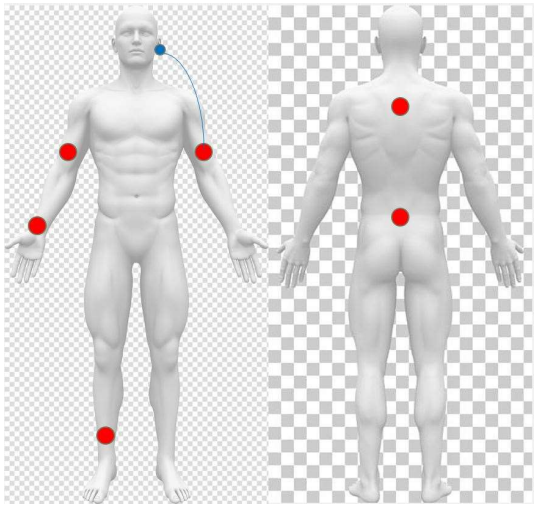


Figure 5.1 Shimmer Sensor Placement

Non-wearable sensors, including the force plate and load cell, have also been used in this study to measure external force exerted on the human body. The load cell has been utilised for all tasks. However, the force plate has not been used during the Two-handed Trolley Pushing/Pulling task, because it is not able to measure ground reaction force while participants walk around the experiment room.

5.2.3 Experiment Procedure

Four repetitive tasks have been designed in this experiment to simulate daily aviation factory work, including Two-handed Box Lifting, Two-handed Box Carrying, Two-handed Box Pushing/Pulling, and Two-handed Trolley Pushing/Pulling. Details for each task have been described in more detail below:

1. **Two-handed Box Lifting (see Figure 5.2a - 5.2e):** Participants lift a box from the floor to waist height, endure for approximately 3 seconds, and lower it down to the floor.

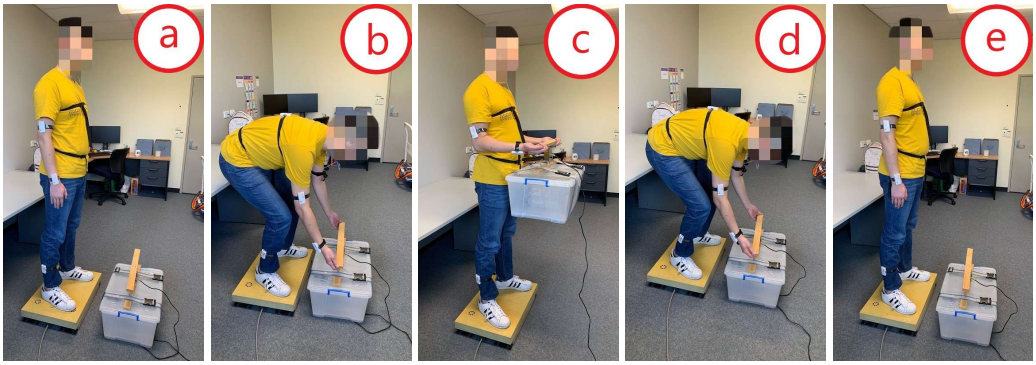


Figure 5.2 Two-handed Box Lifting: a) Stand upright b) Lift the box with two hands c) Endure for approximately 3 seconds d) Lower the box e) Stand upright

2. **Two-handed Box Carrying (see Figure 5.3a - 5.3e):** Participants carry a box with two hands and move it from one chair to another. Two chairs have been secured at each side of the participant, with 60cm from the centreline of the body.

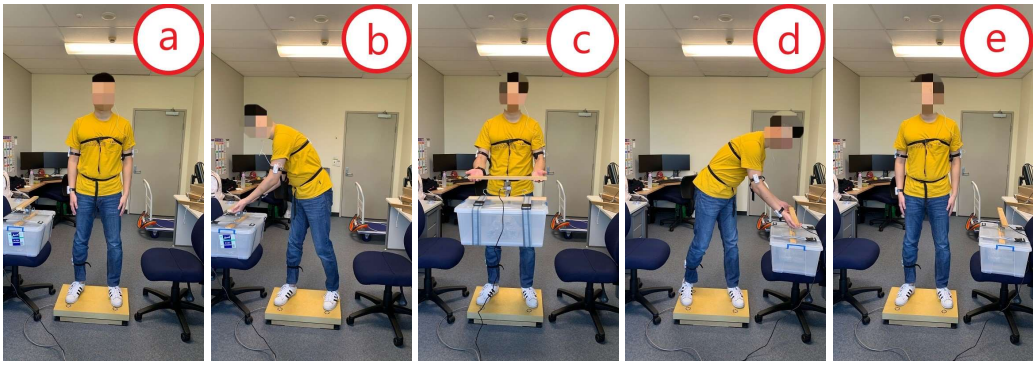


Figure 5.3 Two-handed Box Carrying: a) Stand upright b) Carry the box with two hands c) Move the box from one chair to another d) Lower down the box e) Stand upright

3. **Two-handed Box Pushing/Pulling (see Figure 5.4a – 5.4e):** Participants push a box with two hands further away from their bodies for approximately 60cm and pull it back. The box has been placed on a table at a height of 90cm. There is a handle affixed to the box for pushing/pulling.

Figure 5.4 Two-handed Box Pushing/Pulling: a) Stand upright b) Push the box away with two hands c) Reach the end d) Pull the box back with two hands e) Stand upright

4. **Two-handed Trolley Pushing/Pulling (see Figure 5.5a – 5.5d):** Participants walk and push a trolley away with two hands for approximately 5m and pull it back. The handle has been attached to the box, which has been fastened to the trolley. The handle height for the trolley is 80cm.

Figure 5.5 Two-handed Trolley Pushing/Pulling: a) Stand upright b) Push the trolley with two hands c) Pull the trolley with two hands d) Stand upright

Figure 5.6 displays the experiment procedure for each task. At the beginning of each task, participants have been asked to rest for 5 minutes. They then performed two repetitions of each task utilising loads from light to heavy weights and completed the Borg Scale Questionnaire [24] at the beginning and end of each repetition, in order to record their subjective rating of perceived exertion.

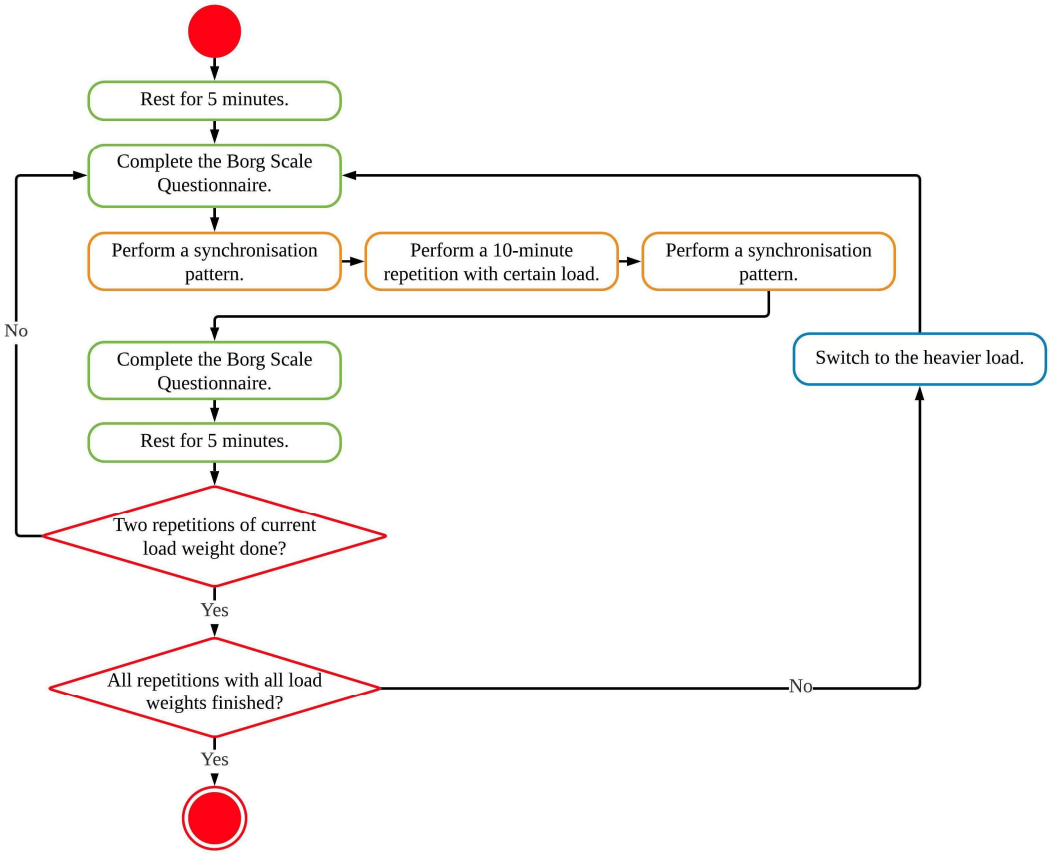


Figure 5.6 Experiment Procedure per Task

There are generally five levels of physical demand, including sedentary, light, medium, heavy, and very heavy work. Each level is defined by the load weights being handled. In addition, based on the experiment design, participants in the experiment performed physical tasks repetitively and constantly. Therefore, in order to incur different fatigue levels, as well as to protect participants safety, the load weights have been designed and listed below, which span between light, medium, and heavy work:

1. Two-handed Box Lifting: 3kg and 5kg.
2. Two-handed Box Carrying: 3kg and 5kg.
3. Two-handed Box Pushing/Pulling: 3kg, 10kg, and 20kg.
4. Two-handed Trolley Pushing/Pulling: 3kg, 40kg, and 80kg.

Each repetition took approximately 10 minutes and participants were required to conduct the task repetitively during this period. During each repetition, participants could use their preferred posture and speed to perform tasks. They were also allowed to stop if they could not sustain the task or due to unexpected issues. A 5-minute resting period was also provided after each repetition. After each task, participants were asked to fully rest and recover prior to commencing the next task. Lastly, as repetitive lifting/lowering may pose significant and continuous pressure to certain joints, such as the waist, to protect participants, the task sequence has been interwoven, which have been listed below:

1. Two-handed Trolley Pushing/Pulling
2. Two-handed Box Carrying
3. Two-handed Box Pushing/Pulling
4. Two-handed Box Lifting/Lowering

The entire experiment lasted for approximately 6 hours for each participant, with lifting/lowering and carrying tasks taking about 60 minutes to finish, whilst the pushing/pulling task took around 90 minutes to finish. As an incentive for taking part into the experiment, each participant was paid a $50 gift card after completing all the tasks.

5.2.4 Data Pre-Processing

As declared in Chapter 4, recorded body motion, heart rate, and external force data have been pre-processed before analysis. This includes data synchronisation, segmentation, resampling, and filtering. Both time- and frequency-domain features have then been extracted from the data. Overall, 1056 and 11 time-domain features have been extracted from body motion and heart rate data per task, respectively. For external force, 11 time-domain features have been extracted from the Two-handed Trolley Pushing/Pulling task, as the force plate has not been included in this task, whilst 121 time-domain features have been extracted from external force measurements for other tasks. Furthermore,

120 frequency-domain features have been extracted from body part acceleration per task. Therefore, 1198 features have been extracted for the Two-handed Trolley Pushing/Pulling task, whilst 1308 features have been extracted for other tasks.

The initial analysis has been conducted on a per-task basis, whilst the body fatigue classification has been performed on both a per-task and overall basis. To mitigate individual differences between participants, the dataset has been normalized per repetition, per person, per task and combined into four datasheets for each of the four tasks, in order to conduct task-specific analysis. Moreover, four datasheets have also been combined into one datasheet for the overall fatigue analysis, which utilised the data from a combination of all tasks.

After feature extraction, data labelling has then been performed on each datasheet. In this study, resting periods were excluded for data analysis. Therefore, samples in datasheets were labelled as either *high* or *low* fatigue. As mentioned in the Data Analysis section, the analysis was conducted on both per-task and overall basis. Therefore, Figure 5.7 displays the number of samples after data labelling for task-specific datasheets. The majority class (*high* fatigue for Two-handed Box Lifting and *low* fatigue for other tasks) were randomly undersampled to balance it with the minority classes.

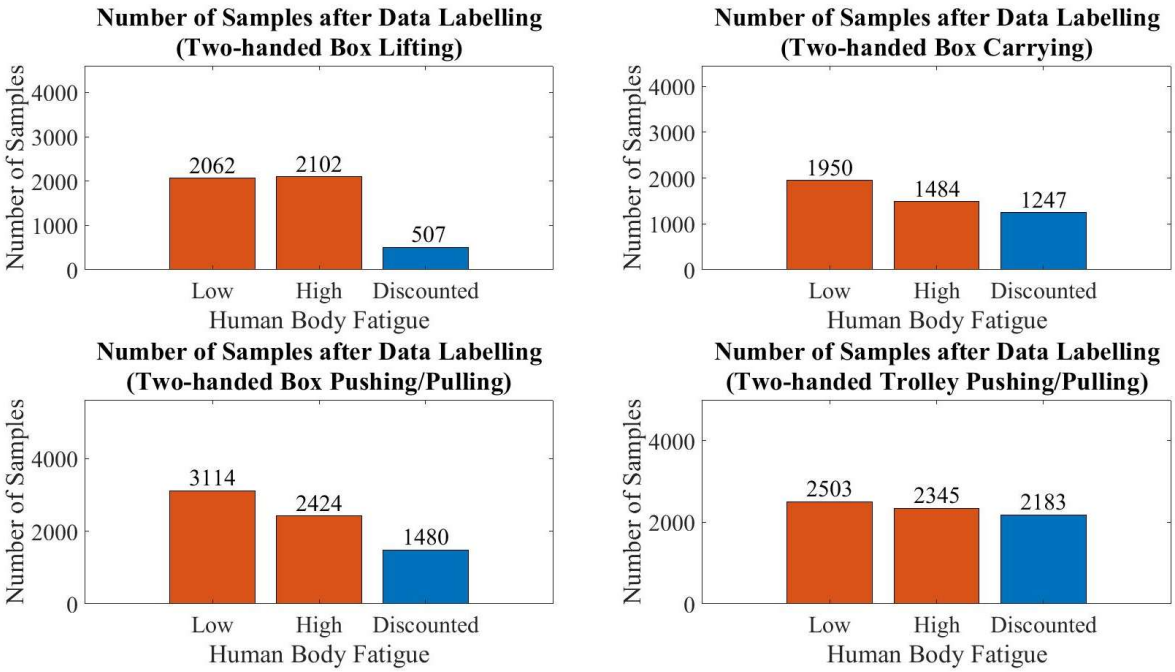


Figure 5.7 Number of Samples after Data Labelling per Task

Figure 5.8 shows the number of samples after data labelling for the overall datasheet from all tasks. *Low* fatigue has been the majority class, which was undersampled randomly to balance it with *high* fatigue class.

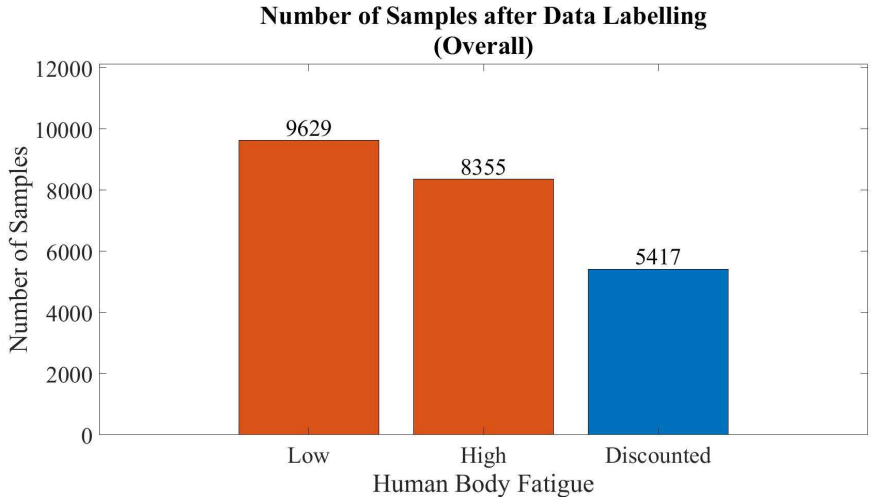


Figure 5.8 Number of Samples after Data Labelling for All Tasks

The purpose of conducting both per-task and overall analysis for body fatigue classification is to evaluate the machine learning model performance on both a specific task and overall task level, to investigate the generality among all tasks. For instance, in the case that the overall model performance remains similar or surpasses the task-specific model performance, it is unnecessary to conduct task-specific fatigue analysis in the future, as the task needs to be recognized first in the real-world application, which significantly increases the computational cost. Additionally, the performance of different sensor systems for classifying fatigue have also been assessed, including wearable, non-wearable, and full (wearable and non-wearable) sensor systems. Therefore, for each sensor system, three feature selection algorithms have first been implemented, which generated three optimal feature sets. For the non-wearable sensor system, approximately 15, 15, and 40 features have been selected by the ReliefF-10, ReliefF-30, and PCA, respectively. Additionally, for the wearable and full sensor systems, there are between 50 and 100 features in the ReliefF-10 and ReliefF-30 feature sets, and between 400 and 500 features in the PCA feature set. Three machine learning approaches have then been applied on each feature set for the binary body fatigue classification. Finally, the performances of different sensor systems, feature selection algorithms, and machine learning approaches have been systematically evaluated to construct the optimal system for the body fatigue classification.

5.3 Multiple Linear Regression Results

The first stage of the analysis included using a multiple linear regression (MLR) model with two purposes. The first is to illustrate that features extracted using a 30-second window with 50% overlap produce the optimal performance for future data processing, and the second one is to demonstrate the relationship between fatigue and different types of sensor measurements [134].

In the analysis of illustrating the best window size, a 50% overlapping window, with sizes of 2-seconds, 10-seconds, 20-seconds, 30-seconds, 60-seconds, 90-seconds, 120-seconds, 150-seconds, and 180-seconds have been selected to extract features [44]. As the experiment has been conducted in a laboratory environment, without any external stimulation, heart rate has been mapped to the body fatigue level, as a high heart rate can be linked with high body fatigue level [135].

The analysis has been performed for each task, with mean heart rate acting as the dependent variable. Independent variables have been selected from the feature set using Correlation-based Feature Selection (CFS) with best first search [136], which aims to find the best feature subset that is highly correlated with the dependent variable, and uncorrelated to each other.

Adjusted R^2 score and Root-mean-square error (RMSE) have been selected to evaluate the performance of the feature sets with different window sizes. An adjusted R^2 Score measures the percentage of variation from the dependent variable that can be explained by the independent variables, which is also penalised by the number of independent variables. RMSE measures how well the predicted data fits the reference data, which is calculated by the standard deviation of the prediction errors. Figure 5.9 and Figure 5.10 illustrate the adjusted R^2 scores and RMSEs versus window sizes. As illustrated in Figure 5.9, the adjusted R^2 score increases as the window size increases. In contrast, Figure 5.10 illustrates that as the window size increases, RMSE decreases. The reason for these changes is that feature sets with small window sizes capture the high frequency component from the signal, while feature sets with larger window sizes tend to reveal the overall trend of the signal. As heart rate changes slowly and gradually, feature sets with larger window sizes can effectively remove the high frequency noise and achieve better prediction performance. Results from both figures also demonstrate that when the window size is greater than 30 seconds, the adjusted R^2 score and RMSE do not significantly increase and decrease, respectively.

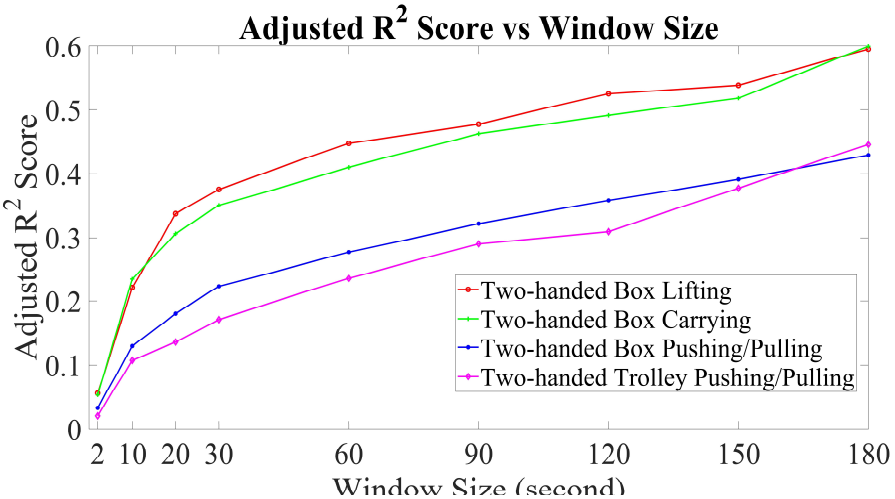


Figure 5.9 Adjusted R^2 Scores vs Window Size

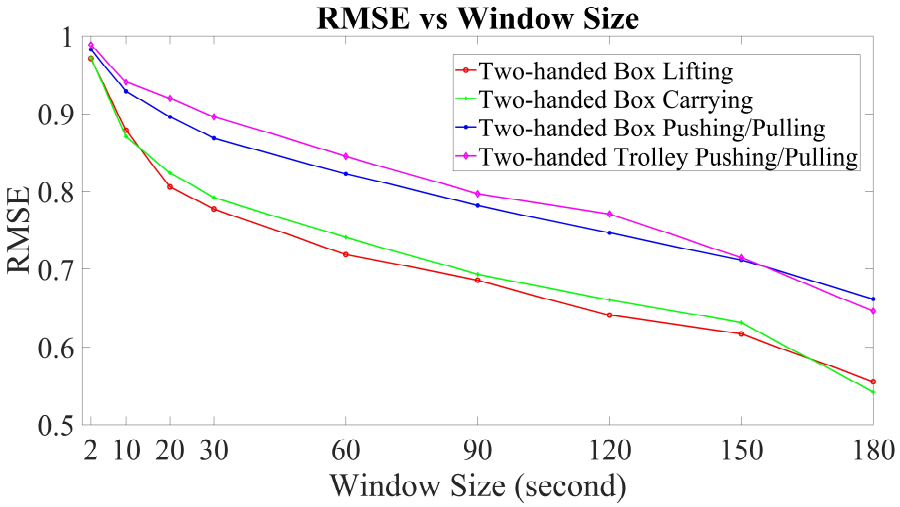


Figure 5.10 RMSE vs Window Size

In addition, Figure 5.11 illustrates the number of samples for each feature set of each task. Results illustrate that the Two-handed Box Lifting and Two-handed Box Carrying tasks have a similar number of samples, whilst a similar relationship applies for Two-handed Box Pushing/Pulling and Two-handed Trolley Pushing/Pulling tasks. These results demonstrate that when the window size increases above 30 seconds, the number of samples decreases significantly. This indicates that the MLR model may not be credible due to an insufficient number of samples.

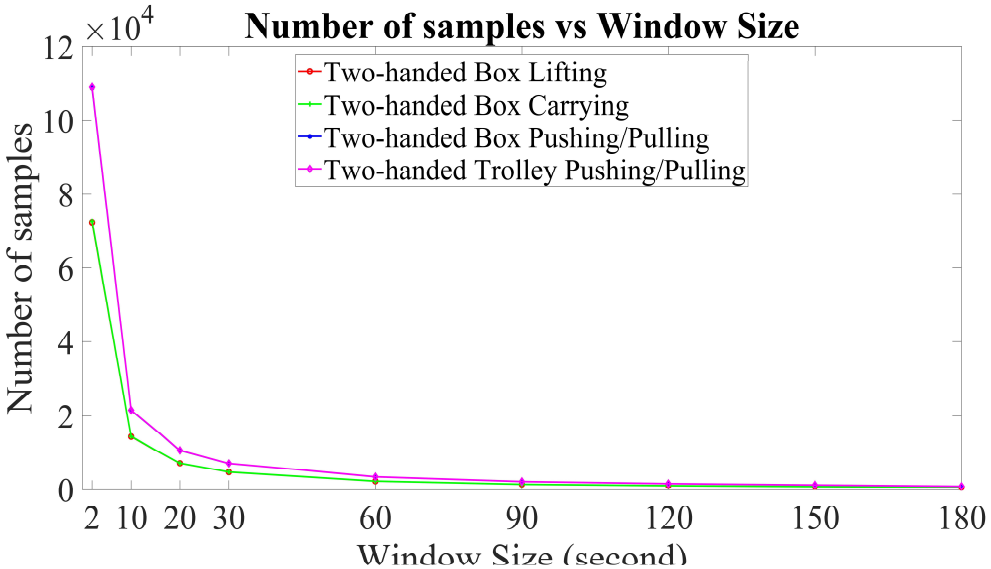


Figure 5.11 Number of Samples vs Window Size

As such, by taking the adjusted R^2 score, RMSE, and the number of samples for each window size into consideration, features extracted using a 30-second window with 50% overlapping area produced the optimal performance and can be applied to future data processing.

The second purpose for the MLR analysis is to demonstrate the relationship between body fatigue and sensor measurements. The analysis has been conducted in three stages to assess the effectiveness of different sensor systems in modelling fatigue and include using, 1) wearable sensors, 2) non-wearable sensors, and 3) both wearable and non-wearable sensors. This analysis has been conducted on a per-task basis. Table 5.1 illustrates the number of features used in each analysis, with 1,187 features coming from the wearable sensor system. As the force plate has not been utilised in the Two-handed Trolley Pushing/Pulling task, 11 and 121 features have been included in the non-wearable sensor analysis for the Two-handed Trolley Pushing/Pulling task and other tasks, respectively. Additionally, the un-normalised data have been used in this analysis. Therefore, four datasheets have been created for four tasks by combining the un-normalised feature sets from all participants.

Table 5.1 Number of Features for Each Sensor System

Sensor Systems	Number of Predictors
Wearable Sensor System	1187
Non-wearable Sensor System	11 (Two-handed Trolley Pushing/Pulling) 121 (Other tasks)
Full (Wearable and Non-wearable) sensor System	1198 (Two-handed Trolley Pushing/Pulling) 1308 (Other tasks)

In MLR analysis, the p-value is an indicator that is used to evaluate the correlation between each independent variable and the dependent variable. In other words, the smaller the p-value is, the more likely that there is a relationship between the dependent and independent variables. Therefore, three MLR models have been constructed first with fatigue as the dependent variable, and features from each sensor setup as the independent variables. Backward elimination has then been applied to each MLR model, which iteratively removes those independent variables with the highest p-value, until the p-values for all independent variables are below a threshold. The threshold has been set up to 0.05 [137]. In addition, similar as the analysis for the optimal window size illustration, the performance of the MLR models have been evaluated using the Adjusted R^2 Score and RMSE measures.

Figure 5.12 and 5.13 demonstrates the task specific MLR model performances based on the Adjusted R^2 Scores and RMSE, respectively. Results illustrated that the non-wearable system has the lowest performance over the three systems, with the highest Adjusted R^2 Score of 0.5853 and lowest RMSE of 1.1861 from the Two-handed Box Pushing/Pulling task. As the Adjusted R^2 Score results are lower than 0.6, this illustrates that only using the non-wearable sensors is not a sufficient predictor of fatigue. In contrast, the wearable sensors demonstrate a significant improvement to the performance of the MLR model, over the non-wearable sensor system, with highest Adjusted R^2 Score of 0.9179 and lowest RMSE of 0.5603, which have been accomplished during the Two-handed Box Carrying and Two-handed Box Pushing/Pulling tasks, respectively. Therefore, the results illustrate that the features from the wearable sensor system are better predictors of fatigue than those from the non-wearable sensor system.

Lastly, the full sensor system, which uses both wearable and non-wearable sensors, demonstrates negligible improvement over the wearable sensor system. The highest Adjusted R^2 Score of 0.9259 from the Two-handed Box Carrying task and lowest RMSE of 0.4972 from the Two-handed Box Pushing/Pulling task demonstrate only a 0.008 improvement over the wearable sensor setup. Results proved that combining both wearable and non-wearable sensors only marginally improved the performance of the MLR model but did not hinder performance.

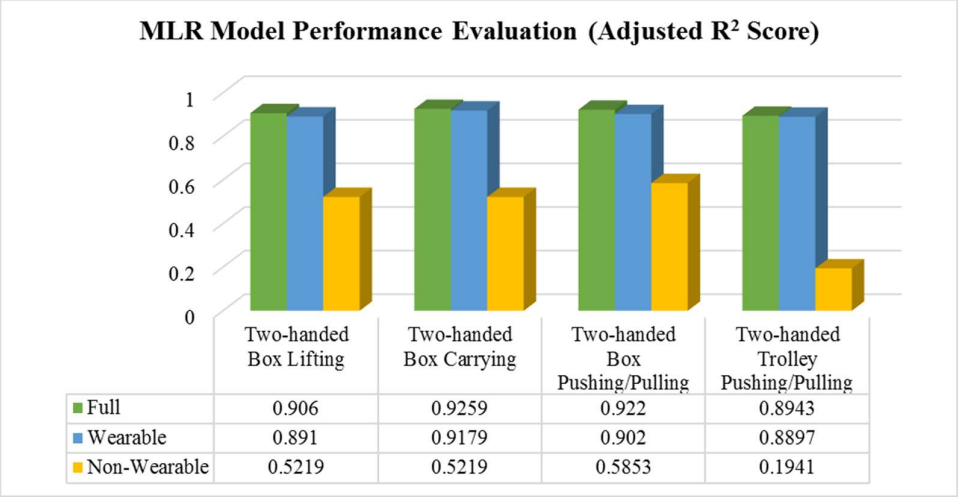


Figure 5.12 Adjusted R^2 Score MLR Model Performance Evaluation

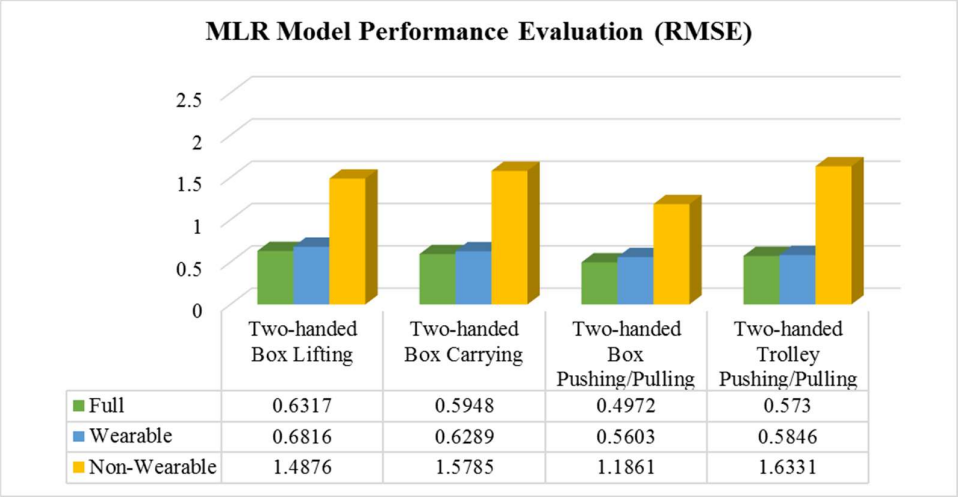


Figure 5.13 RMSE MLR Model Performance Evaluation

5.3.1 Summary

In summary, the MLR results illustrated that features extracted using a 30-second window with 50% overlapping area will be used for further analysis, and there is a strong relationship between body motion, heart rate, external force, and body fatigue. In comparison to other systems [28, 29, 32, 33], our setup has utilised both wearable and external predictors, which demonstrates an improved model for predicting fatigue. Although non-wearable sensors have been included in other works, they are mostly used in conjunction with cameras in a laboratory environment [55]. A benefit of these results is that non-wearable (external) sensors have been used in conjunction with wearable sensors to undertake a more comprehensive analysis, which has demonstrated improved results over using these sensors in isolation. We have also evaluated the performance between wearable and external sensors

and have statistically demonstrated that the wearable sensor system outperforms the non-wearable external sensors. This indicates that the wearable sensors are greater predictors of fatigue. The following section builds on these results and demonstrates the outcomes of classifying the data to detect fatigue.

5.4 Body Fatigue Classification Results

After demonstrating that our system is feasible in detecting fatigue, this section details the results of body fatigue classification and evaluates the performance of different sensor systems, feature selection algorithms, and machine learning approaches on classifying body fatigue.

5.4.1 Task-specific Model Performance Evaluation

As discussed in Chapter 4, three feature selection algorithms have been implemented to create optimal feature sets. The optimal classification performances have been achieved using the PCA feature set. Therefore, the results utilising the PCA feature sets have been described in this section, whilst the results of the ReliefF-10 and ReliefF-30 are displayed in Appendix 4-8.

5.4.1.1 Two-handed Box Lifting

For the Two-handed Box Lifting task, after applying PCA, 39, 393, and 419 features have been selected for the non-wearable, wearable, and full sensor systems, respectively. The k-value for the kNN models have been set to 5 for all sensor systems. For the P-SVM models, degrees and scales have been selected as 3 and 0.1 for all sensor systems. Lastly, costs have been configured as 0.25 for the non-wearable and full sensor systems, and 0.5 for the wearable sensor systems.

Figure 5.14 illustrates the performance of the models based on the non-wearable sensor system. Results demonstrate the highest accuracy of 63.67% from the P-SVM model, the highest F1 and AUC of 64.07% and 67.95% from the kNN model, and lowest error rate of 36.11% from the kNN model. The LDA model has the poorest performance, with an accuracy, F1, and AUC down to approximately 40% and error rate up to 60.64%. Therefore, the kNN and P-SVM models have similar performances using features from the non-wearable sensor system, followed by the LDA model.

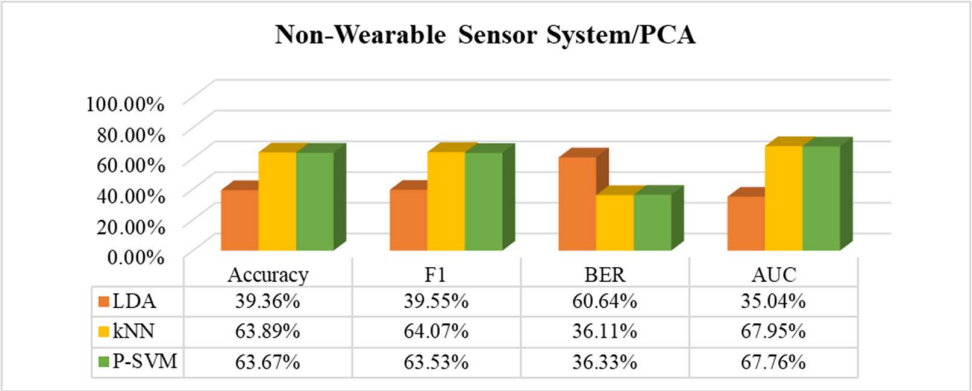


Figure 5.14 Two-handed Box Lifting Machine Learning Model Performance using the Non-Wearable Sensor System and PCA Feature Selection

Figure 5.15 demonstrates the performance of the models utilising features from the wearable sensor system. The performance of the P-SVM model outperforms the kNN and LDA models, with the highest accuracy, F1, and AUC of 95.85%, 95.86%, and 99.14%, and lowest error rate of 4.15%. The kNN model achieves an accuracy and error rate of 75.24% and 24.76%, whilst the LDA model performs the worst, with an error rate of 78.06%.

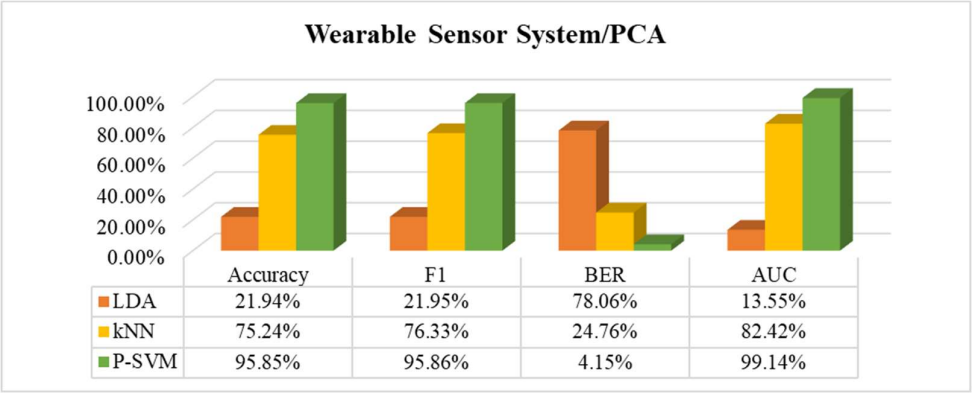


Figure 5.15 Two-handed Box Lifting Machine Learning Model Performance using the Wearable Sensor System and PCA Feature Selection

Figure 5.16 demonstrates the performance of the models based on the full sensor system. The P-SVM model outperformed the kNN and LDA models, with the highest accuracy, F1, and AUC of 96.23%, 96.24%, and 99.33%, and lowest error rate of 3.77%. The kNN model achieved an accuracy and F1 of approximately 75%. Moreover, the performance of the LDA model has decreased significantly compared to other two models, with an accuracy down to 21.23% and error rate as high as 78.77%.

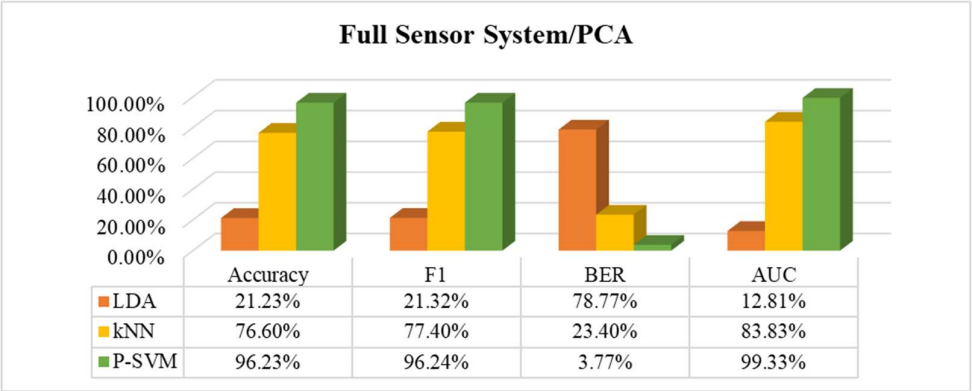


Figure 5.16 Two-handed Box Lifting Machine Learning Model Performance Evaluation using the Full Sensor System and PCA Feature Selection

In summary, considering the optimal performance for each sensor system, using the non-wearable sensor system alone cannot detect fatigue well during the Two-handed Box Lifting task. In addition, the wearable sensor system has demonstrated a significantly better performance than the non-wearable sensor system, indicating that it is more capable of classifying body fatigue. Furthermore, the full system with both wearable and non-wearable sensors illustrated marginal improvement to the classification performance over the wearable sensor system. Therefore, both non-wearable and wearable sensors can contribute to body fatigue classification during the Two-handed Box Lifting task.

5.4.1.2 Two-handed Box Carrying

For the Two-handed Box Carrying task, the numbers of features after applying PCA were similar to those from the Two-handed Box Lifting task, which are 36, 412, and 434 for the non-wearable, wearable, and full sensor systems respectively. The k-values for the kNN models have been selected as 5 for all sensor systems. The degrees, scales, and costs for the P-SVM models have been configured as 3, 0.1, and 0.5 for the non-wearable and wearable sensor systems, and 2, 0.1, and 1 for the full sensor system.

Figure 5.17 demonstrates the performance of the models based on the non-wearable sensor system. The P-SVM model outperforms the kNN and LDA models, with highest accuracy, F1, and AUC of 60.52%, 60.62%, and 64.17%, and lowest error rate of 39.48%. The performance of the kNN model decreases by 3% compared to the P-SVM model, with an accuracy and error rate of 57.35% and 42.65%. The LDA model still performs the worst, with an accuracy, F1, and AUC down to approximately 40%, and error rate up to 57.52%.

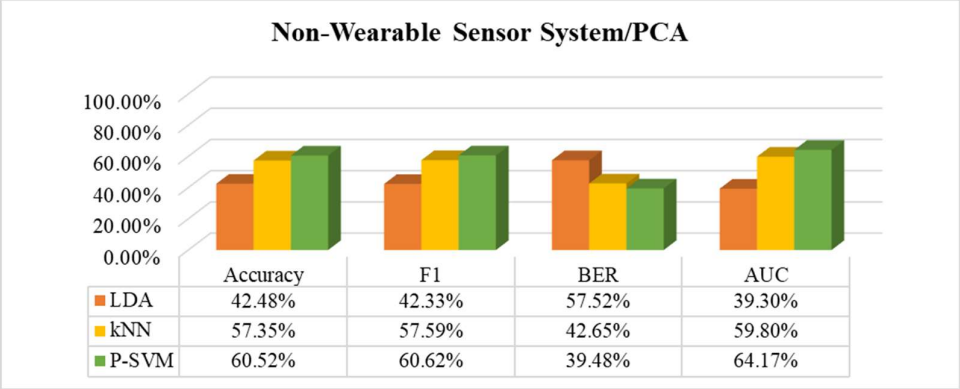


Figure 5.17 Two-handed Box Carrying Machine Learning Model Performance using the Non-Wearable Sensor System and PCA Feature Selection

Figure 5.18 illustrates the performance of models utilising features from the wearable sensor system. The highest accuracy has been achieved by the P-SVM model, followed by the kNN and LDA models, with values of 95.29%, 73.60%, and 28.46%. Similar trend applies for the F1 and AUC. In contrast, the lowest error rate has been achieved by the P-SVM model, followed by the kNN and LDA models, with values of 4.71%, 26.40%, and 71.54% respectively.

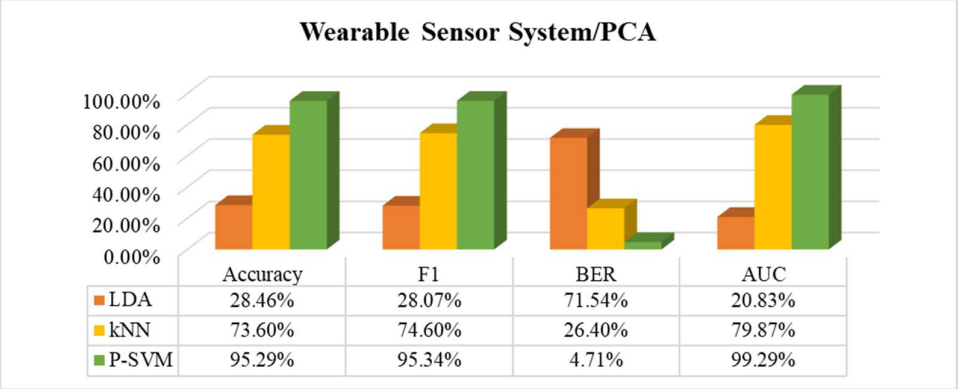


Figure 5.18 Two-handed Box Carrying Machine Learning Model Performance using the Wearable Sensor System and PCA Feature Selection

Figure 5.19 demonstrates the performance of models based on the full sensor system. The P-SVM model outperforms the kNN and LDA models significantly, with an accuracy, F1, error rate, and AUC of 95.43%, 95.47%, 4.57%, and 99.16% respectively. The performance of the kNN model is in the middle of the P-SVM and LDA models, with an accuracy of 73.91% and error rate of 26.09%. Lastly, the LDA model performs the worst, with an error rate up to 72.21%.

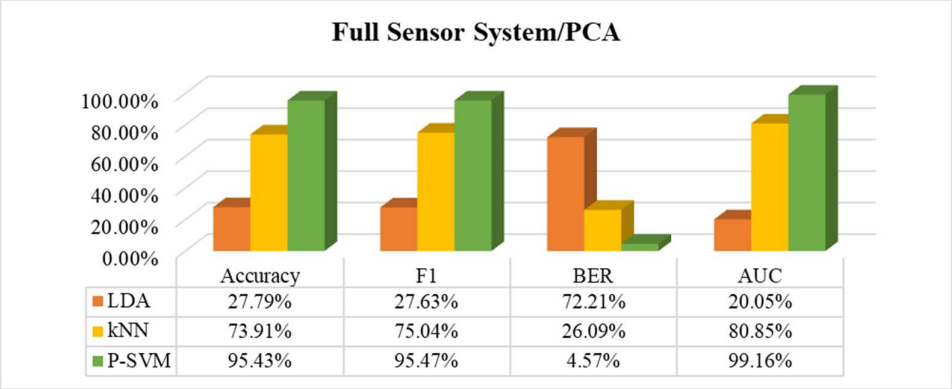


Figure 5.19 Two-handed Box Carrying Machine Learning Model Performance using the Full Sensor System and PCA Feature Selection

In summary, during the Two-handed Box Carrying task, the P-SVM models perform significantly better than the kNN and LDA models utilising the PCA feature sets for all sensor systems. Additionally, the wearable sensor system has been proven to be more capable of classifying fatigue compared to the non-wearable sensor system. Furthermore, incorporating both wearable and non-wearable sensors into the system shows negligible improvement on the body fatigue classification.

5.4.1.3 Two-handed Box Pushing/Pulling

For the Two-handed Box Pushing/Pulling task, 34, 415, and 439 features have been selected by PCA for the non-wearable, wearable, and full sensor systems. The k-values for the kNN models have been selected as 5 for all sensor systems. The degrees of the P-SVM models have been configured as 2 for the non-wearable and full sensor systems, and 3 for the wearable sensor system, whilst the scales have been selected as 0.1 for all sensor systems. Lastly, the costs for the P-SVM models have been configured as 0.25 for the non-wearable and wearable sensor systems, and 0.5 for the full sensor system.

Figure 5.20 demonstrates the performance of models using features from the non-wearable sensor system. The highest accuracy has been achieved by the P-SVM model, followed by the kNN and LDA models, with values of 67.77%, 64.82%, and 43.77% respectively. A similar trend applies for the F1 and AUC. Additionally, the trend reverses for the error rate, with highest and lowest values of 56.23% and 32.23% from the LDA and P-SVM models, respectively.

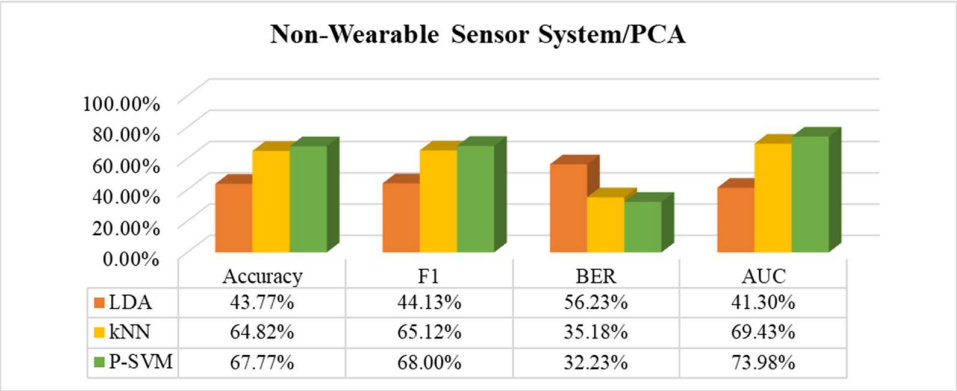


Figure 5.20 Two-handed Box Pushing/Pulling Machine Learning Model Performance using the Non-Wearable Sensor System and PCA Feature Selection

Figure 5.21 demonstrates the performance of models based on the wearable sensor system. The P-SVM model achieves a significantly better performance than the kNN and LDA models, with an accuracy, F1, and AUC of 96.10%, 96.13%, and 99.36%, and an error rate of 3.90%, whilst the LDA model has the lowest performance, with the lowest accuracy and highest error rate of 33.67% and 66.33%. Furthermore, the performance of the kNN model is about 20% lower than the P-SVM model and 40% higher than the LDA model.

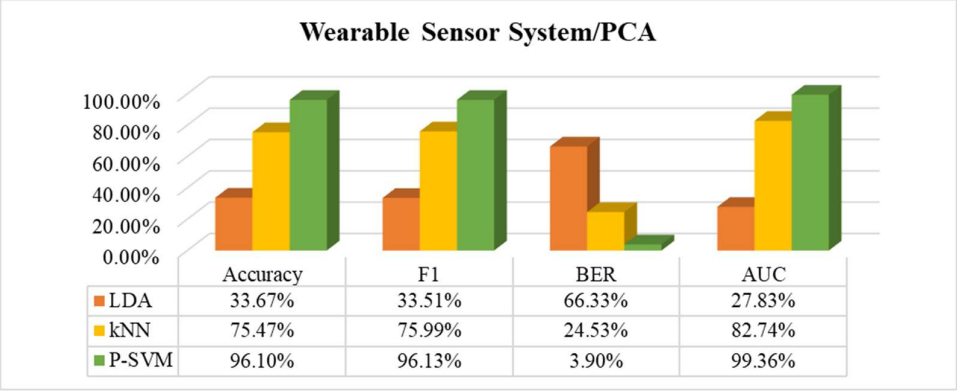


Figure 5.21 Two-handed Box Pushing/Pulling Machine Learning Model Performance using the Wearable Sensor System and PCA Feature Selection

Figure 5.22 demonstrates the performance of the models based on the full sensor system. The highest accuracy, F1, AUC, and lowest error rate have been achieved by the P-SVM model, with values of 96.97%, 96.99%, 99.61%, and 3.03% respectively. In addition, the LDA model still performs the worst, with the highest error rate of 66.98%.

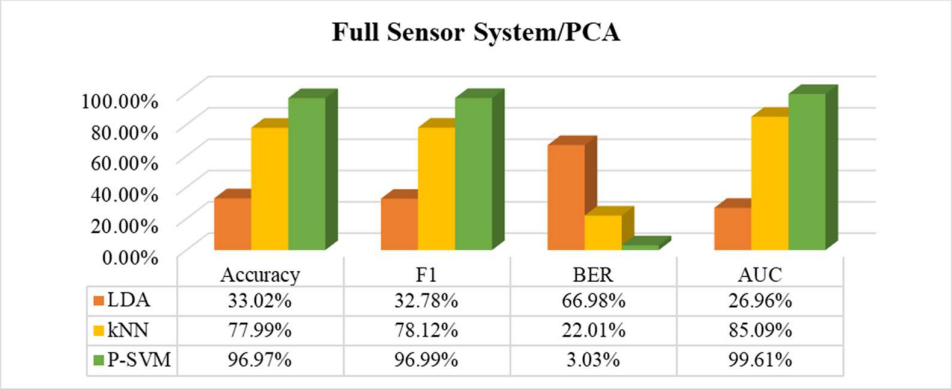


Figure 5.22 Two-handed Box Pushing/Pulling Machine Learning Model Performance Evaluation using the Full Sensor System and PCA Feature Selection

In summary, the performance of the P-SVM model surpasses other two models significantly for all sensor systems. Furthermore, the highest accuracy achieved by the non-wearable sensor system is less than 70%, illustrating that using non-wearable sensors alone cannot detect fatigue. Additionally, the machine learning models based on the full sensor system show negligible improvement over those from the wearable sensor system, indicating that combining both wearable and non-wearable sensors can better classify fatigue during the Two-handed Box Pushing/Pulling task.

5.4.1.4 Two-handed Trolley Pushing/Pulling

During the Two-handed Trolley Pushing/Pulling task, since the force plate was not included, the number of features in non-wearable sensor system is small compared with other tasks. Hence, 7, 442, and 447 features have been selected by PCA for the non-wearable, wearable, and full sensor systems. The k-values for the kNN models have been selected as 7 for the non-wearable sensor system, and 5 for the wearable and full sensor systems. In addition, the degrees of the P-SVM models have been selected as 1 for the non-wearable sensor system, and 3 for the wearable and full sensor systems. Lastly, the scales and costs for the P-SVM models have been configured as 0.1 and 1 for all sensor systems.

Figure 5.23 illustrates the performance of models based on the non-wearable sensor system. All models perform poorly due to the insufficient number of features. The P-SVM model achieves the highest performance, with an accuracy, F1, error rate, and AUC of 53.82%, 52.74%, 46.18%, and 55.55%. The kNN model performs similarly as the P-SVM model, whilst the LDA model performs the worst, with an accuracy, F1, error rate, and AUC of 45.95%, 46.18%, 54.05%, and 44.26% respectively.

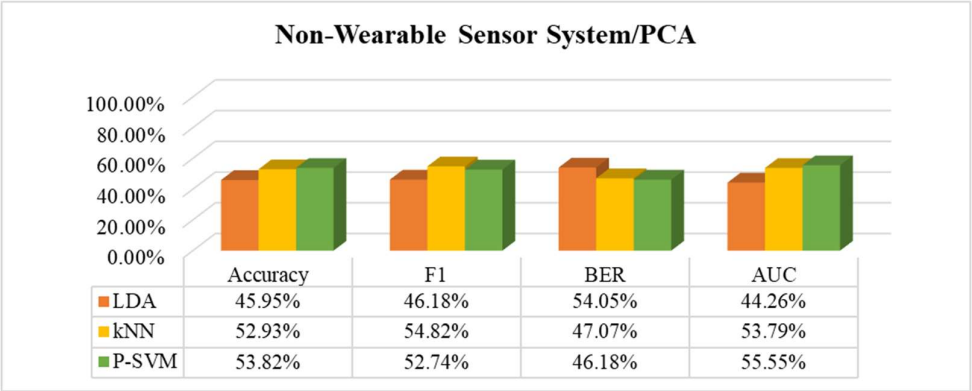


Figure 5.23 Two-handed Trolley Pushing/Pulling Machine Learning Model Performance using the Non-Wearable Sensor System with PCA Feature Selection

Figure 5.24 illustrates the performance of models based on the wearable sensor system. The P-SVM model outperforms the other two models significantly, with the highest accuracy, F1, and AUC of 97.09%, 97.09%, and 99.64%, and lowest error rate of 2.91%. The kNN model performs worse than the P-SVM model, with an accuracy of 71.74% and error rate of 28.26%. Lastly, the LDA model performs the worst, with lowest accuracy, F1, and AUC of 33.38%, 33.66%, and 27.56%, and highest error rate of 66.62%.

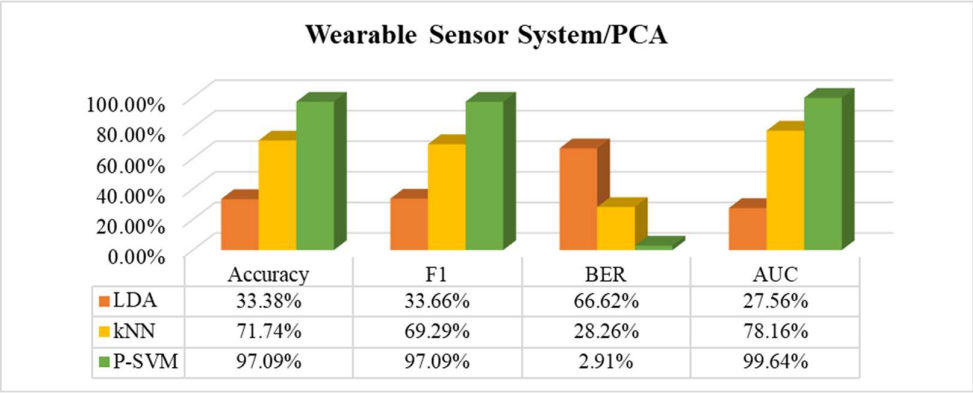


Figure 5.24 Two-handed Trolley Pushing/Pulling Machine Learning Model Performance using the Wearable Sensor System and PCA Feature Selection

Figure 5.25 demonstrates the performance of models based on the full sensor system. The P-SVM model outperformed the other two models significantly, with an accuracy, F1, error rate, and AUC of 97.17%, 97.17%, 2.83%, and 99.66%, whilst the performance of the kNN model decreased by approximately 30% compared with the P-SVM model, with an accuracy and F1 of approximately 70%, error rate of 28.06%, and AUC of 78.36%. Lastly, the performance of the LDA model is the worst over three models, with an error rate of 66.71%.

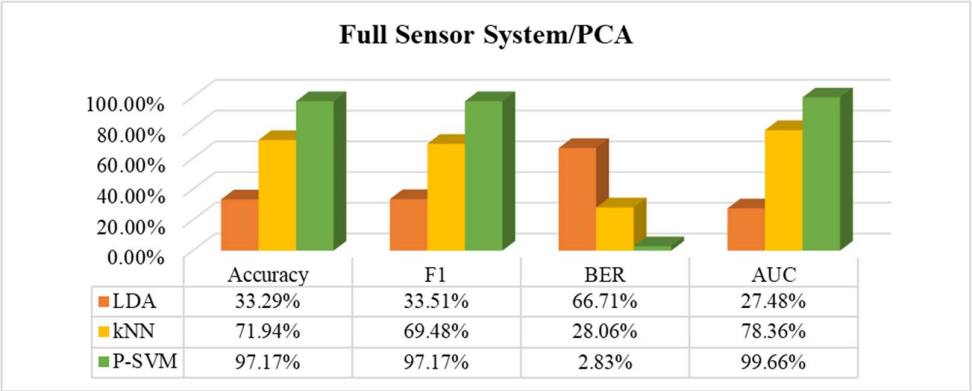


Figure 5.25 Two-handed Trolley Pushing/Pulling Machine Learning Model Performance using the Full Sensor System and PCA Feature Selection

In summary, considering the insufficient number of features from the non-wearable sensor system, results illustrated that using a non-wearable sensor system alone cannot detect fatigue properly during the Two-handed Trolley Pushing/Pulling task. Additionally, as the number of features from wearable sensors are significantly greater than that from the non-wearable sensor system, the performance between the wearable and full sensor systems remains unchanged. Therefore, the results illustrated that although combining both wearable and non-wearable sensors can slightly improve the fatigue classification performance during the Two-handed Trolley Pushing/Pulling task, using wearable sensor system alone can still achieve reasonable classification accuracy.

5.4.2 Overall Model Performance Evaluation

In this section, data from each task have been combined into one datasheet to conduct binary body fatigue classification, with 41, 502, and 538 features being selected by the PCA for the non-wearable, wearable, and full sensor systems respectively. The k-values for the kNN models and the degrees for the P-SVM models have been selected as 5 and 3 for all sensor systems. In addition, the scales for the P-SVM models have been configured as 0.01 for the non-wearable sensor system, and 0.1 for the wearable and full sensor systems. Lastly, the costs for the P-SVM models have been selected as 1 for the non-wearable and full sensor systems, and 0.25 for the wearable sensor system.

Figure 5.26 demonstrates the performance of models based on the non-wearable sensor system. The kNN and P-SVM models have similar performance, with an accuracy, F1, error rate, and AUC of approximately 59%, 60%, 41%, and 62% respectively. The LDA model still performs the worst, with an accuracy, F1, error rate, and AUC of 45.86%, 46.02%, 54.14%, and 43.99%.

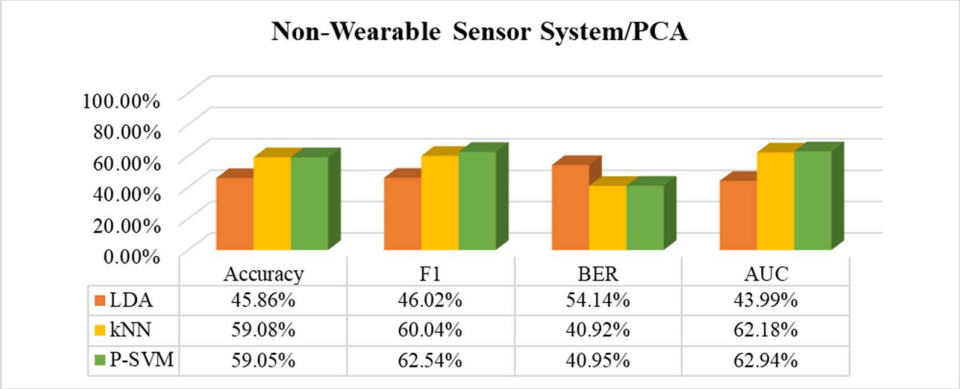


Figure 5.26 Overall Machine Learning Model Performance using the Non-Wearable Sensor System and PCA Feature Selection

Figure 5.27 demonstrates the performance of models utilising features from the wearable sensor system. The performance of the P-SVM model surpassed the kNN model, with an accuracy, F1, error rate, and AUC of 93.18%, 93.22%, 6.82%, and 98.05%. The kNN model achieved a performance in the middle of the P-SVM and LDA models, with an accuracy, F1, and AUC of between 70% and 80%, and an error rate of 27.88%. Moreover, the LDA model performs the worst, with an accuracy, F1, and AUC less than 40%, and error rate greater than 60%.

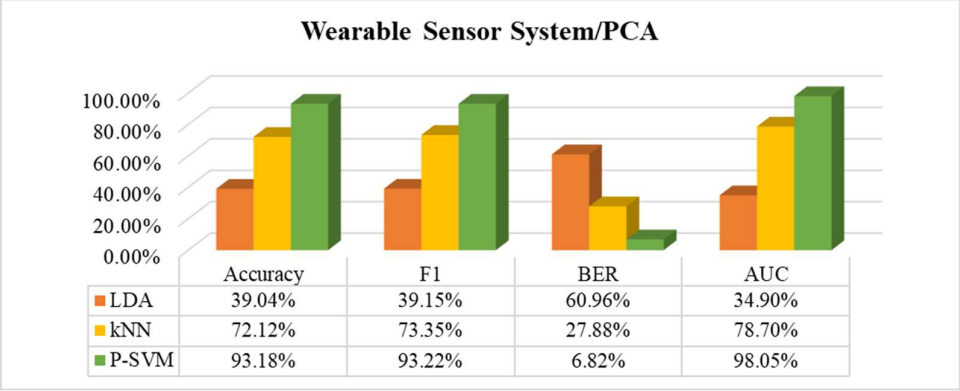


Figure 5.27 Overall Machine Learning Model Performance using the Wearable Sensor System and PCA Feature Selection

Figure 5.28 demonstrates the performance of models based on the full sensor system. The P-SVM model outperforms the kNN and LDA models significantly, with an accuracy, F1, error rate and AUC of 94.15%, 94.18%, 5.85%, and 98.49%. The kNN model performs 20% lower than the P-SVM model, with an accuracy, F1, error rate, and AUC of 74.10%, 74.50%, 25.90%, and 80.78%. Lastly, the LDA model performs the worst, with the lowest accuracy, F1, and AUC of 38.54%, 38.65%, and 34.16%, and highest error rate of 61.46%.

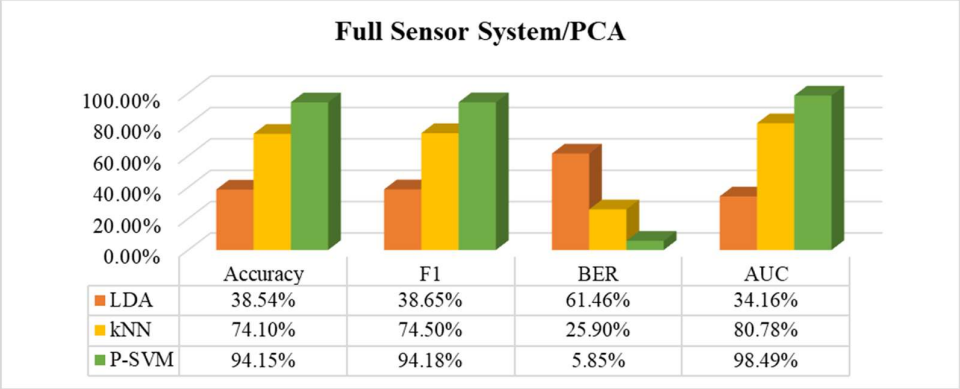


Figure 5.28 Overall Machine Learning Model Performance using the Full Sensor System and PCA Feature Selection

In summary, for the overall data analysis with a combination of all tasks, the P-SVM models have the optimal classification performance for all sensor systems, followed by the kNN and LDA models. Additionally, the optimal performance of the full sensor system displays a negligible improvement over that from the wearable sensor system. Therefore, combining both wearable and non-wearable sensors in one system is beneficial in classifying fatigue.

5.4.3 Summary

This analysis conducted binary body fatigue classification using three machine learning approaches on both a per-task and overall basis. This has been undertaken using three types of sensor systems, including non-wearable, wearable, and the full (non-wearable and wearable) system. The performance of each model, from each feature set, has been compared in order to establish which type of data and algorithms are best for performing fatigue classification. This is necessary so that more accurate models can be developed that do not include redundant data streams that contribute very little to the overall result. The results demonstrate that P-SVM consistently outperformed other algorithms, whilst LDA was the worst. This can be attributed to the design of the algorithms. LDA was found to not handle a complicated feature set with hundreds of features well. In contrast, P-SVM is a complex machine learning method, indicating that it is capable of solving difficult classification problems. However, kNN is a trade-off between LDA and P-SVM, as it has a lower computational cost than P-SVM, without diminishing performance.

The non-wearable sensor features set results demonstrated that P-SVM and kNN perform similarly, followed by LDA. As the model performances are relatively poor, using non-wearable sensors alone may not be appropriate to conduct binary body fatigue classification. In addition, for both the per-

task and overall analysis, the wearable sensor system shows significant improvement over the non-wearable sensor system, demonstrating that the wearable sensor system is more accurate in classifying body fatigue than the non-wearable sensor system. Furthermore, by comparing the performances of the P-SVM models over the three feature sets, results illustrated that the inclusion of the non-wearable features only increased performance by 1% over the wearable features. This demonstrates that wearable sensors alone are sufficient for detecting fatigue. Taking the high infrastructure cost of non-wearable sensor deployment into account, removing non-wearable sensors from the system will not sacrifice substantial model performance.

Finally, the highest accuracies achieved by task-specific models have been 96.23%, 95.43%, 96.97%, and 97.17% for the Two-handed Box Lifting, Two-handed Box Carrying, Two-handed Box Pushing/Pulling, and Two-handed Trolley Pushing/Pulling tasks respectively, whilst these values are comparable with the accuracy achieved by the overall models, which is 94.15%. Therefore, results illustrated that the body fatigue classification can still accomplish reasonable performance with a combination of all tasks, which decreases the computational cost and increases the fatigue classification efficiency. Therefore, the following analysis has been done on an overall basis by combining all tasks' data.

5.5 Sensor Optimization Results

The initial sensor setup for this study included both wearable and non-wearable sensors in order to cover the movement of the whole body. The collected data and the initial results have then been used as a baseline for assessing body fatigue. However, such sensor deployment is not feasible for a real-world application, because including many sensors in one system may lead to high computational cost, increase obtrusiveness and discomfort, cause privacy and security issues, and hinder the classification model performance due to redundant information. Therefore, this section covers the process of sensor optimisation that aims to minimise the number of sensors used in the system without sacrificing model performance. Figure 5.29 illustrates the sensor optimisation procedure, coupled with the corresponding reasons. There are five major steps in this procedure, which removes non-wearable sensors, heart rate measurement, IMUs on Upper Spine and Right Arm, as well as computationally intensive features in sequence. Details for each step will be discussed in the following paragraphs, with results demonstrating that removing these sensors from the system will not sacrifice significant classification model performance.

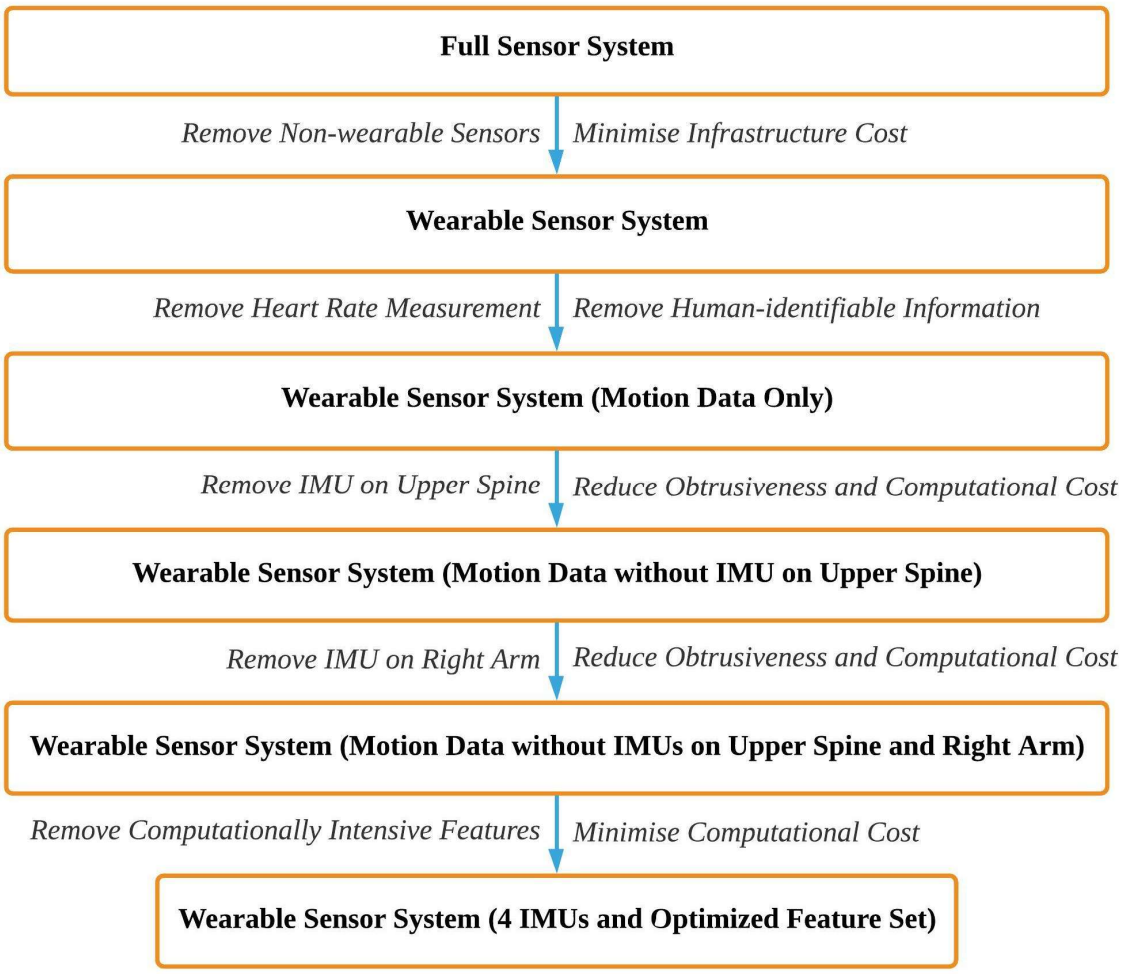


Figure 5.29 Sensor Optimisation Procedure

As illustrated in Section 5.3 and 5.4, the wearable sensor system significantly outperformed the non-wearable system when classifying fatigue. Furthermore, the addition of the non-wearable sensors to the wearable sensors only illustrated a slight improvement to the model's performance. Considering the high infrastructure cost of deploying non-wearable sensors in an aviation factory environment, as well as their negligible contribution to classify body fatigue, the non-wearable sensors have been removed from the system, and the wearable sensor system has been used as the initial setup for the sensor optimisation. Furthermore, as the performance of the task-specific and overall models are consistent, the sensor optimisation has been conducted with a combination of data from all physical tasks.

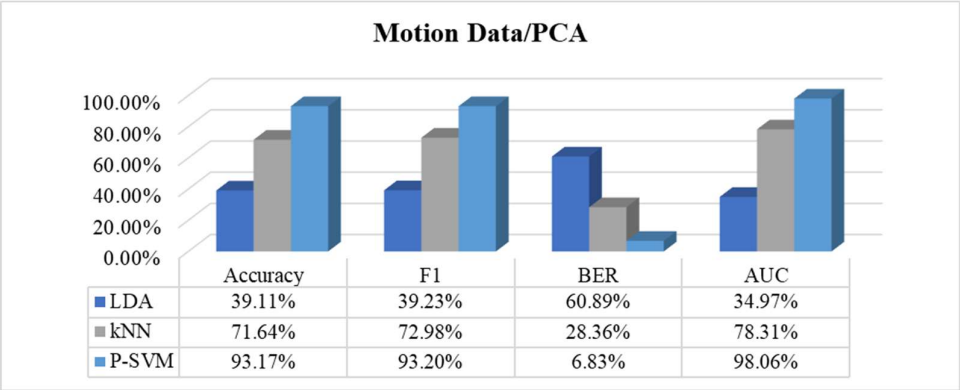
In addition, considering the highest body fatigue classification performance, the PCA feature selection algorithm works significantly better than the ReliefF-10 and ReliefF-30 during the sensor

optimisation procedure. Therefore, the results will be displayed based on the PCA feature sets, and the results utilising the ReliefF-10 and ReliefF-30 feature sets are displayed in Appendix 9.

5.5.1 Physiological Data Optimisation

While heart rate is an indicator of fatigue, the accuracy of its estimation can be adversely influenced by motion artefacts caused by body movements, especially when measured by a PPG sensor, because the body movements may cause the movement of the PPG sensor on the skin [114, 119, 120, 138-140]. In addition, recording, transmitting, and sharing personal health information, such as physiological data, can pose a significant security and privacy issue to a factory-wide system implementation, because the data may be disclosed to illegal or unauthorised entities/persons and used for other purposes rather than health monitoring [141-143]. Therefore, the first step of the sensor optimisation procedure has been to evaluate classification performance of the wearable sensor system with/without physiological data.

Figure 5.30 illustrates the performance of models using only motion data, with heart rate having been removed from the system. There are 500 features being selected by the PCA. The k-value for the kNN model has been selected as 5, whilst the degree, scale, and cost of the P-SVM model have been selected as 3, 0.1, and 0.5, respectively. The performance of the P-SVM model surpasses two models significantly, with highest accuracy, F1, and AUC of 93.17%, 93.20% and 98.06%, and lowest error rate of 6.83%. The kNN model performs worse than the P-SVM model, with an accuracy, F1, error rate, and AUC of 71.64%, 72.98%, 29.36%, and 78.31%. Lastly, the LDA model performs the worst, with lowest accuracy, F1, and AUC less than 40%, and an error rate of 60.89%.



	Accuracy	F1	BER	AUC
LDA	39.11%	39.23%	60.89%	34.97%
kNN	71.64%	72.98%	28.36%	78.31%
P-SVM	93.17%	93.20%	6.83%	98.06%

Figure 5.30 Machine Learning Model Performance using the Motion Data System and PCA Feature Selection

Overall, the optimal performance using the motion data has been achieved by the P-SVM model, with an accuracy and error rate of 93.17% and 6.83%, which are consistent with values from the wearable sensor systems, which are 93.18% and 6.82%. Hence, results illustrated that removing physiological data from the system does not significantly decrease the model performance of the binary body fatigue classification.

5.5.2 Motion Data Optimisation

After removing the physiological data, the next step is to minimise the number of IMUs strapped to the human body, whilst maintaining a high classification performance. Affixing many sensors to the human body may lead to obtrusiveness and discomfort of factory workers, which may cause a series of problems, such as decrease in efficiency and loss of attention. Additionally, as each IMU generates a large amount of data, which requires a significant computational cost for data processing, the number of sensors needs to be minimised to eliminate redundant data and reduce computational cost.

The motion data optimisation process consists of looking at the correlation between the IMUs, the machine learning model performance, and the computational cost of feature extraction. One IMU has been removed from the system at a time, and the model performance for the trimmed system has then been compared with the original one, in order to achieve an acceptable level of classification performance using the minimum number of sensors.

Since each IMU records body part acceleration and angular velocity with different coordinate systems, it is impractical to calculate the correlation of each axis between IMUs directly, such as the x, y, and z axes. Therefore, their magnitudes have been used to calculate the correlation between each sensor. Table 5.2 illustrates the correlation of the acceleration and angular velocity magnitudes between each IMU, which has been calculated as the mean value of the correlation per repetition, per participant, for all tasks. Values greater or equal to 0.6 are considered to be a strong correlation, which have been highlighted in the table. For both acceleration and angular velocity magnitudes, three combinations of IMUs have strong correlations and have been highlighted, which are Right Arm and Left Arm, Right Arm and Right Wrist, and Hip and Upper Spine. The highest values for the acceleration and angular velocity correlations both happen with the combination of Hip and Upper Spine, which are 0.76 and 0.76, respectively, whilst the values are between 0.6 and 0.7 for other two combinations. All other correlations between different IMUs are below 0.6.

Table 5.2 Acceleration and Angular Velocity Magnitude Correlation between IMUs

Acceleration Magnitude						
Body Areas	Left Arm	Right Wrist	Right Arm	Hip	Right Shank	Upper Spine
Left Arm	1	0.51	**0.64**	0.33	0.18	0.47
Right Wrist		1	**0.67**	0.18	0.11	0.27
Right Arm			1	0.34	0.19	0.47
Hip				1	0.31	**0.76**
Right Shank					1	0.25
Upper Spine						1
Angular Velocity Magnitude						
Body Areas	Left Arm	Right Wrist	Right Arm	Hip	Right Shank	Upper Spine
Left Arm	1	0.52	**0.66**	0.35	0.21	0.37
Right Wrist		1	**0.62**	0.35	0.24	0.33
Right Arm			1	0.36	0.26	0.38
Hip				1	0.37	**0.76**
Right Shank					1	0.34
Upper Spine						1

Therefore, either the Hip or Upper Spine can be removed from the system. According to feedback from the participants, the IMU at the Upper Spine was extremely uncomfortable during the experiment, as it needed to be strapped to the upper chest. Additionally, it continuously and gradually slid downwards due to the repetitive body motion. Hence, the IMU from the Upper Spine has been removed first, and the model performance for the trimmed system has then been evaluated.

Figure 5.31 illustrates the performance of models based on the motion data system without an IMU on the Upper Spine. After applying PCA, 428 features have been selected. The k-value for the kNN model has been selected as 5, whilst the degree, scale, cost of the P-SVM model have been selected as 3, 0.1, and 0.5, respectively. The highest accuracy has been achieved by the P-SVM model, with the value of 90.95%, which is 19% and 50% higher than values from the kNN and LDA models. A similar trend has been followed by the F1 and AUC, with highest values of 90.98% and 96.92% from the P-SVM model, and lowest values of 40.22% and 36.38% from the LDA model. Lastly, the highest error rate has been from the LDA model, which is 59.90%, followed by the kNN and P-SVM models.

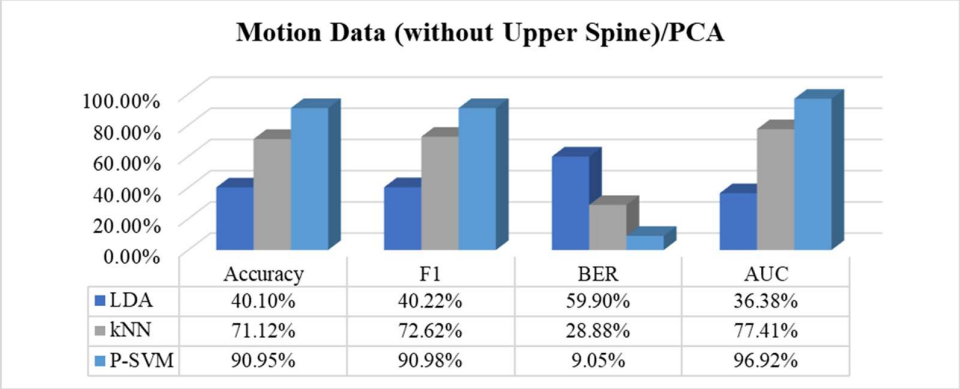
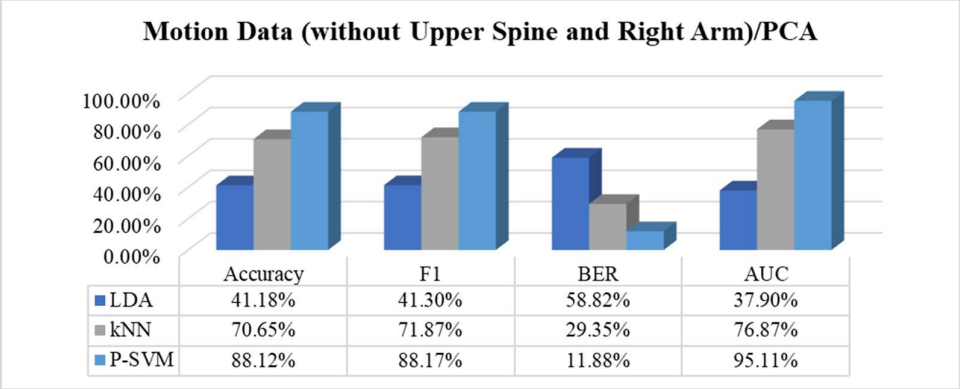


Figure 5.31 Machine Learning Model Performance using the Motion Data System (without Upper Spine) and PCA Feature Selection

In summary, the classification performance decreased by approximately 3% after sensor optimisation. Therefore, results illustrated that removing the IMU on the Upper Spine from the system did not sacrifice significant model performance for the body fatigue classification.

As illustrated in Table 5.2, the other two combinations of IMUs with high correlations are Left Arm and Right Arm, and Right Arm and Right Wrist. Since all tasks in this study have been designed as symmetric movements, there may be some overlapping information recorded by IMUs on the Left Arm and Right Arm. Therefore, one of them is a possible candidate to be removed from the system without losing too much performance. In addition, as there is another IMU on the Right Wrist, removing the IMU from the Left Arm may make the system unable to record full body motion. Therefore, the next step of the sensor optimisation has been to eliminate the IMU from the Right Arm. The machine learning model performance has then been evaluated before and after sensor optimization, to prove that the system is still feasible.

Figure 5.32 demonstrates the performance of models based on the motion data system without IMUs on the Upper Spine and Right Arm. After feature selection, 353 features have been selected. The k-value for the kNN model has been selected as 5, whilst the degree, scale, and cost of the P-SVM model has been selected as 3, 0.1, and 1. The P-SVM model outperforms the kNN and LDA models by approximately 18% and 47%, with highest accuracy, F1, and AUC of 88.12%, 88.17%, and 95.11%, and lowest error rate of 11.88%. The kNN model performs worse than the P-SVM model, with an accuracy and error rate of 70.65% and 29.35% respectively. Lastly, the LDA model still performs the worst, with an error rate of 58.82%, indicating that it is not able to classify most samples correctly.



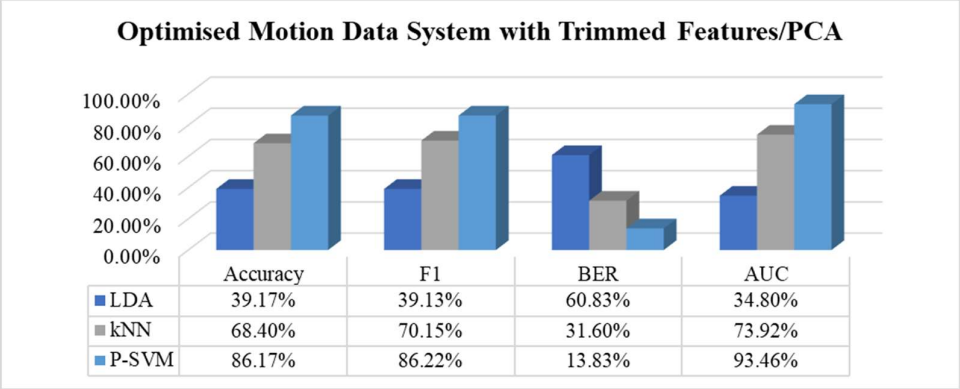
	Accuracy	F1	BER	AUC
LDA	41.18%	41.30%	58.82%	37.90%
kNN	70.65%	71.87%	29.35%	76.87%
P-SVM	88.12%	88.17%	11.88%	95.11%

Figure 5.32 Machine Learning Model Performance using the Motion Data System (without Upper Spine and Right Arm) and PCA Feature Selection

In conclusion, by comparing the motion data system with/without data from the Right Arm, the optimal model performance decreased by less than 3%, illustrating that removing the IMU from the Right Arm will not harm the system performance tremendously. Therefore, the IMU from the Right Arm has been removed from the system, and four IMUs have been kept in the system, which are located in the Left Arm, Right Wrist, Hip, and Right Shank.

Including four IMUs in the system can minimise obtrusiveness, as well as maintain a reasonable level of performance. However, to achieve a real-time system, the computational cost needs to be further reduced. Therefore, the last step for the motion data optimisation has been to trim computationally intensive features, such as frequency domain features and autocorrelation. Frequency domain feature extraction requires the calculation of FFT and PSD, whilst autocorrelation calculates the correlation between the current and previous window.

Figure 5.33 displays the performance of models based on the optimized motion data system with trimmed features. After applying PCA, 266 features have been selected in the optimal feature set. The k-value for the kNN models have been configured as 5, and the degree, scale, and cost of the P-SVM model have been selected as 3, 0.1, and 0.5, respectively. The highest accuracy has been achieved by the P-SVM model, with a value of 86.17%, followed by the kNN and LDA models. A similar trend applied for the F1 and AUC, with highest values of 86.22% and 93.46% from the P-SVM model, and lowest values of 39.13% and 34.80% from the kNN model. In addition, the lowest error rate has also been achieved by the P-SVM model, followed by the kNN and LDA models, with values of 13.83%, 31.60%, and 60.83% respectively.



	Accuracy	F1	BER	AUC
LDA	39.17%	39.13%	60.83%	34.80%
kNN	68.40%	70.15%	31.60%	73.92%
P-SVM	86.17%	86.22%	13.83%	93.46%

Figure 5.33 Machine Learning Model Performance using the Optimised Motion Data System with Trimmed Features and PCA Feature Selection

Overall, the highest accuracy achieved is similar to the performance of the system before trimming features, indicating that the computationally intensive features can be removed from the system while the machine learning model performance can be maintained.

5.5.3 Summary

In summary, the goal of the sensor optimisation stage is to mitigate privacy issues, minimise obtrusiveness during physical tasks, reduce computational cost, as well as maintain the system's feasibility and performance. As such, the physiological PPG sensor has first been removed, followed by the two IMUs from the Upper Spine and Left Arm. Finally, computationally intensive features have been removed from the system, including frequency domain features and autocorrelation. Results illustrated that a suitable level of classification can still be achieved after optimising the system.

5.6 Discussion

In this study, four repetitive physical tasks have been designed to simulate daily aviation factory work, including Two-handed Box Lifting, Two-handed Box Carrying, Two-handed Box Pushing/Pulling, and Two-handed Trolley Pushing/Pulling. Data have been recorded from both wearable and non-wearable sensors, including body motion, external force exerted on the human body, and physiological data, such as heart rate. Multimodal data have then been pre-processed and both time- and frequency-domain features have been extracted. Self-reported ratings of fatigue have been acquired from the participants at the beginning and end of each repetition and a change score has been calculated. The multiple linear regression analysis has been conducted using a multiple linear regression analysis, with fatigue as the output, and features from different sensors as the input.

Results demonstrated that there is a strong relationship between body fatigue and body-related information. Fatigue has then been classified as a binary classification problem using a personalised threshold per participant, per task, to label the data. Three feature sets, including non-wearable, wearable, and full (wearable and non-wearable) sensor feature sets, have been created, to evaluate the optimal features for binary body fatigue classification on both a per-task and overall basis. Redundant features have then been removed by three feature selection algorithms, including ReliefF-10, ReliefF-30, and PCA. In addition, three machine learning approaches have been applied to each optimised feature set, including LDA, kNN, and P-SVM. The performance of each combination of feature selection and machine learning approaches has then been evaluated to select the optimal one for each feature set. Furthermore, the results of the sensor optimisation process have also been displayed. Sensor optimisation has been conducted to prepare the system for a more realistic factory application, aiming at minimising obtrusiveness, computational cost, and privacy issues, as well as maintaining performance.

Classification has been undertaken on both a per-task and overall basis. As to the non-wearable sensor system, results illustrated that the P-SVM and kNN models had similar performance, whilst the LDA models had the worst performance. The reason is that the numbers of features in feature sets from the non-wearable sensor system were insufficient. Therefore, it is difficult to identify clusters of different fatigue levels for the kNN model and to identify a clear boundary between each class for the P-SVM model. Additionally, although the feature sets from the non-wearable sensor system were relatively small, there is still not a clear linear relationship between features and fatigue. Hence, the LDA model cannot accurately detect fatigue. For both wearable and full sensor systems, the P-SVM model outperformed the kNN and LDA models significantly. This could be because P-SVM is a more complex machine learning approach than kNN and LDA, which is able to solve complex classification problems within a relatively large feature space. Moreover, as the LDA models tended to deduce a linear relationship between inputs and output, they were not able to solve complex problems, such as body fatigue.

In addition, the performance of three types of sensor systems have been evaluated. This is essential, because the results can help in understanding the contribution of each sensor system towards body fatigue assessment and in developing a more detailed and optimised model without redundant information. Results illustrated that the wearable sensor system performed significantly better than the non-wearable sensor system on classifying fatigue, with an improvement in accuracy of approximately 30%. Furthermore, the full sensor system only resulted in a slight improvement over

the wearable sensor system, with an increase in accuracy of less than 3%. As a result, using only wearable sensors has proven to be satisfactory in undertaking fatigue classification. Results also illustrated that the machine learning model performances were consistent between task-specific and the overall analysis, implying that the analysis can be performed without identifying the task beforehand.

Lastly, optimisation of the sensor system has been performed utilising data from a combination of all tasks. The heart rate data has first been removed from the system due to privacy issues of recording physiological data in an aviation factory environment. Motion data from the Upper Spine and Right Arm have then been eliminated due to the strong correlations between each IMU. Furthermore, computationally intensive features, such as frequency-domain features and autocorrelation, have finally been removed from the system to minimise the computational cost and prepare the system for real-time fatigue detection. Results illustrated that classification accuracy was still 86.17%, as most discarded information was redundant. In the optimised sensor system, four IMUs have been retained on different body areas, including Left Arm, Right Wrist, Hip, and Right Shank. The optimised system can be worn for an extended period of time. In the meantime, it will not distract workers or pose additional risks to the human body during work.

Table 5.3 illustrates a comparison between this research and previous studies to demonstrate the major advantages of the developed scheme, compared to the existing ML-based methods. Previous research has been undertaken that conducts ergonomic assessment using joint angles derived from IMUs [26, 39-41, 104]. These joint angles can be used to identify improper postures directly through an array of thresholds. Additionally, they can also act as the input to ergonomic assessment tools, such as the RULA and REBA, to calculate ergonomic risk scores. However, during dynamic physical tasks, posture is not the only determinant for identifying ergonomic risks. Instead, body movement is another dominant causal factor of WMSDs. Therefore, this study is targeted at understanding fatigue through the nature of body-related information, including body motion, external force, and heart rate. Heart rate has been widely measured in previous studies using ECG sensors, which requires attaching multiple electrodes to the chest [28, 29]. Although it can provide a more accurate heart rate signal, the high obtrusiveness of the electrodes makes it uncomfortable and difficult to wear for a long time. Instead, this study used a PPG sensor on the ear lobe to record a raw PPG signal and derive heart rate. In comparison, the PPG ear clip is more comfortable and can be worn for a longer time without distracting workers. Additionally, although the PPG signal has been measured from the ear lobe in this study, similar data can easily be recorded from the wrist using smartwatches, as PPG sensors are

one of the most common functionalities embedded in modern smartwatches. Furthermore, non-wearable sensors, such as the force plate and load cell, have also been widely implemented in ergonomic analysis. However, most studies utilised non-wearable sensors to conduct ergonomic analysis in conjunction with camera-based motion capture systems in a laboratory environment [30, 32, 55, 95]. Limited work has been done in performing ergonomic analysis with a combination of both wearable and non-wearable sensors. Therefore, this study analysed fatigue by incorporating both wearable and non-wearable sensors in one system. Furthermore, their contributions towards fatigue assessment have also been evaluated.

Table 5.3 Comparison between Previous Studies

Article	Sensor Measurements	Output	Classification Accuracy
This Study	IMUs – Body Motion PPG – Heart Rate Force Plate – Ground Reaction Force Load Cell – Hand Exertion Force	Body Fatigue (Borg Scale/Personalised Threshold)	94.15%
[22] (2017)	PPG – Heart Rate EEG – Brainwave Signal Skin Temperature	Body Fatigue (Borg Scale/Fixed Threshold)	82.60%
[26] (2017)	IMUs – Joint Angles	Awkward Postures	95%
[28] (2013)	IMUs – Torso Acceleration ECG – Heart Rate, Breathing Rate Location	Awkward Postures	90%
[29] (2017)	IMUs – Body Motion ECG – Heart Rate	Body Fatigue (Borg Scale/Fixed Threshold)	79% (Specificity)
[30] (2009)	Motion Capture System – Body Posture Force Plate – Ground Reaction Force	Awkward Postures	-
[32] (2018)	Motion Capture System – Body Posture Smart Insole – Ground Reaction Force	Joint Energy Consumption	-
[39] (2017)	IMUs – Joint Angles	Maximum Holding Time	-
[40] (2017)	IMUs – Joint Angles	Awkward Postures	-
[41] (2013)	IMUs & Goniometers – Joint Angles	RULA Score	-
[55] (2016)	Load Cell – Hand Exertion Force VICON – Body Motion	Manual Force Strength	-
[44] (2019)	PPG – Heart Rate EDA – Skin Conductance Skin Temperature	Physical Demand (Energy Expenditure Program)	90%
[95] (2012)	Motion Capture System – Body Posture Force Plate – Ground Reaction Force	Lumbar External Loading	-
[104] (2017)	IMUs – Joint Angles	Awkward Postures	-
[45] (2020)	IMUs – Body Motion ECG – Heart Rate	Body Fatigue (Borg Scale/Fixed Threshold)	87.90%

The highest accuracies achieved during task-specific analysis were 93.23%, 95.43%, 96.97%, and 97.17% for the Two-handed Box Lifting, Two-handed Box Carrying, Two-handed Box Pushing/Pulling, and Two-handed Trolley Pushing/Pulling tasks, respectively, whilst the highest accuracy achieved with the overall analysis was 94.15%. The classification accuracies for the machine learning approach in this study have illustrated significant improvement over previous work.

Studies in [22, 44] reported classification accuracies of 82.60% and 90%, respectively, and both of them utilised physiological data alone to conduct fatigue assessment. In this study, multiple types of data have been included in the system, which provided more aspects of information related to body fatigue. Furthermore, Maman et al. [45] conducted a physical demand assessment study, with a classification accuracy of 87.9%. Four IMUs and one heart rate monitor were included in the study. In contrast, more sensors have been included in our system, with six IMUs, a heart rate monitor, a force plate, and a load cell, which can monitor body-related information in a more detailed and comprehensive manner, which can therefore increase the model performance. In addition, as the classification accuracy from the overall model has been similar to those from task-specific models, results illustrated that the body fatigue can still be detected using data from a combination of all tasks.

This study conducted fatigue detection by processing direct sensor measurements to in-depth body related information. By monitoring fatigue, workers will be able to proactively adjust their working habits, in order to minimise risks of suffering WMSDs. Moving forward, a real-time fatigue assessment system will be constructed, with real-time data collection, analysis, logging, and visualisation. Both workers and factory operation's awareness of workers' body conditions can be considerably improved by visualising workers' body-related information. In the short-term, once workers realise that they have maintained a high fatigue for a long duration, they can take corresponding measures to lower their fatigue level, such as taking a break, adjusting improper postures, and drinking some water. On the other hand, factory operations can also provide workers safety instructions directly if realizing that they have been working with high fatigue for an extended period of time. From a long-term aspect, continuous monitoring of fatigue for workers in a large aviation factory can help factory operations to optimise task allocation, calibrate task sequence, and redesign tasks if needed. For instance, if workers in a certain workplace have reported high fatigue during most of the time per shift, their tasks may need to be redesigned or the task sequence needs to be adjusted to prevent WMSDs. Finally, workers can develop more healthy and safe working habits by understanding their body conditions, whilst the factory efficiency and productivity can also be improved with more scientific task design, job allocation, and less occupational health and safety cost.

5.7 Chapter Summary

WMSDs are a significant issue for the aviation manufacturing industry, which negatively affects both workers' health and factory efficiency and productivity. This study performed both body fatigue classification using machine learning approaches, utilising both wearable and non-wearable sensors

to measure body motion, external force exerted on the human body, and physiological data, such as heart rate. In addition, sensor deployment optimisation has also been conducted in this study to minimise the number of sensors, whilst maintaining classification performance. An experiment has been undertaken by thirty participants, who undertook four tasks to simulate repetitive physical tasks experienced during daily aviation factory work. Results of multiple linear regression analysis between fatigue and body-related information illustrated that the maximum Adjusted R^2 Score of 0.9259 and lowest RMSE of 0.4972 were achieved. Furthermore, the results of binary classification illustrated accuracies up to 97.17% and 94.15% for task-specific and overall analysis, respectively, utilising a combination of wearable and non-wearable sensors, which demonstrated a significant improvement over the current state-of-the-art. Lastly, a systematic sensor deployment optimisation approach has been proposed. Results illustrated that the accuracy maintained a suitable level of 86.17% after eliminating redundant information from the system.

Chapter 6

Study Two: Development of a Real-time Fatigue Assessment System

6.1 Introduction

As discussed in Chapter 5, the first study focused on evaluating different sensor systems and machine learning approaches for classifying fatigue during repetitive physical tasks. Additionally, the sensor deployment has also been optimised to remove redundant sensors. The second study in this research, which builds on the first, has been undertaken to conduct real-time fatigue classification during aviation manufacturing tasks in a workshop, in order to simulate an aviation factory environment. The study has been performed for three objectives:

1. Construct a real-time system capable of detecting fatigue in a factory-like environment with aviation manufacturing tasks.
2. Evaluate the performance of different feature selection and machine learning approaches on multi-level body fatigue classification.
3. Validate the hardware performance of the real-time system.

6.2 Methodology

Based on the results from the first study, the ErgoRelief framework has been further implemented and tested in this study. However, due to COVID-19, it was not possible to travel interstate to conduct the experiment in a real aviation factory. Therefore, the experiment environment has been recreated in a local workshop with operating machines and moving objects to simulate the aviation factory (see Figure 6.1). Experiment tasks include operating an orbital buffer machine and pushing/pulling a heavy trolley load, which have been designed based on real aviation manufacturing tasks, involving hand tool utilisation and primary part assembly [5]. Furthermore, based on the sensor optimisation results from Chapter 5, only body motion data have been recorded in this study from four Shimmer3^{TM} IMUs, including body part acceleration and angular velocity. Participant details, system overview, experiment procedure, and data analysis approach have been introduced in following sections.

Figure 6.1 Simulated Experiment Environment

6.2.1 Participants

Thirty participants have undertaken the experiment, with an age range from 20 to 45 years (mean = 27.6, standard deviation = 6.29). Participants did not have any history of cardiovascular illness and physical injury before taking part in the experiment. Prior to taking part in the experiment, they were required to undertake and pass a workplace area induction quiz, including emergency evacuation procedures, public amenities, and first aid stations. The experiment has been approved by the University of Queensland Ethics Sub-Committee.

6.2.2 System Implementation

Figure 6.2 illustrates an overview of the real-time wearable sensor-based fatigue assessment system. During the data collection stage, a local PC continuously retrieves data from each IMU through Bluetooth. Data have then been pre-processed in the local PC, including data synchronisation, segmentation, resampling, filtering, and feature extraction. After that, body fatigue was derived through machine learning models, which has then been logged to a time-series database using Wi-Fi along with body part acceleration and angular velocity. InfluxDB was selected as the time-series database in this study [144].

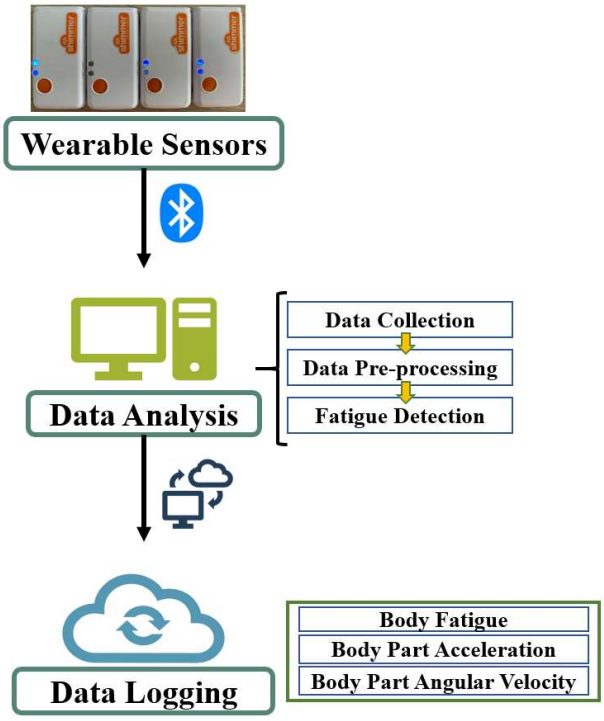


Figure 6.2 System Plan

Figure 6.3 demonstrates the sensor placement in this study. Four IMUs have been strapped to the Left Arm, Right Wrist, Hip, and Right Shank. Each IMU measures three-axis acceleration and angular velocity. Data have been both streamed to a PC through Bluetooth and stored in the on-board SD card. The SD card data have been exported after each participant completed the experiment.

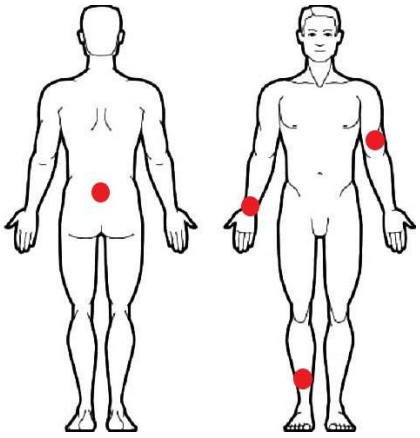


Figure 6.3 Sensor Placement

Figure 6.4 illustrates the data processing procedure. In this study, the system is composed of two aspects: a real-time fatigue assessment system and machine learning model training system. The real-

time fatigue assessment system includes real-time data collection and fatigue classification. In addition, the machine learning model training system trained, updated, and evaluated the model performances on classifying fatigue after each participant finishing the experiment, utilising the data recorded on the on-board SD card. Data have been pre-processed first following the procedure detailed in Section 4.3, including data synchronisation, segmentation, resampling, and filtering. Features have been extracted from the data following the procedure discussed in Section 4.4. For the real-time fatigue assessment system, the feature set has been input into the model directly to classify Rest/Low/High body fatigue. For the machine learning model training system, after participants finishing the experiment, each sample in the feature set has been labelled by the subjective rating of fatigue, and all participants' feature sets have been combined into one datasheet. The numbers of samples in each class have then been balanced through RUS and multiple feature selection algorithms have been applied on the datasheet for feature optimisation. Finally, three machine learning approaches have been implemented on optimised feature sets. Their performances have been evaluated and the optimal one has been selected to classify fatigue for the following participant.

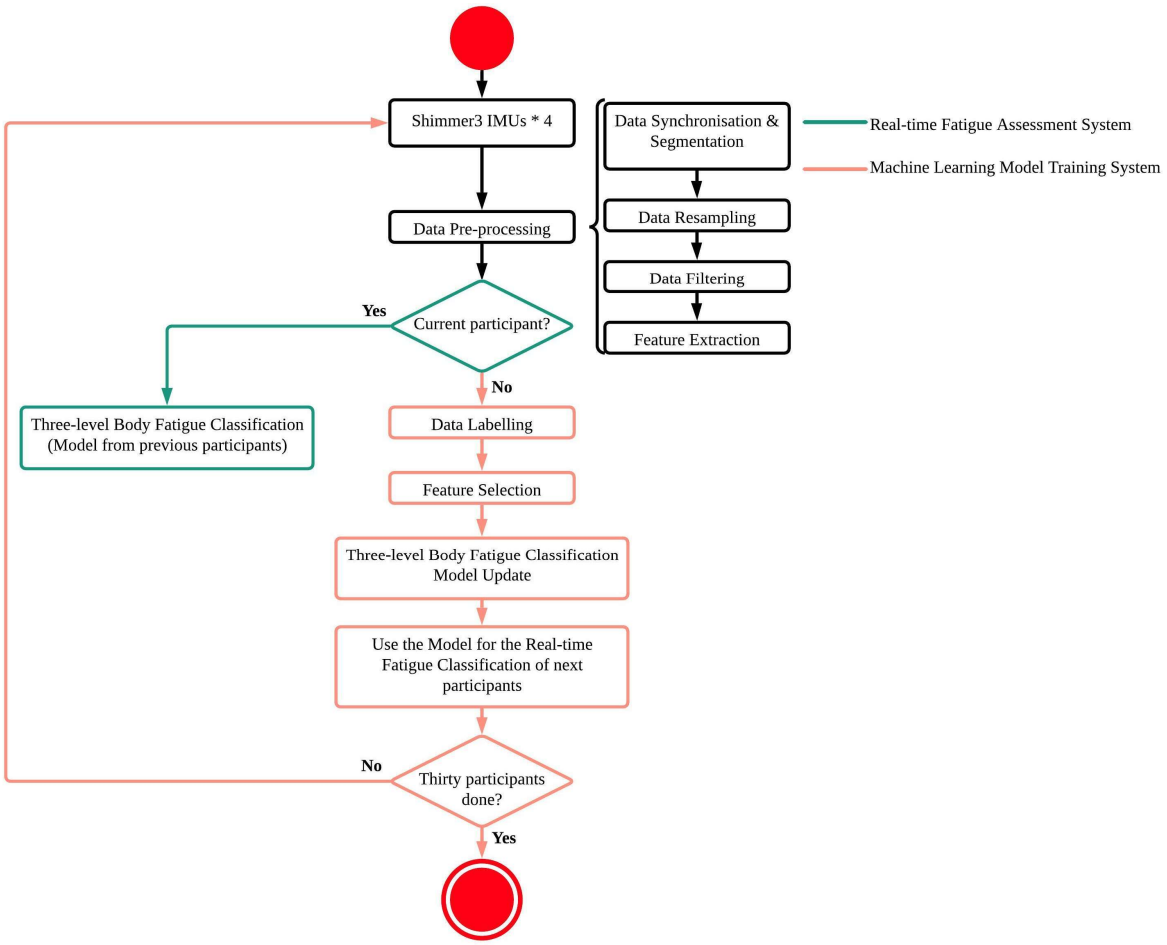


Figure 6.4 Data Processing Procedure

6.2.3 Experiment Procedure

In this study, we aimed to simulate real complex aviation-like tasks, including operating hand tools, and assembling different structures. As such, orbital buffer operation has been selected as a task because it is commonly used in real factory environments and it is a safe and easy to use piece of equipment that does not require complicated training. Two physical tasks have been designed to simulate daily aviation manufacturing work, including operating an orbital buffer machine and pushing/pulling a heavy trolley load. Details of each task have been elaborated below:

1. **Operating an Orbital Buffer Machine (see Figure 6.5):** Participants power the orbital buffer on, operate it with two hands to polish the timber surface up/down and left/right continuously for 5 minutes, and power the orbital buffer off.

Figure 6.5 Operating an Orbital Buffer Machine

2. **Pushing/Pulling a Heavy Trolley Load (see Figure 6.6a – 6.6d):** Participants load 16 boxes from the floor to the trolley one at a time with two hands, with a total weight varying between 40kg and 80kg, pull and push the trolley around a circular path from and to the origin for approximately 20m, and unload boxes from the trolley to the floor one at a time with two hands.

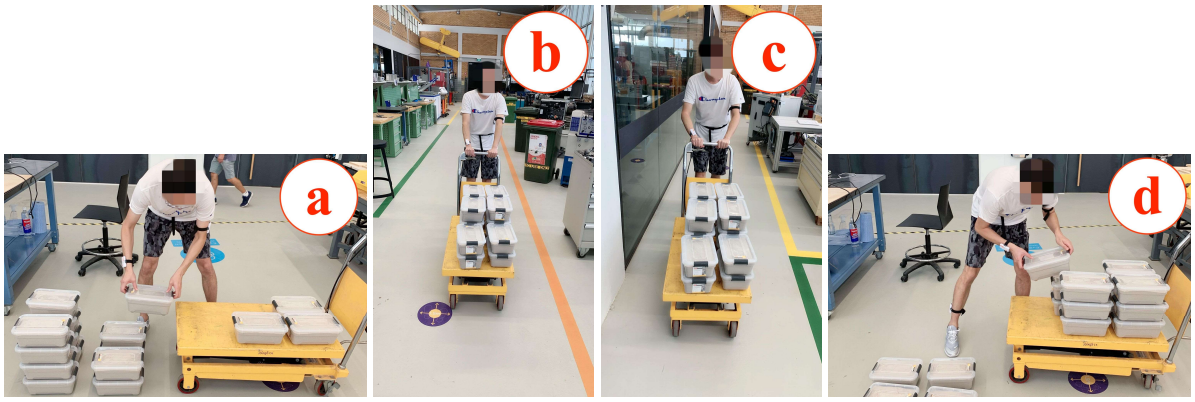


Figure 6.6 Pushing/Pulling a Heavy Trolley Load: a) Load 16 boxes one at a time from the floor to the trolley with two hands. b) Pull the trolley c) Push the trolley d) Unload 16 boxes one at a time from the trolley to the floor with two hands.

Figure 6.7 illustrates the experiment procedure followed by each participant. There are four repetitions in the experiment, with each repetition lasting for approximately 20 minutes, followed by a 10-minute resting period. Each repetition composed two 10-minute loops. During each loop, participants have been required to conduct each of two tasks consecutively. At the beginning and end of each repetition, participants have been asked to complete the Borg Scale Questionnaire to indicate their subjective rating of fatigue. Additionally, similarly to the first study, they have also been required to perform a synchronisation pattern. During each repetition, apart from the task requirements, participants have been allowed to conduct the experiment in their most comfortable speed and posture. They have also been allowed to communicate with the researcher for any unexpected issues.

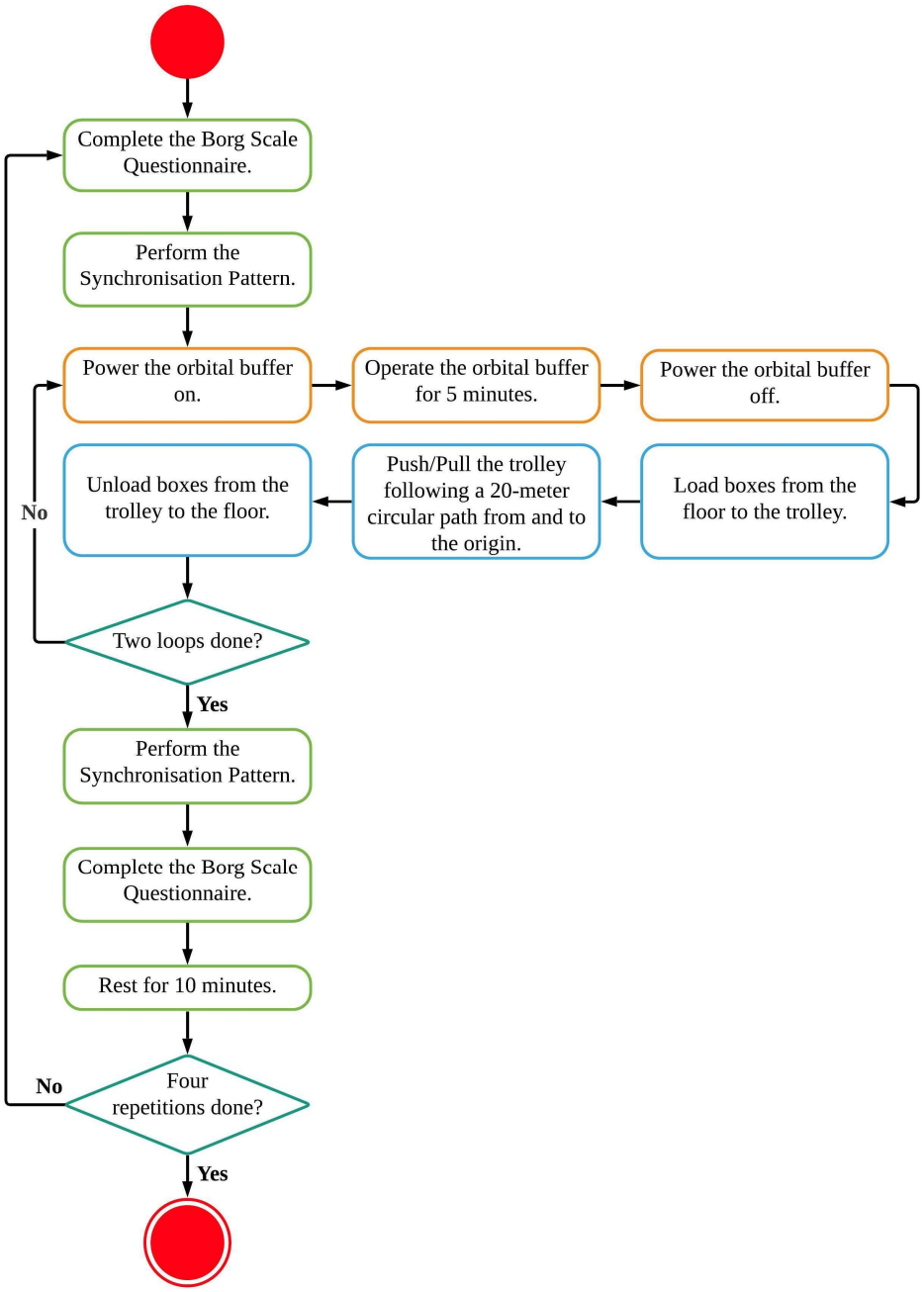


Figure 6.7 Experiment Procedure

Table 6.1 illustrates the weights for the orbital buffer and boxes, which have been changed for each repetition to incur different feelings towards fatigue. In this experiment, the orbital buffers weighed 3kg and 7kg for the first and last two repetitions, whilst the box weight has been interwoven, with 2.5kg at the first and third repetitions, and 5kg at the second and fourth repetitions. These weights are in line with industry equivalents. The changes in load weights between repetitions aimed at incurring different feelings towards body fatigue for each participant. In addition, the total weight for 16 boxes

is 40kg or 80kg for each of four repetitions, which is consistent with the load weight during the Two-handed Trolley Pushing/Pulling task from the Study One.

Table 6.1 Orbital Buffer and Box Weights per Repetition

Repetitions	Orbital Buffer Weight	Box Weight
1	3kg	2.5kg
2	3kg	5kg
3	7kg	2.5kg
4	7kg	5kg

The experiment lasted for approximately 2 hours for each participant. Each participant has been paid by a $30 gift card to compensate his/her time for attending the experiment.

6.2.4 Data Processing

As illustrated in Section 4.3, the first step of data processing has been data pre-processing, including data synchronisation, segmentation, resampling, and filtering. For the data synchronisation and segmentation, the synchronisation pattern at the beginning and end of each repetition has been utilised to segment data for the machine learning model training based on the acceleration peaks. In contrast, for the real-time fatigue assessment system, the PC continuously received data packets via a Bluetooth connection, from each Shimmer3 IMU, which contained a timestamp, acceleration, and angular velocity values. Since each IMU has a different starting time offset, each sample has then been synchronised globally. After that, the data for all IMUs have been segmented using a 30-second window with a 15-second step, which have then been used for further analysis. Figure 6.8 displays the code snippet for receiving data through Bluetooth streaming and synchronising data. A MATLAB library has been provided by the Shimmer [102] for Bluetooth data streaming. Therefore, from line 1 to 17, Shimmer3 sensor has been initialised and configured properly, including enabling Bluetooth data streaming, setting up sampling rate, and enabling UNIX timestamp. After that, from line 20 to 28, a while loop has been created to keep obtaining sensor measurements, with a UNIX timestamp for each sample in the packet being generated using the function from line 30 to 49. In line 34, the time difference between adjacent samples has been calculated based on the internal timestamps. In addition, when receiving a packet, its last sample has been assigned with the current UNIX timestamp, which has been extracted in line 38. Hence, a for loop has been constructed to assign the UNIX timestamp to each sample using the time differences calculated from internal timestamps. Finally, the UNIX timestamps for all samples have been added to the packet. The code snippet illustrated in

Figure 6.8 has been utilised for all Shimmer3 sensors in the system. After assigning each sample with a global UNIX timestamp, previous 30-second data from all sensors can be extracted using UNIX timestamps, with a 15-second step.

```
% Shimmer Sensor Initialization
shimmer_sensor = ShimmerHandleClass(sensor_port);
SensorMacros = SetEnabledSensorsMacrosClass;

while (~shimmer_sensor.connect)
    warning("Shimmer sensor connection failed");
end

if ~shimmer_sensor.setinternalboard('GSR')
    warning("Shimmer sensor set internalboard failed");
end
shimmer_sensor.setsamplingrate(fs);
shimmer_sensor.enabletimestampunix(1);

while (~shimmer_sensor.startdatalogandstream)
    warning("Shimmer sensor start failed");
end

% A while loop for keeping obtaining data from the sensor through Bluetooth
data = [];
while (True)
    [calibratedData,signalName,~,~]=shimmer_sensor.getdata('c');
    if size(calibratedData, 1) > 0
        [calibratedData] = assign_unix_time(calibratedData);
        data = [data; calibratedData];
    end
    pause(0.2) % pause for 0.2 second
end

function [calib_data] = assign_unix_time(calib_data)

    % Calculate the time difference between each sample in the Bluetooth
    % packet using sensor's internal timestamp.
    time_diff = diff(calib_data(:, 1));
    % Extract the unix timestamp from the last packet. When receiving the
    % last packet, its last sample will be assigned by the current unix
    % timestamp.
    timestamp = calib_data(end, 2);

    % Loop through each sample and assign them with unix timestamps based
    % on the time differences of the internal timestamps.
    for i = size(time_diff, 1):-1:1
        prev_timestamp = timestamp(end, 1) - time_diff(i);
        timestamp = [timestamp; prev_timestamp];
    end

    % Add unix timestamps for all samples to the packet.
    calib_data(:, 2) = flip(timestamp);
end
```

Figure 6.8 Bluetooth Data Synchronisation and Segmentation Code

After data synchronisation and segmentation, both acceleration and angular velocity data from each IMU are resampled and filtered. As illustrated in the sensor optimisation section in Chapter 5, time-domain features have then been extracted from the body part acceleration, acceleration jerk, angular velocity, and angular acceleration, including 10^{th}, 25^{th}, 50^{th}, 75^{th}, 90^{th} percentiles, mean, standard

deviation, inter-quartile range, kurtosis, and mean absolute deviation. In total, 640 features have been extracted from the system.

After feature extraction, for the real-time fatigue assessment system, features have been input into the machine learning model directly to conduct body fatigue classification. For the machine learning model training system, samples in the feature set have been labelled as *Rest*, *Low*, and *High* fatigue following the procedure introduced in Section 4.5. Samples belonging to the resting period have been labelled as *Rest*, whilst each repetition's data have been labelled as either *High* or *Low* fatigue depending on the change in subjective fatigue level.

Each participant's data was combined into one datasheet after feature extraction. The next step has been to balance the numbers of samples from each class and perform feature selection. Figure 6.9 illustrates the number of samples after data labelling for thirty participants. The *High* fatigue class has the lowest number of samples, with a value of 3996. Therefore, RUS has been implemented on the *Rest* and *Low* fatigue classes, in order to balance their numbers of samples with the *High* fatigue class.

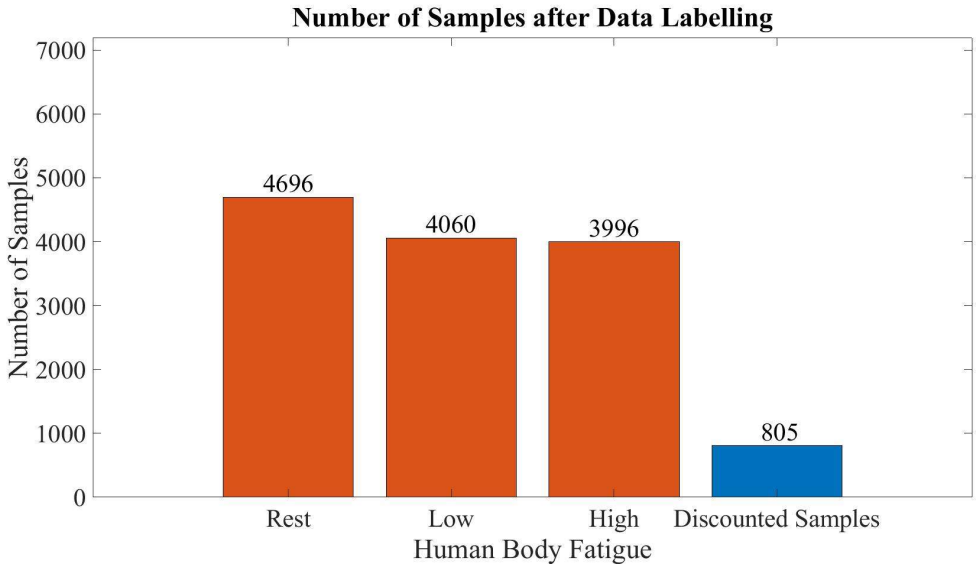


Figure 6.9 Number of Samples after Data Labelling

Compared to the first study, PCA has been removed from the feature selection algorithm in this study, as it cannot be implemented for real-time data analysis. Therefore, ReliefF-10 and ReliefF-30 have been implemented as the feature selection algorithms in this study. The feature selection has been performed after each participant completes the experiment, with approximately 20 features being maintained in the optimal feature set to minimize computational cost and achieve satisfactory

machine learning model performance. Therefore, the feature set has been updated based on all previous participants' data. Furthermore, similarly to the first study, LDA, kNN, and P-SVM have been selected as machine learning approaches to conduct three-level body fatigue classification. Hence, six models have been constructed after each participant finishes the experiment. Their performances have been evaluated and the optimal one has been used to classify fatigue for the following participant.

6.3 Results

6.3.1 Hardware Performance Evaluation

The hardware performance for the system has mainly been evaluated based on two parameters: battery lifetime and packet receiving rate, which correspond to system sustainability and stability respectively. Battery lifetime is one of the most important indicators illustrating whether the system can provide durable data streaming, whilst the packet receiving rate demonstrates the ratio between the number of successfully received samples and the total number of samples, which indicates whether the system can receive most packets successfully.

As declared in the Shimmer3™ User Manual [109], 3.2V corresponds to 0% of the battery voltage. Therefore, the battery percentage has been calculated based on Equation 6.1, where the maximum voltage has been obtained at the beginning of data recording, when Shimmer3™ IMUs have been fully charged.

$$Battery\ Percentage = \frac{Current\ Voltage - 3.2V}{Maximum\ Voltage - 3.2V} * 100\% \tag{6.1}$$

Figure 6.10 illustrates the battery lifetime vs time, with dotted lines showing the battery percentage for each IMU, and a solid line indicating the average battery percentage for all IMUs. The data have been continuously recorded for 5.5 hours. Considering that workers will not continuously work for more than four or five hours during a day without a lunch break, 5.5 hours data recording can ensure that the data recording is able to cover the whole shift. For all IMUs, the battery percentages decreased from 100% to between 60% and 70%, with the highest and lowest values of 68.76% and 62.05% respectively. Additionally, the mean battery percentage decreased from 100% to 65.03% after 5.5 hours data recording. Therefore, results demonstrated that the system could record data continuously for an extended period of time.

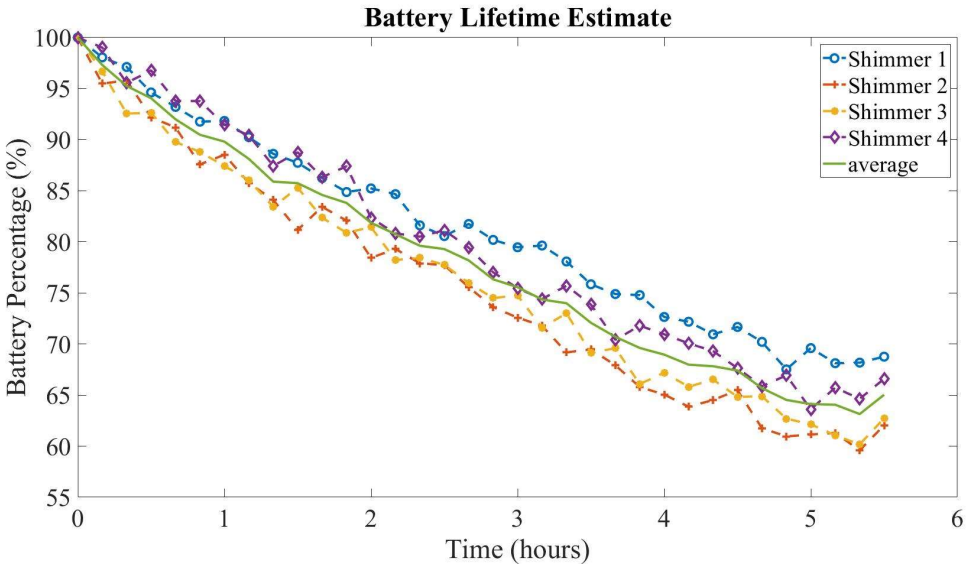


Figure 6.10 Battery Percentage vs Time

During the experiment, the distance between each IMU and local PC was less than 5 meters. In this condition, the mean packet receiving rate over all participants is 99.73%. As declared by Mansfield and Antonakos [145], the packet receiving rate should be maintained above 90% in order to ensure the data streaming quality. Therefore, the results indicated that the system is capable of collecting a sufficient number of samples to conduct body fatigue classification.

6.3.2 Machine Learning Model Training

By observing the highest accuracy achieved for each participant, the ReliefF-30 outperforms ReliefF-10 on feature selection. Therefore, the classification results based on the ReliefF-30 feature sets are discussed in this section, whilst the results based on the ReliefF-10 feature sets are displayed in Appendix 10. Furthermore, as illustrated in Chapter 5, since LDA always has the worst performance over all machine learning approaches, it has been removed from this study. The classification results during the machine learning model training act as the baseline to compare their performance with real-time fatigue assessment.

Figure 6.11 to 6.14 demonstrate the machine learning model performance per participant. Figure 6.11 illustrates the model accuracies per participant. The P-SVM models perform better than the kNN models for the first 14 participants, with values of around 80% and 75%. However, the results reverted for the last 16 participants. The kNN models perform better than the P-SVM models, with values of approximately 85% and 75% respectively.

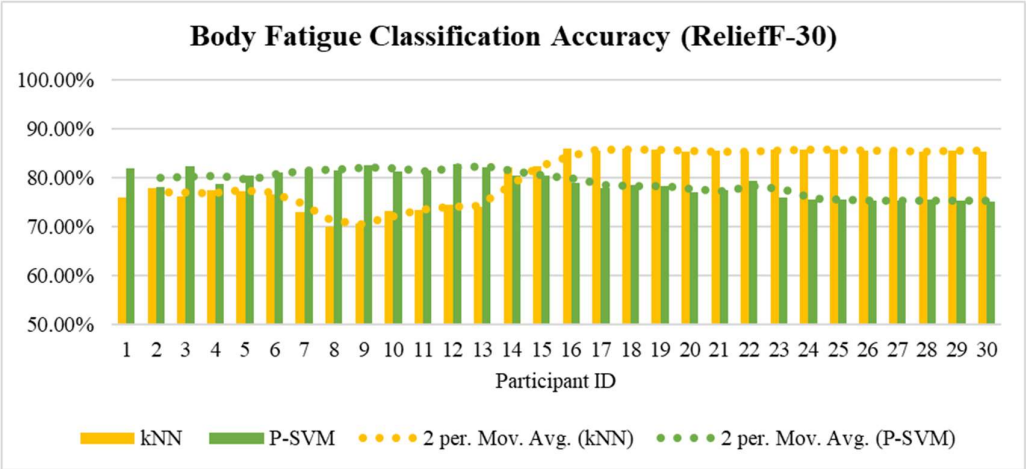


Figure 6.11 Body Fatigue Classification Accuracy

Figure 6.12 displays the model F1s for each participant. The results are similar to those from the model accuracies. The P-SVM models outperform the kNN models with relatively small data size, while the kNN models achieve better performance than the P-SVM models when data size increases.

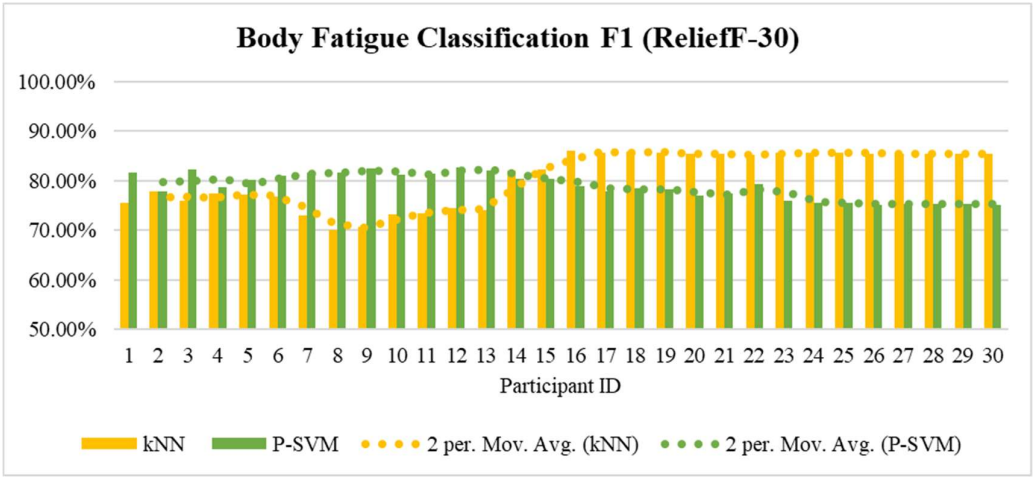


Figure 6.12 Body Fatigue Classification F1

The trend of model AUCs is similar to those from the accuracies and F1 (see Figure 6.13). For the first 14 participants, the kNN models have relatively lower AUCs, with values of approximately 85%, whilst the P-SVM models have better performances, with values of approximately 93%. For the last 16 participants, the kNN models perform the best, followed by the P-SVM models, with values of around 95% and 90%, respectively.

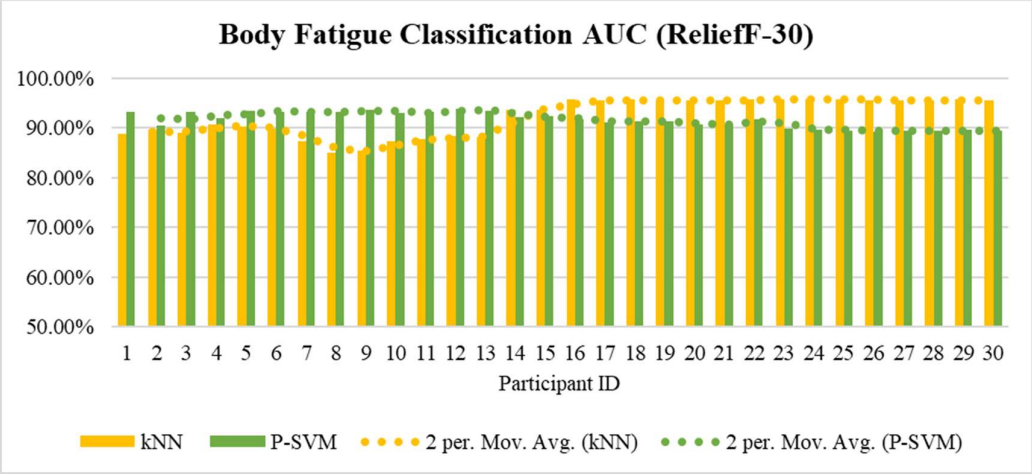


Figure 6.13 Body Fatigue Classification AUC

Figure 6.14 demonstrates the model error rates per participant. The P-SVM models have lower error rates than the kNN models for the first 14 participants, with values of around 20% and 25% respectively, while the kNN models surpass the P-SVM models for the remaining participants, with values of approximately 15% and 25%.

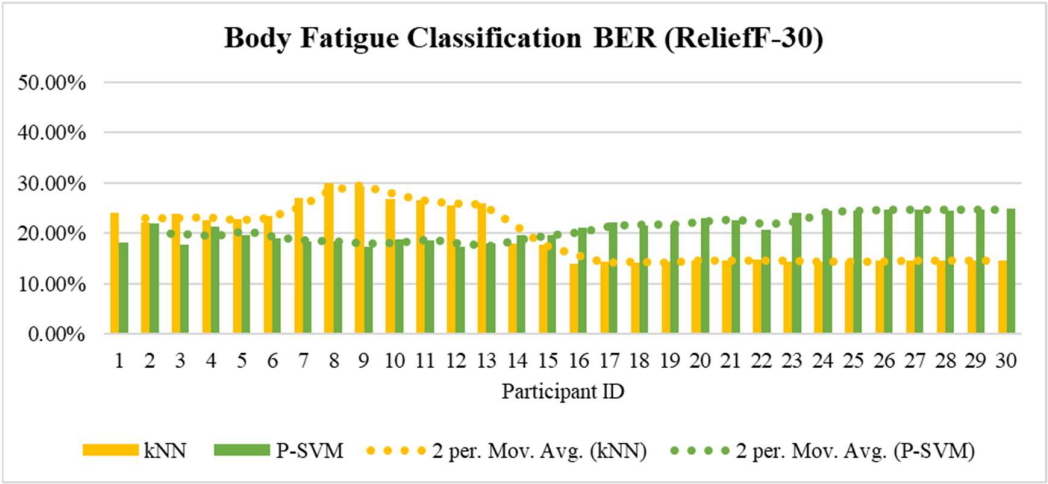


Figure 6.14 Body Fatigue Classification BER

In summary, the kNN models are more capable of handling relatively large data set than the P-SVM models, while the P-SVM models can achieve better performance with small datasets. Lastly, by systematically investigating the model performances, the optimal offline model for each participant can achieve an accuracy and F1 of approximately 85%, AUC of approximately 95%, and error rate down to around 15%.

6.3.3 Real-time Body Fatigue Assessment

Figure 6.15 evaluates the model performances of real-time body fatigue classification through accuracy, F1, and error rate. As illustrated in the machine learning model training results, the optimal model for each participant kept changing based on the results from previous participants. For instance, for the first 14 participants, since the P-SVM models outperformed the kNN models, new participants therefore used the P-SVM model to conduct real-time body fatigue assessment. In contrast, for the following participants, the performance of the kNN model surpassed the P-SVM model. Hence, the kNN model has then been selected for the following participants. The model for the first participant has the poorest performance, with an accuracy, F1, and error rate of 36.59%, 37.31%, and 63.21% respectively. The reason is that the data is not sufficient to train the model for the first participant. After the first participant, the model performance starts to fluctuate among participants, with highest accuracy, F1, and lowest error rate of 81.41%, 76.90%, and 23.11% respectively from participant number 25. Overall, the mean accuracy, F1, and error rate achieved by all participants are 65.93%, 61.80%, and 36.20%. As the model tended to classify fatigue for the current participant through previous participants' data, the model performance fluctuates among participants. In addition, the results for the real-time body fatigue classification are not as good as the machine learning model training results, since the machine learning model training and evaluation have been conducted by incorporating current participant's data.

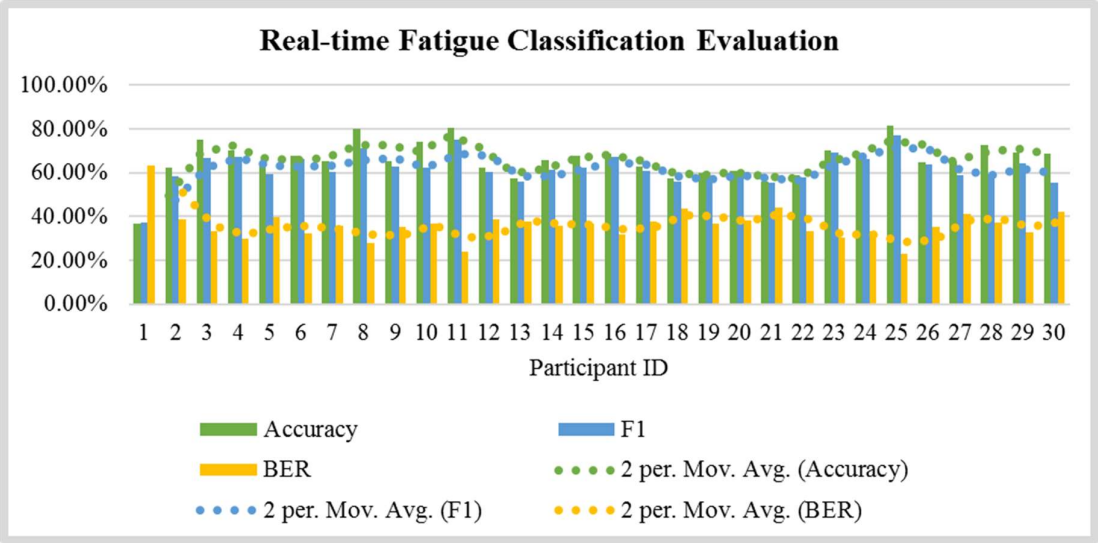


Figure 6.15 Real-time Fatigue Classification Evaluation

Figure 6.16 demonstrates the AUC values for each fatigue level during real-time body fatigue classification for each participant, and the AUC values have been calculated on a "one-versus-all" basis. The AUC values for the *High* class started from 37.48% for the first participant and increased

gradually to between 70% and 90% for the following participants, with a mean value of 72.36% for all participants. A similar trend applied for the AUC values of the *Low* class, with minimum and maximum values of 45.35% and 92.05% respectively. Lastly, the AUC values of the *Rest* class are always greater than those from the other two fatigue levels, with a mean value of 98.87%, which makes sense as there is a significant difference in body motion between the resting and task periods.

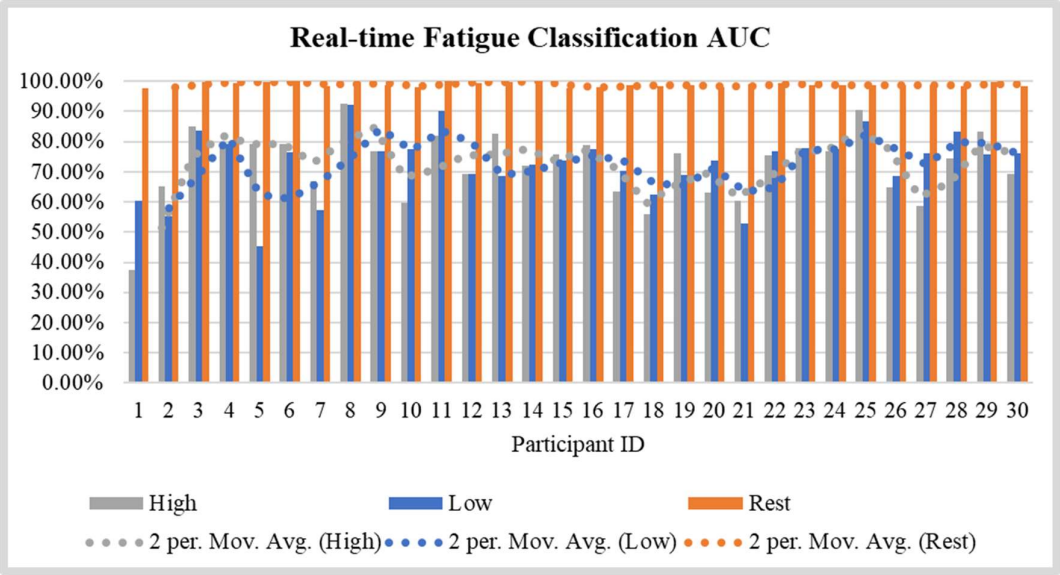


Figure 6.16 Real-time Fatigue Classification AUC

Figure 6.17 shows an example of real-time fatigue assessment for a participant, with blue and orange dots representing the predicted and reference fatigue levels, respectively. There are three body fatigue levels in this figure, including *Rest*, *Low*, and *High*. For this participant, the first three repetitions have been labelled as *Low* fatigue, whilst the last repetition has been labelled as *High* fatigue. In addition, the resting periods have been labelled as *Rest* after each repetition. As it can be seen, the real-time fatigue assessment model can detect most samples during the resting periods, which is consistent with the results displayed in Figure 6.16. Furthermore, during each repetition, fatigue levels for more than half of samples can be detected correctly. However, some samples have been misclassified as *High* or *Low*, as the real-time fatigue assessment was conducted using the previous participants' data. Overall, the results of Figure 6.17 were consistent with the results in Figure 6.16, with similar AUC values of the *High* and *Low* fatigue levels, and highest AUC value of the Rest fatigue level.

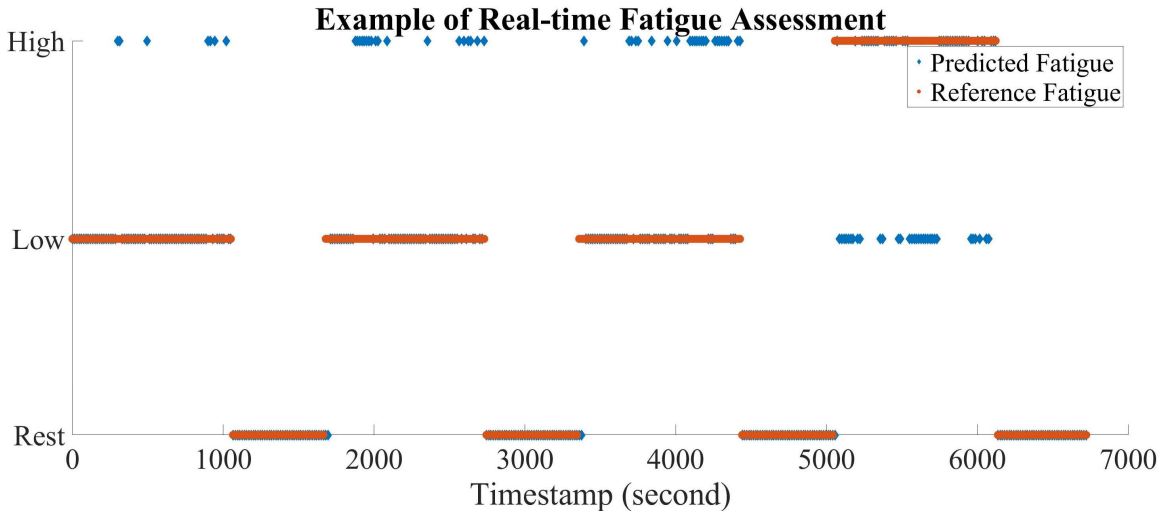


Figure 6.17 Example of Real-time Fatigue Assessment

6.4 Discussion

In this study, a real-time wearable sensor-based fatigue assessment system has been constructed, including real-time data collection, analysis, and logging. To verify the system's stability and durability and evaluate the performance of a three-level body fatigue classification, an experiment has been conducted to simulate aviation manufacturing tasks in a factory-like environment, including operating an orbital buffer machine and pushing/pulling a heavy trolley load. The system has been designed and tested in order to be implemented in an aviation manufacturing environment. The experiment environment, task design, scenarios and load weights have been selected based on input from our industry partner. Body motion data have been recorded from four IMUs on different body areas, which act as the input for classifying fatigue. The recorded data was pre-processed, and time-domain features have been extracted. Similarly, to the first study, each repetition's data have been labelled based on the subjective rating of fatigue level. Additionally, data from the resting periods have also been labelled as *Rest*. Each participant's data was combined, and two feature selection algorithms have been applied on the dataset for feature optimization, including ReliefF-10 and ReliefF-30. Two machine learning approaches, including kNN and P-SVM have then been implemented utilising each feature set to conduct three-level body fatigue classification. Their performances have been systematically evaluated, and the model with optimal performance has been selected to classify fatigue in real time for the following participant. Furthermore, the performance of hardware system on wireless data streaming has also been evaluated through battery lifetime and packet receiving rate to demonstrate that the system is stable and durable for deploying in an aviation factory environment.

For the hardware system performance, after 5.5-hours of data recording, the battery percentages decreased from 100% to between 60% and 70% for all IMUs. Furthermore, the mean packet receiving rate of all participants has been greater than 99%, illustrating that most packets can be received by the local PC through Bluetooth data streaming. Therefore, results illustrated that the system could operate continuously and stably, which is suitable for deploying in an aviation factory environment. As to the three-level body fatigue classification, results illustrated that the P-SVM models outperformed the kNN models with a relatively small data size, while the kNN models accomplished a better performance than the P-SVM models when data size increases. The reason is that for the small data size, the P-SVM models have been able to recognize clear boundaries between classes, whilst the kNN models did not have sufficient data to identify clusters for each class. However, as the data size increases, the kNN models started having enough data to classify clusters, which had a slightly better performance than the P-SVM models. After recording data for all participants, results have shown that the kNN model has the optimal performance, followed by the P-SVM models.

This study designed and evaluated a real-time wearable sensor-based fatigue assessment system, targeted at conducting three-level body fatigue classification during aviation manufacturing tasks in a factory environment. Four contributions have been made in this study compared to the current state-of-the-art, which are:

1. Development of a real-time wearable sensor-based fatigue assessment system.
2. Evaluating the hardware stability and sustainability of wireless data streaming, in order to demonstrate that the system applicability to deploying in an aviation factory environment.
3. Exploring the machine learning model performance on analysing fatigue during aviation-based factory tasks.
4. Comparing the performance of different feature selection and machine learning approaches to select the optimal combination for three-level body fatigue classification.

Previous studies have been targeted at performing ergonomic analysis on static or repetitive physical tasks, such as lifting, pushing/pulling, squatting, kneeling, and working overhead [25, 26, 40]. Although these tasks form a basis for real complex factory work, it is still necessary to evaluate the system performance during complex factory-like tasks. Therefore, the main advantages of the proposed system are that it is focused on detecting body fatigue classification by simulating common factory tasks, such as hand tool operation and object manipulation. Moreover, most studies have been completed in a strictly controlled laboratory environment, in order to avoid additional noise and

collect clean data. However, the effect of a factory environment on ergonomic monitoring systems needs to be further explored, as many additional impact factors can be applied to the system, such as surrounding noise, signal interference, and heavy traffic, which can hinder the system performance. However, due to the effect of COVID-19, it has been impractical to conduct the experiment in a real factory environment. Hence, the experiment in this study has been conducted in a workshop, which simulated an aviation factory environment. The performances of the hardware system and machine learning approaches have then been evaluated, making the results more realistic. In addition, although many ergonomic monitoring systems have been designed and tested in real time [28, 39, 41, 42, 48], limited work has been done on evaluating the hardware performance, as stable and durable wireless data streaming is as important as precise ergonomic analysis. Therefore, this study evaluated the quality of Bluetooth data streaming through battery lifetime and packet receiving rate, illustrating the system was capable of conducting real-time body fatigue assessment reliably and continuously. Lastly, rather than using either physiological data alone or a combination of motion and physiological data to conduct ergonomic analysis, results of this study illustrated that using body motion alone can still provide accurate body fatigue classification.

The real-time wearable sensor-based fatigue assessment system designed in this study contributes significantly to body fatigue assessment, which converts body motion to fatigue through machine learning approaches. WMSDs can then be prevented by continuously monitoring the change and accumulation of body fatigue levels. Moving forward, the system can be extended with more types of inputs to conduct ergonomic assessment, such as environmental factors and lifestyle. For instance, surrounding noise, sleeping quality, and medication history can also play an essential role in estimating fatigue. Additionally, by recording body motion, body posture reconstruction can be performed, which provides an alternative way of performing ergonomic analysis. Furthermore, the current system was focused on factory workers. However, after resolving the privacy issue of sharing body-related information, the system can be extended to factory-wide, incorporating both aviation manufacturing workers and factory operations. Therefore, factory operations can manage the factory's daily routine more efficiently. In the meantime, workers are able to receive not only insight about their body conditions, but also safety advice from factory operations. This will help workers developing a healthier and safer working habit in the long term. Finally, the system in this study is focused on analysing fatigue during aviation manufacturing tasks. However, the workflow can be extended to more varieties of tasks in aviation fabrication, assembly, and maintenance, which will effectively prevent WMSDs and increase factory productivity and efficiency.

6.5 Chapter Summary

Both workers' health and the efficiency and safety of aviation manufacturing industries can be negatively affected by WMSDs. To minimize the chance of suffering WMSDs, a real-time wearable sensor-based fatigue assessment system has been constructed. Body motion from different body areas have been collected by the system to classify fatigue. An experiment has been conducted in a workshop to simulate a factory environment, with tasks being designed based on the real aviation manufacturing work, including operating an orbital buffer machine and pushing/pulling a heavy trolley load. Results of hardware performance analysis illustrated that most packets can be received by the system successfully through Bluetooth streaming, and the sensor battery can support durable and continuous data streaming for more than 5 hours. Furthermore, the results of the real-time body fatigue assessment and classification model training demonstrated the highest accuracies of 81.41% and 85.37% respectively, illustrating that the system is able to conduct accurate body fatigue classification using body motion only.

Chapter 7

Conclusions and Future Work

In this chapter, a summary and conclusion of this thesis is presented. The research contributions and future work are discussed.

7.1 Introduction

WMSDs have been a significant issue for both workers and the aviation industry, as they will not only pose a long-term adverse effect to workers' health, but also decrease factory productivity and generate excessive costs for occupational health and safety. Therefore, corresponding measures need to be taken to monitor WMSDs closely, in order to prevent them prior to the injury. Hence, a wearable sensor-based ergonomic monitoring system has been developed in this thesis, utilising body fatigue as the indicator towards WMSDs, as workers with the accumulation of fatigue are highly anticipated to suffer WMSDs.

7.2 Main Findings

The ErgoRelief framework has been proposed as a real-time wearable sensor-based fatigue assessment system. The system comprises three components, including data collection, analysis, and visualisation, as well as two stakeholders, aviation manufacturing workers and factory operations. By implementing this framework, workers can visualise their body-related information, which can alert them to the considerable accumulation of fatigue. In the meantime, by visualising data from either a certain worker or a group of workers, factory operations can manage the manufacturing plan more efficiently. In addition, bi-directional communication can be established between workers and factory operations through smart devices, such as smartphones, which allows workers to receive safety advice and report unexpected issues from and to factory operations. Compared with previous studies, this framework incorporated both wearable and non-wearable sensors to conduct body fatigue assessment, making the results more comprehensive, since both biometric and external factors have been taken into consideration. In addition, rather than providing simple and straightforward feedback to workers, this framework included factory operations as another stakeholder. Factory operations are able to visualise workers' body fatigue more systematically and provide workers professional advice to control their body fatigue.

As multiple streams of data have been collected in the ErgoRelief framework, such as body motion, physiological data, and external force, it is essential to design a method to prepare them for body fatigue assessment, due to their sensitivity towards motion artefacts. Therefore, a data processing pipeline has been proposed, including data pre-processing, feature extraction, data labelling, feature selection, and body fatigue classification. The data pre-processing was responsible for cleaning data and aligning various types of data to a consistent time basis, including data synchronisation, segmentation, resampling, and filtering. Features have then been extracted from the data using a sliding window. Each sample in the feature set has been labelled by different fatigue levels through a personalised threshold, and the feature set has been optimised by a series of feature selection algorithms. Finally, the feasibility of the ErgoRelief framework has been demonstrated through the multiple linear regression analysis, and the body fatigue assessment has been conducted through machine learning approaches. By implementing this data processing pipeline, multimodal data can be converted to in-depth body-related information, such as fatigue. Compared with previous studies [29, 44], our data processing pipeline is able to handle data from multiple data sources with independent systems.

To demonstrate the feasibility of the ErgoRelief framework, a study has been conducted to perform body fatigue classification during repetitive physical tasks in a laboratory environment. Both wearable and non-wearable sensors have been utilised in this study. Wearable sensors are responsible for measuring body motion and physiological data, such as heart rate, whilst the non-wearable sensors collected external forces exerted on the human body. The performance of different sensor systems (non-wearable, wearable, and a combination of wearable and non-wearable sensors) on assessing fatigue have been explored, to select the optimal predictors towards body fatigue. Additionally, the performance of different feature selection and machine learning approaches have also been systematically evaluated in this study. Finally, the sensor deployment has been optimised to prepare the system for real-world application, aiming at minimising obtrusiveness, computational cost, avoiding privacy and security issues, as well as maintaining the classification performance. An experiment has been conducted by thirty participants in a laboratory environment, with data being collected while participants were performing repetitive physical tasks. Results of the multiple linear regression analysis illustrated that the highest Adjusted R^2 Score was 0.9259 and the lowest RMSE was down to 0.4972, proving that there is a relationship between body motion, heart rate, external force, and fatigue. Moreover, results of body fatigue classification demonstrated that the wearable sensor system performed significantly better than the non-wearable sensor system, whilst the addition of non-wearable sensors to the wearable sensor system improved the classification performance

slightly. Therefore, considering the high infrastructure cost and impracticalities of deploying non-wearable sensors in a factory environment, non-wearable sensors can be removed from the system without sacrificing significant classification performance. In addition, multiple feature selection and machine learning approaches have been implemented on the feature set to conduct the binary body fatigue classification. Considering the highest accuracy, the ReliefF-10 and ReliefF-30 feature sets have relatively similar classification performance, since ReliefF-10 and ReliefF-30 are the same feature selection approach with a different number of neighbours, and the number of neighbours does not influence the classification result significantly. The PCA feature set has a significantly better performance than the ReliefF-10 and ReliefF-30 feature sets. It is because compared with eliminating redundant features from the feature sets directly using ReliefF feature selection, the PCA projects the original feature set to a new dimension, in order to reduce the dimension of the feature set and preserve the major data variation. For ReliefF-10, ReliefF-30, and PCA feature sets, the LDA models always have the worst performance, because LDA is one of the simplest machine learning approaches, which cannot solve complicated problems, such as body fatigue classification. The kNN and P-SVM models have similar classification performance utilising the ReliefF-10 and ReliefF-30 feature sets, whilst the P-SVM models outperformed the kNN model significantly using the PCA feature set. It is because that the P-SVM model is more complex than the kNN model. Since the PCA feature set has more features than the ReliefF-10 and ReliefF-30 feature sets, the P-SVM models are able to handle the complicated feature set better than the kNN models. The optimal body fatigue classification performance has been achieved by the P-SVM utilising PCA feature sets from the full sensor system with a combination of wearable and non-wearable sensors, with an accuracy as high as 97.17% and 94.15% for task-specific and overall analysis respectively, which demonstrates a significant improvement over previous studies. Hence, these results illustrated that our system could detect body fatigue. Besides, the classification can be done with a combination of all tasks, rather than for each task, which can significantly conserve computational power. Lastly, during the sensor optimisation stage, a physiological sensor, two IMUs on the Upper Spine and Right Arm, as well as computationally intensive features have been eliminated from the system. The classification accuracy for the optimised system has been up to 86.17%, illustrating that it can still conduct body fatigue classification with reasonable performance.

The second study has been undertaken to conduct real-time body fatigue assessment, as well as provide real-time feedback to participants through the mobile application. Body motion data have been recorded from four IMUs on different body areas. An experiment has been conducted with thirty participants in a workshop to simulate aviation manufacturing tasks. Results illustrated that the

highest accuracies of fatigue classification have been 81.41% and 85.37% for the real-time body fatigue assessment and machine learning model training respectively, demonstrating that the system can conduct precise fatigue classification in real time. In addition, machine learning models using the ReliefF-10 and ReliefF-30 feature sets have similar performances in classifying fatigue during real complex aviation-like tasks, because both ReliefF-10 and ReliefF-30 are the same algorithm except using a different number of neighbours. Moreover, the P-SVM model outperformed the kNN model for the first few participants, illustrating that it is able to handle feature sets with a limited number of samples better. However, as the data size continued growing, the performance of the kNN model surpassed the P-SVM model, because the kNN model is able to identify clusters of different classes clearly with more data. Furthermore, the battery lifetime of IMUs decreased from 100% to between 60% and 70% after 5.5 hours data recording, showing that the system can operate for an extended period of time. Finally, the packet receiving rate for each IMU was greater than 99%, illustrating that the system is able to receive most packets successfully through Bluetooth.

In this research, traditional machine learning approaches, such as LDA, kNN, and P-SVM, have been implemented. Compared with deep learning approaches, such as artificial neural networks, which learn features from the data automatically, features need to be extracted prior to the classification for the traditional machine learning approaches [146]. In this research, the extraction and study of specific features was an intentional choice. We constructed the data processing pipeline in order to understand the contribution that each set of features has toward body fatigue assessment. Therefore, by using traditional machine learning approaches, we can better understand the data and corresponding features, which can then help towards optimizing the sensor deployment and preparing the system for a real-world application.

The first study in this research explored the performance of different types of sensors for classifying body fatigue, including both wearable and non-wearable sensors. After the first study, multiple sensors were found to be redundant in assessing body fatigue, including the non-wearable sensors, PPG, as well as two Shimmer3 IMU sensors. Therefore, the optimized sensor system has been further tested in the second study and illustrated that it can still achieve reasonable classification performance. Additionally, the results of the first study illustrated that P-SVM and kNN performed relatively better than LDA. This was consistent in the second study's results, which further demonstrated that during the real-time classification, using different dataset sizes, either kNN or P-SVM achieved high accuracy for multi-level body fatigue classification.

Overall, this research proposed the framework of real-time body fatigue assessment, including data collection, analysis, logging, and visualisation. The implementation of this framework can help workers to adjust their task performance and help aviation operations in optimising task ergonomics and related human factors around manufacturing processes, as well as reducing safety risks. In addition, the development of the data processing pipeline significantly improves the system extensibility, allowing the addition and removal of various types of sensors into the system without interrupting the system operation. Furthermore, the evaluation of the body fatigue classification performance on different sensor systems, feature selection algorithms, and machine learning approaches demonstrates the system feasibility and has constructed the knowledge base for the optimal system and algorithm. The result also helps in optimising the sensor deployment by minimising the number of sensors, as the system should be not only accurate, but also unobtrusive and comfortable when implementing in an aviation factory environment. Lastly, a real-time body fatigue detection system has been constructed with real-time data collection, analysis, and logging, which achieved aspects in the ErgoRelief framework. The detected fatigue levels can then help aviation factory workers to have a deeper understanding about their body conditions, ensure workers' safety, and increase factory efficiency and productivity.

The results of this research can also be contextualised for workers in other industries, as there are consistencies between aviation manufacturing environments and general factory environments, in terms of the tools used and tasks performed. In addition, tasks in both studies were common for all industry workers. Therefore, the real-time wearable sensor-based fatigue evaluation system can be further calibrated and implemented for workers across different industries.

7.3 Limitations

This research represents the first step in the development of a real-time wearable sensor-based fatigue assessment system for use in the context of aviation manufacturing. Although the results are promising, there are some limitations that need to be addressed in future research.

One limitation is that most of the participants that undertook both studies were university staff and students with minimal manufacturing experience. In order to analyse fatigue in an aviation manufacturing environment, further work is need that includes more qualified aviation manufacturing, which would improve the ecological validity of the results. In addition, the second study was intended to be undertaken in a real aviation manufacturing environment with aviation

manufacturing workers. However, due to COVID-19, it was not possible to travel to the factory due to lockdowns, travel restrictions and time constraints. As such, the second study was performed in a workshop to simulate, as realistically as possible, an aviation factory environment. In order to minimize the need for qualified workers, the tasks in this experiment were selected to ensure that participants were proficient in undertaking them with minimal training. Therefore, the findings are still consistent and generalisable as the goal of the studies was to detect fatigue.

Additionally, repetitive physical tasks for both studies have been designed based on real aviation manufacturing tasks. However, due to the high complexity of aviation manufacturing tasks, task repetitiveness may be extremely low. For instance, the duration for a single task may last for a few days. Therefore, rather than analysing repetitive tasks, it is also essential to design another approach to analyse tasks with low repetitiveness and a longer duration.

7.4 Future Work

Based on the limitations mentioned in the previous section, future work still needs to be accomplished before real-world deployment in an operational factory. Firstly, although the prototype for the smartphone feedback application has been developed, its effectiveness for improving workers' awareness about their body conditions has not been evaluated. Therefore, further work is needed in order to gather user feedback and fully explore how the smartphone feedback application can help workers to decrease their fatigue and prevent WMSDs.

Secondly, although data have been analysed in real time, the machine learning models for the three-level body fatigue classification have still been trained after each participant finished the experiment, which is expensive, time-consuming, and interruptive when implemented in an aviation factory environment. Therefore, it is essential to design a mechanism for the real-time machine learning model training. In the future, the classification model should be retrained and updated automatically and simultaneously while workers perform their tasks.

Thirdly, the performance impact of external factors on fatigue analysis needs to be further explored, which can be divided into two aspects: environmental and lifestyle. The environmental factors include surrounding temperature, humidity, noise, and air ventilation, whilst the lifestyle factors contain sleeping quality, diet habits, exercise, and smoking/drinking frequency. Although these factors will not pose an immediate effect on the body fatigue during work, they will have a prolonged negative

effect on the human body, which can lead to the faster accumulation of body fatigue while performing tasks. Therefore, not only the short-term causal factors of body fatigue, such as body motion and external force, but also the long-term factors of fatigue need to be further assessed, in order to provide a more comprehensive fatigue assessment.

In this research, body fatigue has been selected as the indicator towards WMSDs. However, the recorded data can also be used to prevent WMSDs in multiple ways, for example, in the case of full body reconstruction. The body motion data from the IMUs can be transformed into a consistent coordinate system, which can be used to estimate full body postures. By combining the observation of improper postures with fatigue, the linkage between postures and fatigue can be further investigated, which can help workers to adjust their fatigue levels.

Lastly, although the real-time fatigue assessment system has been developed, it has only been tested with a few devices. It is necessary to test and calibrate the system in a factory environment with more devices and a complex network environment, as many additional factors can influence the system stability, such as electricity outages, network outages, and limited bandwidth.

In summary, by accomplishing these future works, a robust real-time factory-wide fatigue assessment system can be established, which can operate continuously for an extended period of time, assess body fatigue through wearable sensors for all factory workers, as well as provide multiple types of feedback to both factory workers and operations, targeting at improving both factory productivity and workers' safety.

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Appendix 1

Study One: Ethics Approval

THE UNIVERSITY OF QUEENSLAND

Sub-Committee Human Research Ethics Approval

Project Title:	Wearable Computing System for Situational Awareness in Smart Factories – 13/09/2019 AMENDMENT
Chief Investigator:	Dr Chelsea Dobbins
Supervisor:	Dr Chelsea Dobbins
Co-Investigator(s):	Mr Guobin Liu, Dr Matthew D'Souza, Mr Ngoc Phuong
School(s):	School of ITEE, The University of Queensland
Approval Number:	2019001586
Granting Agency/Degree:	UQ / Boeing Australia / PhD
Duration:	31 August 2021

Comments/Conditions:
Amendment 13/09/2019:

- Amended Participation Information Sheet and Project Description
- Amendment form_Guobin Liu_signed
- Participant Information Sheet - Guobin Liu
- Project Description - Guobin Liu

Note: if this approval is for amendments to an already approved protocol for which a UQ Clinical Trials Protection/Insurance Form was originally submitted, then the researchers must directly notify the UQ Insurance Office of any changes to that Form and Participant Information Sheets & Consent Forms as a result of the amendments, before action.

Name of responsible Sub-Committee:
University of Queensland Engineering, Architecture and Information Technology, Low & Negligible Risk Ethics Sub-Committee
This project complies with the provisions contained in the *National Statement on Ethical Conduct in Human Research* and complies with the regulations governing experimentation on humans.

Name of Ethics Sub-Committee representative:
Professor Deanna Kemp
Chairperson
University of Queensland Engineering, Architecture and Information Technology, Low & Negligible Risk Ethics Sub-Committee

Signature _____ Date **24/09/2019**

Appendix 2

Study Two: Ethics Approval

THE UNIVERSITY OF QUEENSLAND

Sub-Committee Human Research Ethics Approval

Project Title:	Wearable Computing System for Situational Awareness in Smart Factories – 22/09/2020 AMENDMENT
Chief Investigator:	Dr Chelsea Dobbins
Supervisor:	Dr Chelsea Dobbins
Co-Investigator(s):	Mr Guobin Liu, Dr Matthew D'Souza, Mr Ngoc Phuong, Dr Tyler Okimoto, Dr Nicole Gillespie, Alexandria Macdade
School(s):	School of ITEE, The University of Queensland
Approval Number:	2019001586
Granting Agency/Degree:	UQ / Boeing Australia / PhD
Duration:	31 August 2021

Comments/Conditions:
Amendment 22/09/2020:

- Add Investigators : Dr Tyler Okimoto, Dr Nicole Gillespie, Alexandria Macdade
- Amendment form_Guobin & Alex
- Participant Information Sheet - Guobin & Alex
- Project Description - Guobin & Alex
- Withdrawal Form - Guobin & Alex

Note: if this approval is for amendments to an already approved protocol for which a UQ Clinical Trials Protection/Insurance Form was originally submitted, then the researchers must directly notify the UQ Insurance Office of any changes to that Form and Participant Information Sheets & Consent Forms as a result of the amendments, before action.

Name of responsible Sub-Committee:
University of Queensland Engineering, Architecture and Information Technology, Low & Negligible Risk Ethics Sub-Committee
This project complies with the provisions contained in the *National Statement on Ethical Conduct in Human Research* and complies with the regulations governing experimentation on humans.

Name of Ethics Sub-Committee representative:
Professor Deanna Kemp
Chairperson
University of Queensland Engineering, Architecture and Information Technology, Low & Negligible Risk Ethics Sub-Committee

Signature _____ Date **02/10/2020**

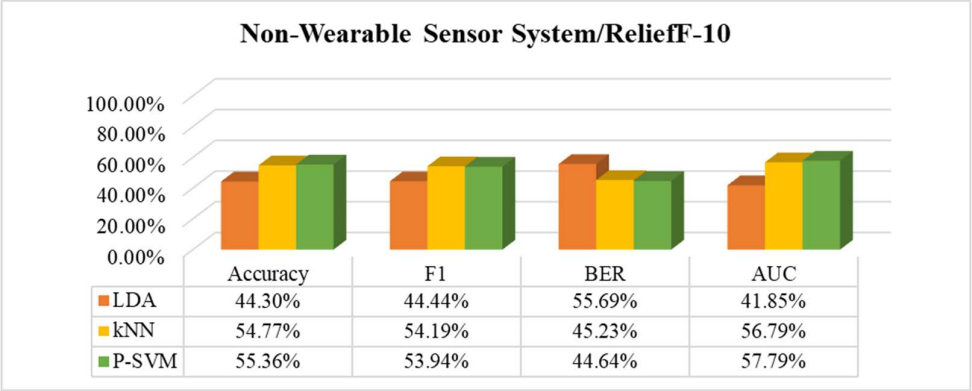
Appendix 3

Borg Rating of Perceived Exertion

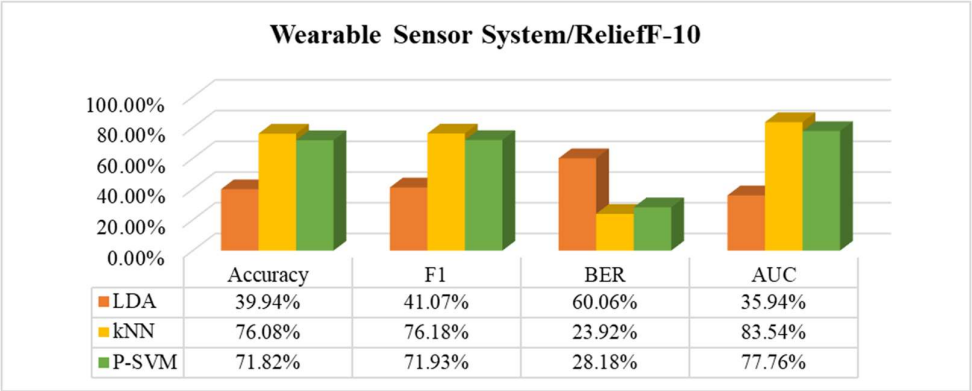
How you might describe your exertion	Borg rating of your exertion	Examples (for most adults <65 years old)
None	6	Reading a book, watching television
Very, very light	7 to 8	Tying shoes
Very light	9 to 10	Chores like folding clothes that seem to take little effort
Fairly light	11 to 12	Walking through the grocery store or other activities that require some effort but not enough to speed up your breathing
Somewhat hard	13 to 14	Brisk walking or other activities that require moderate effort and speed your heart rate and breathing but don't make you out of breath
Hard	15 to 16	Bicycling, swimming, or other activities that take vigorous effort and get the heart pounding and make breathing very fast
Very hard	17 to 18	The highest level of activity you can sustain
Very, very hard	19 to 20	A finishing kick in a race or other burst of activity that you can't maintain for long

Appendix 4

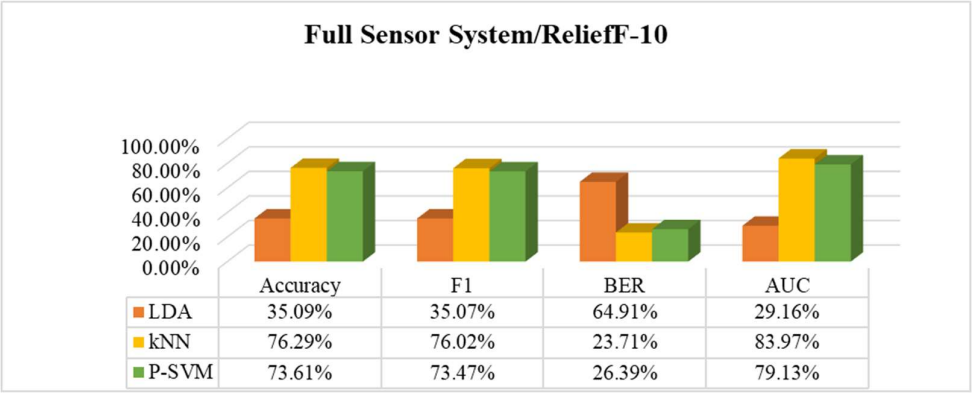
Two-handed Box Lifting Body Fatigue Classification Results (ReliefF-10 and ReliefF-30)



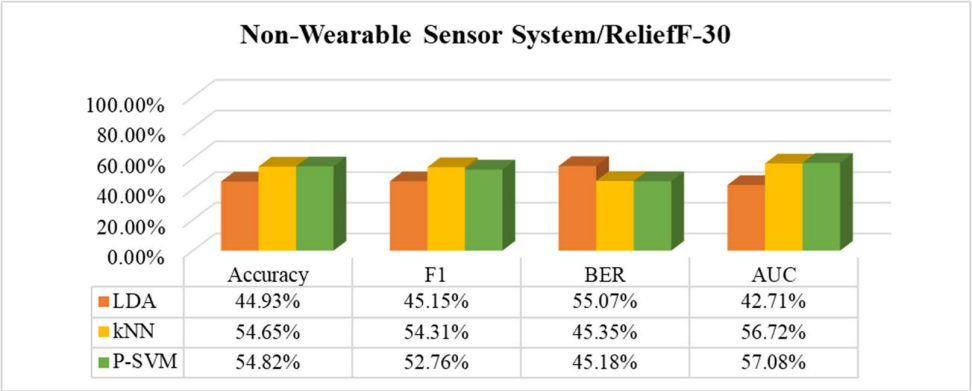
	Accuracy	F1	BER	AUC
LDA	44.30%	44.44%	55.69%	41.85%
kNN	54.77%	54.19%	45.23%	56.79%
P-SVM	55.36%	53.94%	44.64%	57.79%



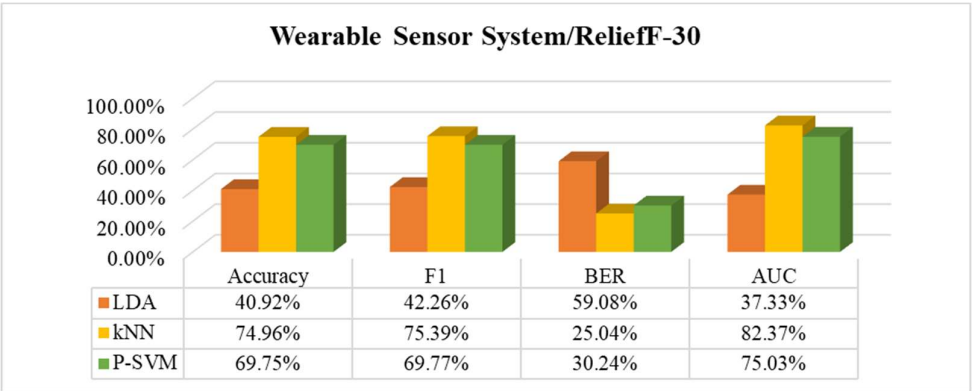
	Accuracy	F1	BER	AUC
LDA	39.94%	41.07%	60.06%	35.94%
kNN	76.08%	76.18%	23.92%	83.54%
P-SVM	71.82%	71.93%	28.18%	77.76%



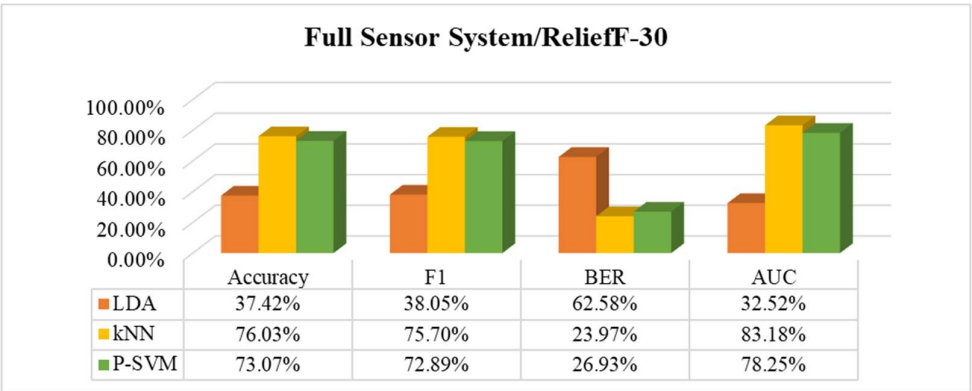
	Accuracy	F1	BER	AUC
LDA	35.09%	35.07%	64.91%	29.16%
kNN	76.29%	76.02%	23.71%	83.97%
P-SVM	73.61%	73.47%	26.39%	79.13%



	Accuracy	F1	BER	AUC
LDA	44.93%	45.15%	55.07%	42.71%
kNN	54.65%	54.31%	45.35%	56.72%
P-SVM	54.82%	52.76%	45.18%	57.08%



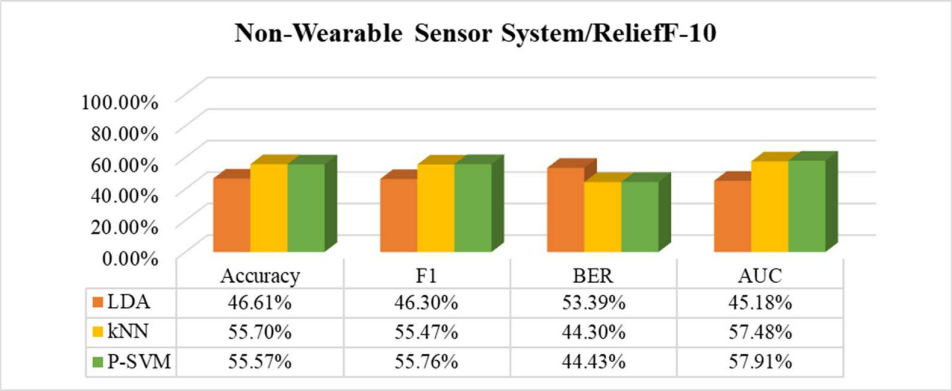
	Accuracy	F1	BER	AUC
LDA	40.92%	42.26%	59.08%	37.33%
kNN	74.96%	75.39%	25.04%	82.37%
P-SVM	69.75%	69.77%	30.24%	75.03%



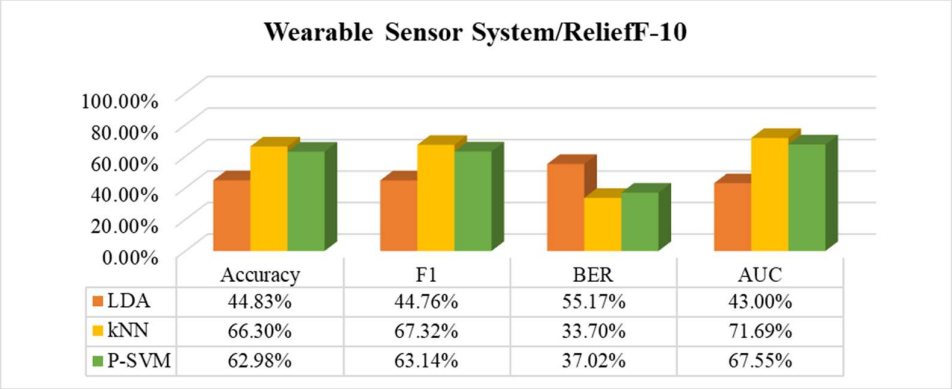
	Accuracy	F1	BER	AUC
LDA	37.42%	38.05%	62.58%	32.52%
kNN	76.03%	75.70%	23.97%	83.18%
P-SVM	73.07%	72.89%	26.93%	78.25%

Appendix 5

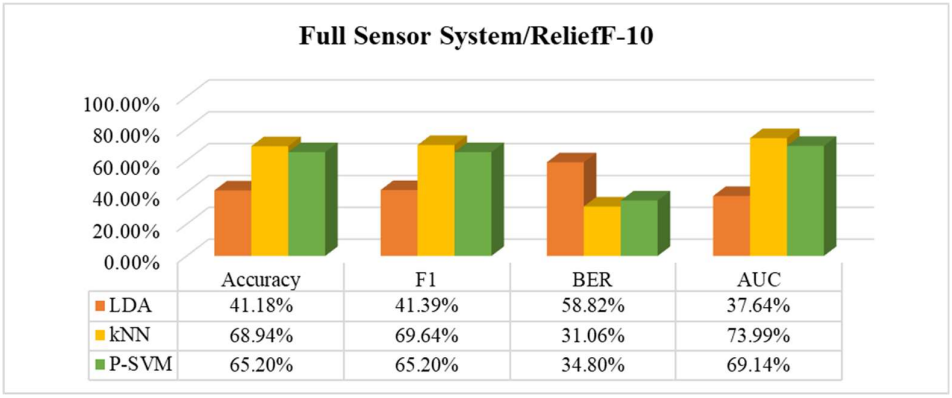
Two-handed Box Carrying Body Fatigue Classification Results (ReliefF-10 and ReliefF-30)



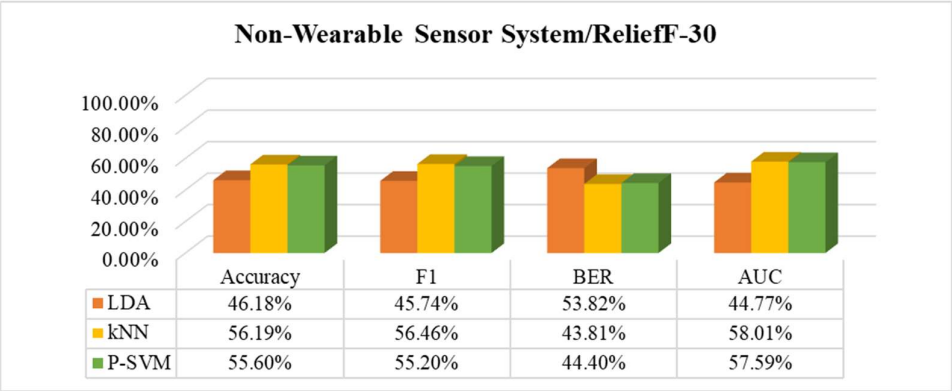
	Accuracy	F1	BER	AUC
LDA	46.61%	46.30%	53.39%	45.18%
kNN	55.70%	55.47%	44.30%	57.48%
P-SVM	55.57%	55.76%	44.43%	57.91%



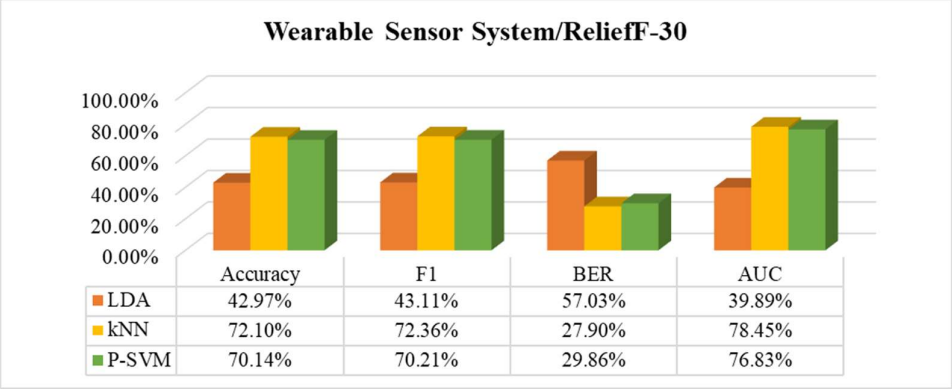
	Accuracy	F1	BER	AUC
LDA	44.83%	44.76%	55.17%	43.00%
kNN	66.30%	67.32%	33.70%	71.69%
P-SVM	62.98%	63.14%	37.02%	67.55%



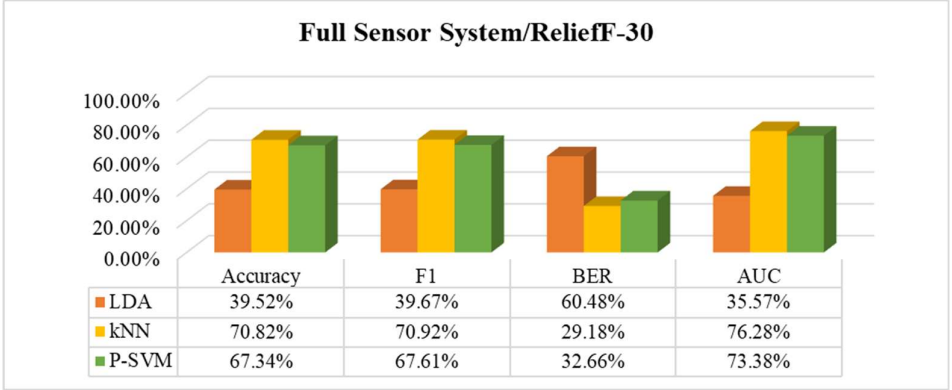
	Accuracy	F1	BER	AUC
LDA	41.18%	41.39%	58.82%	37.64%
kNN	68.94%	69.64%	31.06%	73.99%
P-SVM	65.20%	65.20%	34.80%	69.14%



	Accuracy	F1	BER	AUC
LDA	46.18%	45.74%	53.82%	44.77%
kNN	56.19%	56.46%	43.81%	58.01%
P-SVM	55.60%	55.20%	44.40%	57.59%



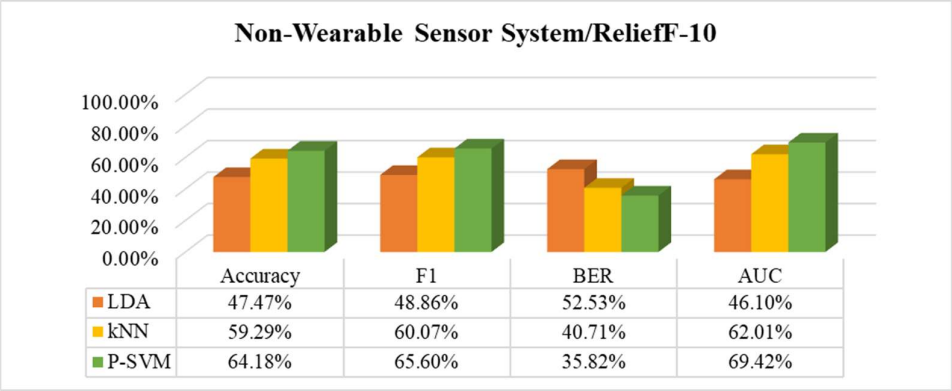
	Accuracy	F1	BER	AUC
LDA	42.97%	43.11%	57.03%	39.89%
kNN	72.10%	72.36%	27.90%	78.45%
P-SVM	70.14%	70.21%	29.86%	76.83%



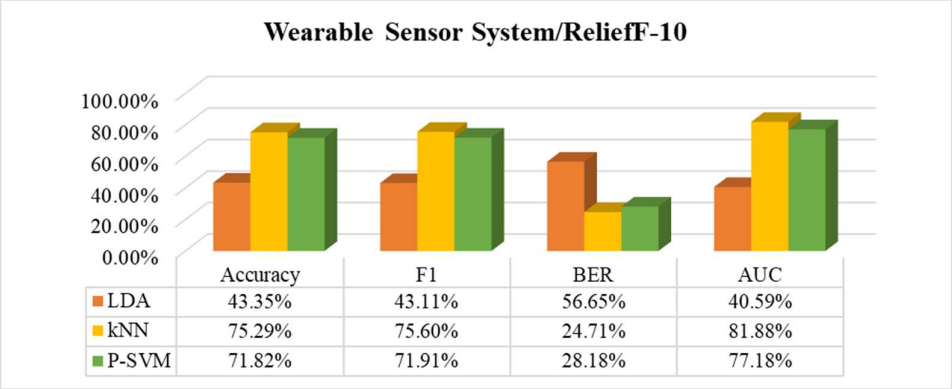
	Accuracy	F1	BER	AUC
LDA	39.52%	39.67%	60.48%	35.57%
kNN	70.82%	70.92%	29.18%	76.28%
P-SVM	67.34%	67.61%	32.66%	73.38%

Appendix 6

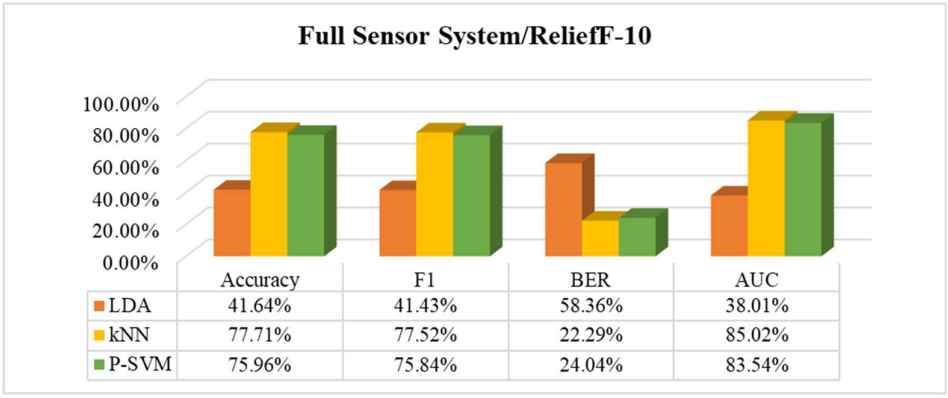
Two-handed Box Pushing/Pulling Body Fatigue Classification Results (ReliefF-10 and ReliefF-30)



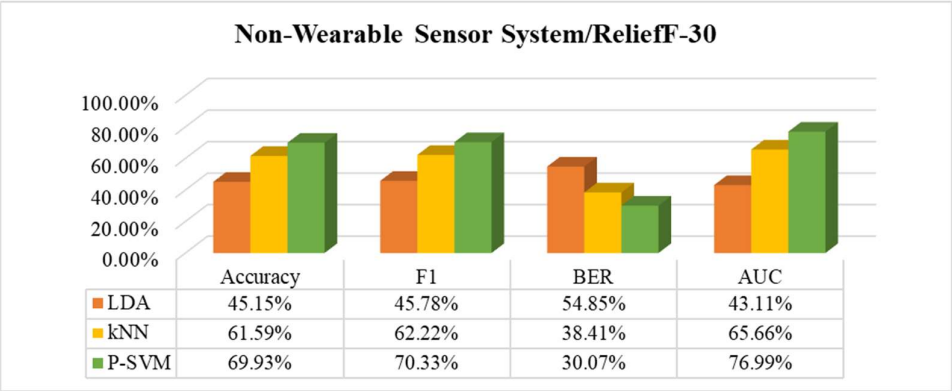
	Accuracy	F1	BER	AUC
LDA	47.47%	48.86%	52.53%	46.10%
kNN	59.29%	60.07%	40.71%	62.01%
P-SVM	64.18%	65.60%	35.82%	69.42%



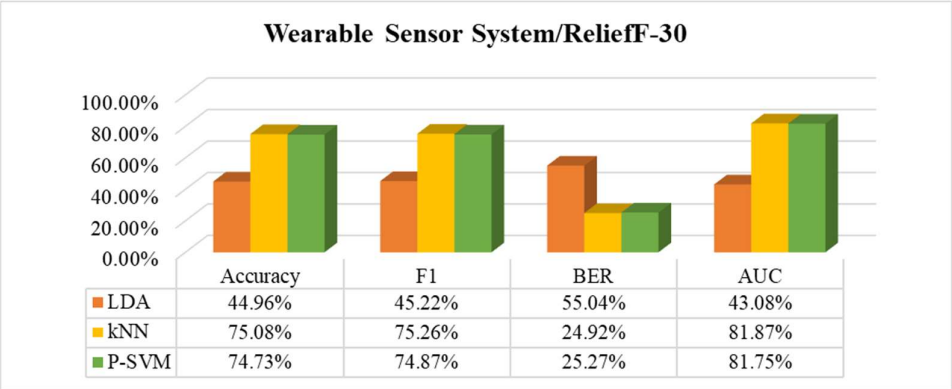
	Accuracy	F1	BER	AUC
LDA	43.35%	43.11%	56.65%	40.59%
kNN	75.29%	75.60%	24.71%	81.88%
P-SVM	71.82%	71.91%	28.18%	77.18%



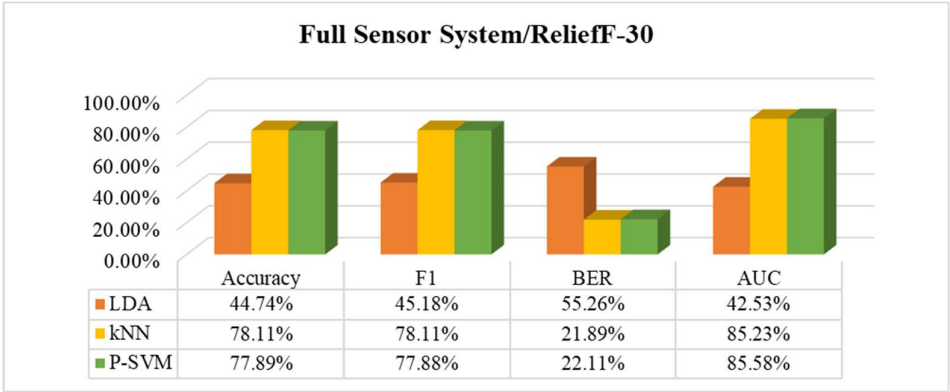
	Accuracy	F1	BER	AUC
LDA	41.64%	41.43%	58.36%	38.01%
kNN	77.71%	77.52%	22.29%	85.02%
P-SVM	75.96%	75.84%	24.04%	83.54%



	Accuracy	F1	BER	AUC
LDA	45.15%	45.78%	54.85%	43.11%
kNN	61.59%	62.22%	38.41%	65.66%
P-SVM	69.93%	70.33%	30.07%	76.99%



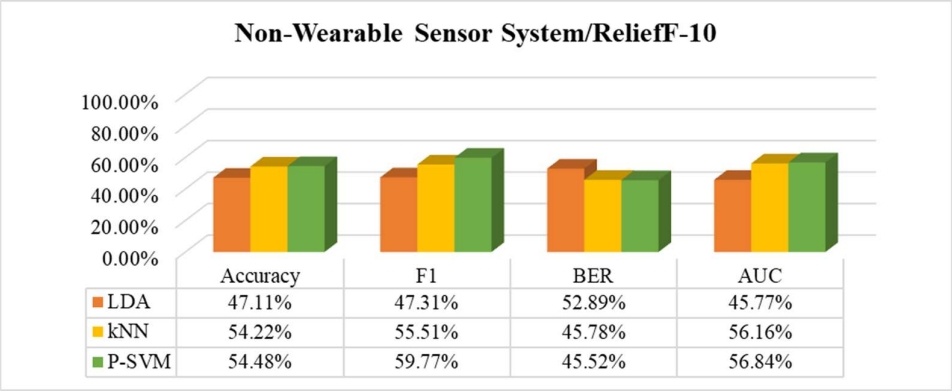
	Accuracy	F1	BER	AUC
LDA	44.96%	45.22%	55.04%	43.08%
kNN	75.08%	75.26%	24.92%	81.87%
P-SVM	74.73%	74.87%	25.27%	81.75%



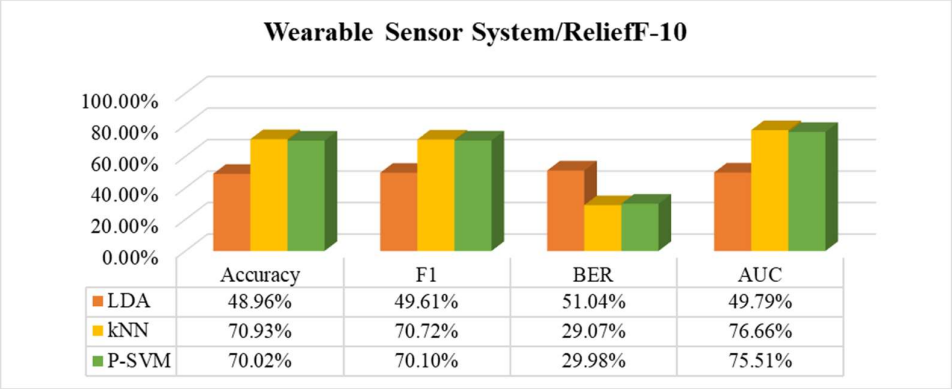
	Accuracy	F1	BER	AUC
LDA	44.74%	45.18%	55.26%	42.53%
kNN	78.11%	78.11%	21.89%	85.23%
P-SVM	77.89%	77.88%	22.11%	85.58%

Appendix 7

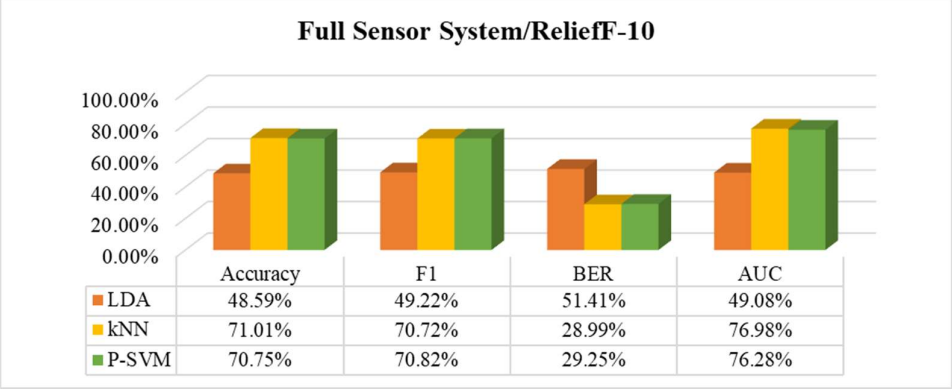
Two-handed Trolley Pushing/Pulling Body Fatigue Classification Results (ReliefF-10 and ReliefF-30)



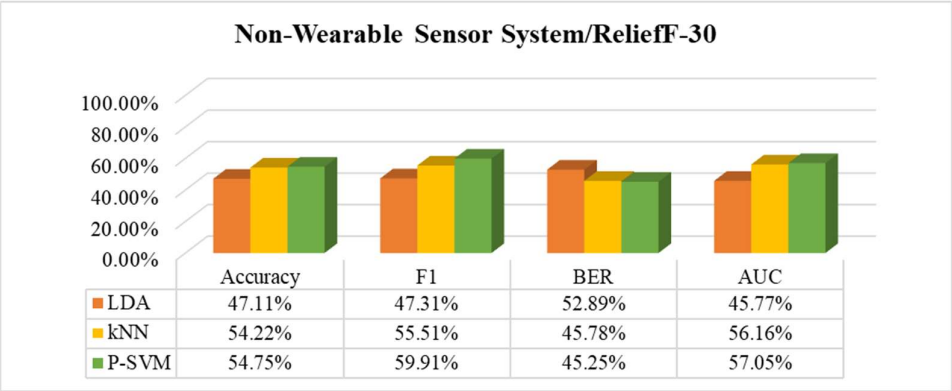
	Accuracy	F1	BER	AUC
LDA	47.11%	47.31%	52.89%	45.77%
kNN	54.22%	55.51%	45.78%	56.16%
P-SVM	54.48%	59.77%	45.52%	56.84%



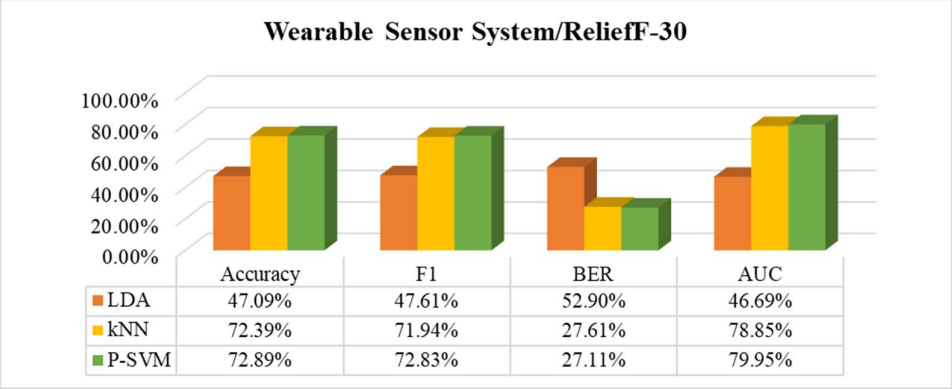
	Accuracy	F1	BER	AUC
LDA	48.96%	49.61%	51.04%	49.79%
kNN	70.93%	70.72%	29.07%	76.66%
P-SVM	70.02%	70.10%	29.98%	75.51%



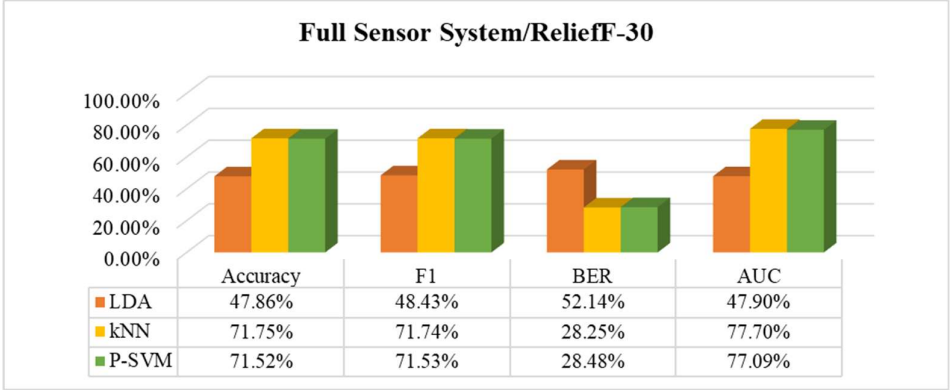
	Accuracy	F1	BER	AUC
LDA	48.59%	49.22%	51.41%	49.08%
kNN	71.01%	70.72%	28.99%	76.98%
P-SVM	70.75%	70.82%	29.25%	76.28%



	Accuracy	F1	BER	AUC
LDA	47.11%	47.31%	52.89%	45.77%
kNN	54.22%	55.51%	45.78%	56.16%
P-SVM	54.75%	59.91%	45.25%	57.05%



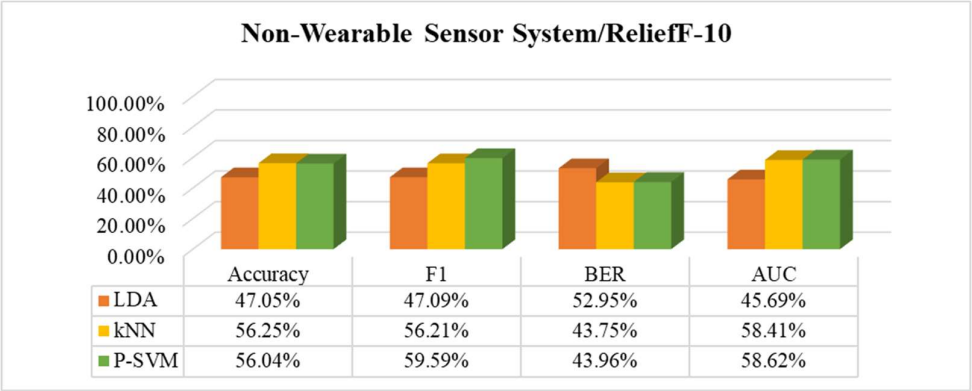
	Accuracy	F1	BER	AUC
LDA	47.09%	47.61%	52.90%	46.69%
kNN	72.39%	71.94%	27.61%	78.85%
P-SVM	72.89%	72.83%	27.11%	79.95%



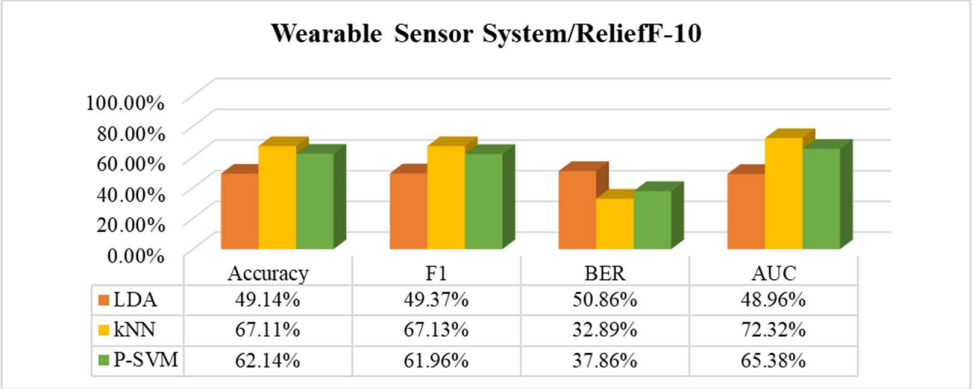
	Accuracy	F1	BER	AUC
LDA	47.86%	48.43%	52.14%	47.90%
kNN	71.75%	71.74%	28.25%	77.70%
P-SVM	71.52%	71.53%	28.48%	77.09%

Appendix 8

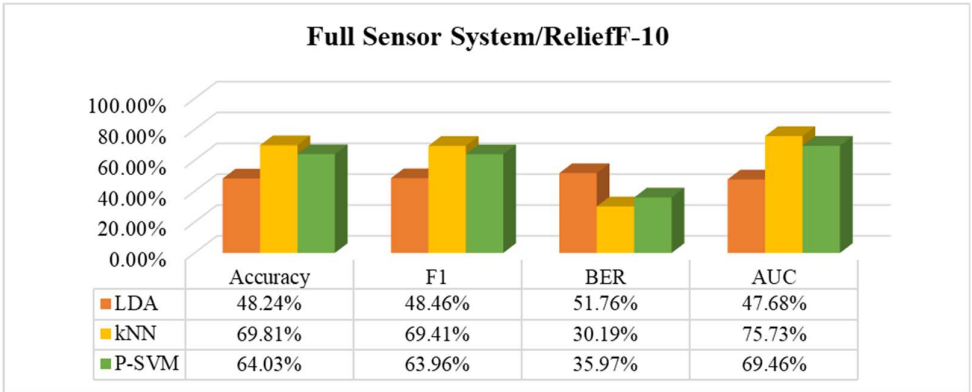
Overall Body Fatigue Classification Results (ReliefF-10 and ReliefF-30)



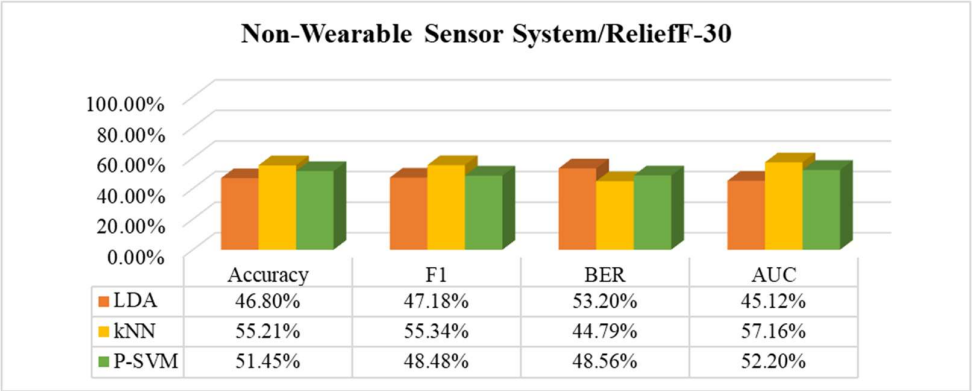
	Accuracy	F1	BER	AUC
LDA	47.05%	47.09%	52.95%	45.69%
kNN	56.25%	56.21%	43.75%	58.41%
P-SVM	56.04%	59.59%	43.96%	58.62%



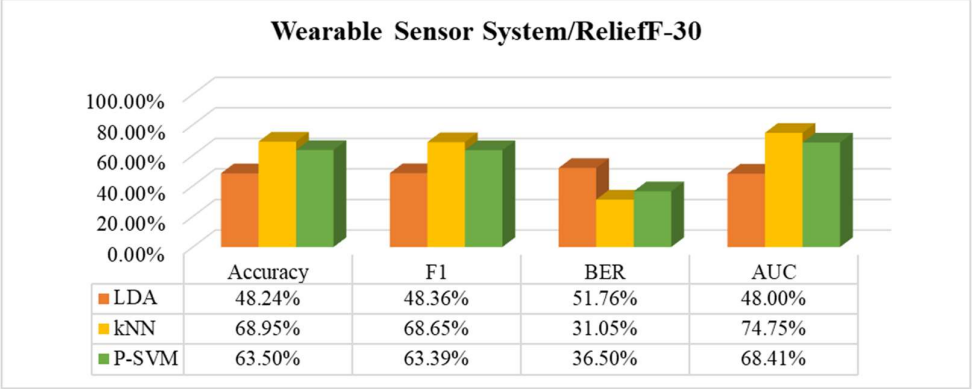
	Accuracy	F1	BER	AUC
LDA	49.14%	49.37%	50.86%	48.96%
kNN	67.11%	67.13%	32.89%	72.32%
P-SVM	62.14%	61.96%	37.86%	65.38%



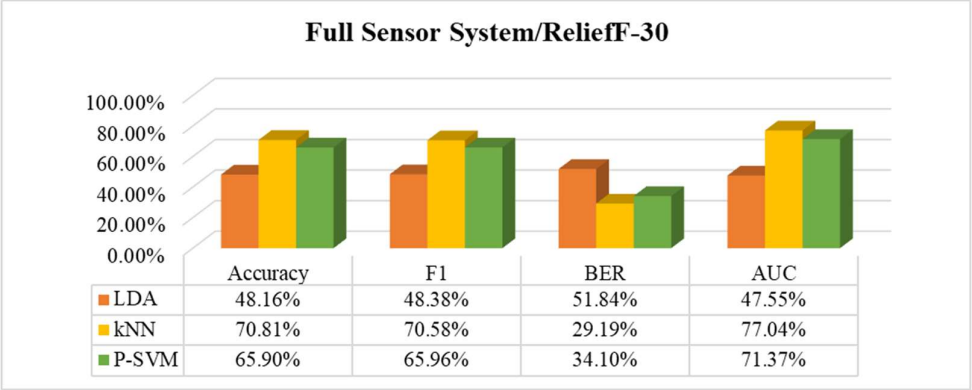
	Accuracy	F1	BER	AUC
LDA	48.24%	48.46%	51.76%	47.68%
kNN	69.81%	69.41%	30.19%	75.73%
P-SVM	64.03%	63.96%	35.97%	69.46%



	Accuracy	F1	BER	AUC
LDA	46.80%	47.18%	53.20%	45.12%
kNN	55.21%	55.34%	44.79%	57.16%
P-SVM	51.45%	48.48%	48.56%	52.20%



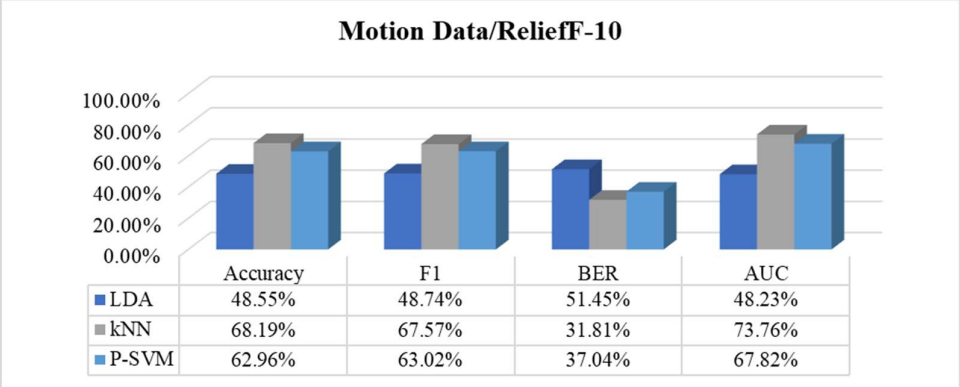
	Accuracy	F1	BER	AUC
LDA	48.24%	48.36%	51.76%	48.00%
kNN	68.95%	68.65%	31.05%	74.75%
P-SVM	63.50%	63.39%	36.50%	68.41%



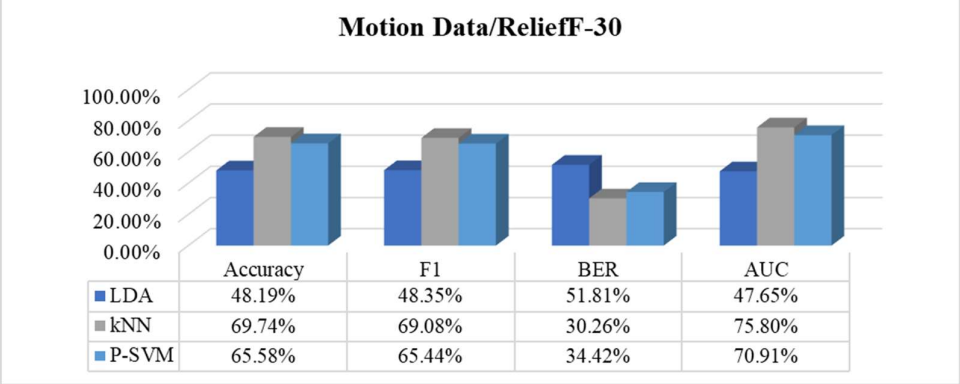
	Accuracy	F1	BER	AUC
LDA	48.16%	48.38%	51.84%	47.55%
kNN	70.81%	70.58%	29.19%	77.04%
P-SVM	65.90%	65.96%	34.10%	71.37%

Appendix 9

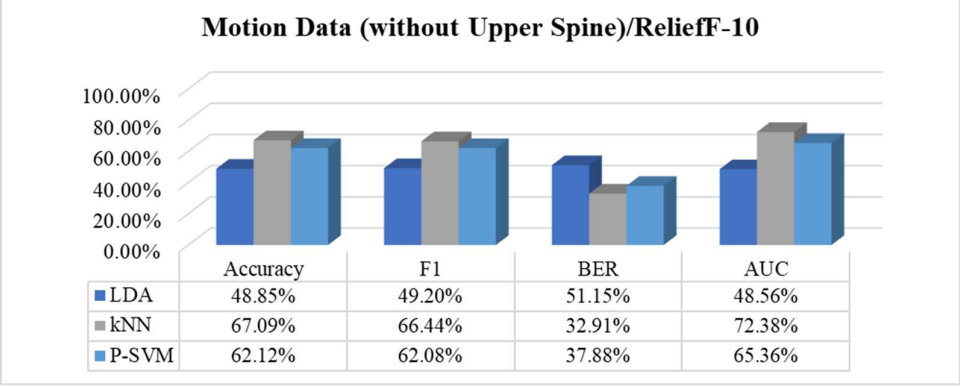
Sensor Optimisation Results (ReliefF-10 and ReliefF-30)



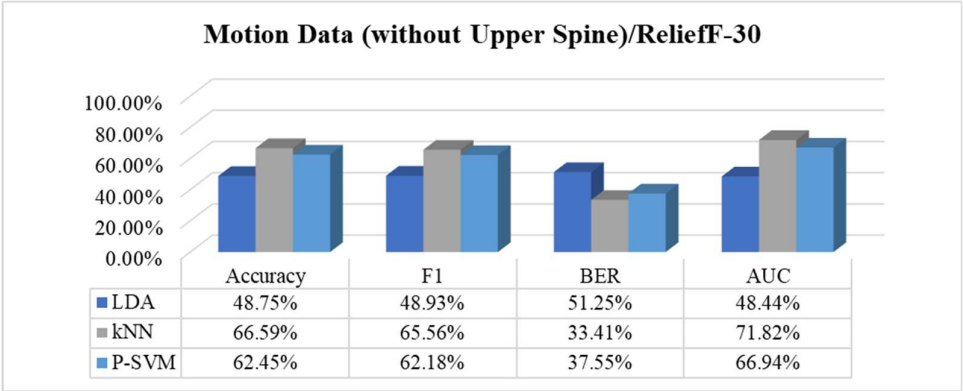
	Accuracy	F1	BER	AUC
LDA	48.55%	48.74%	51.45%	48.23%
kNN	68.19%	67.57%	31.81%	73.76%
P-SVM	62.96%	63.02%	37.04%	67.82%



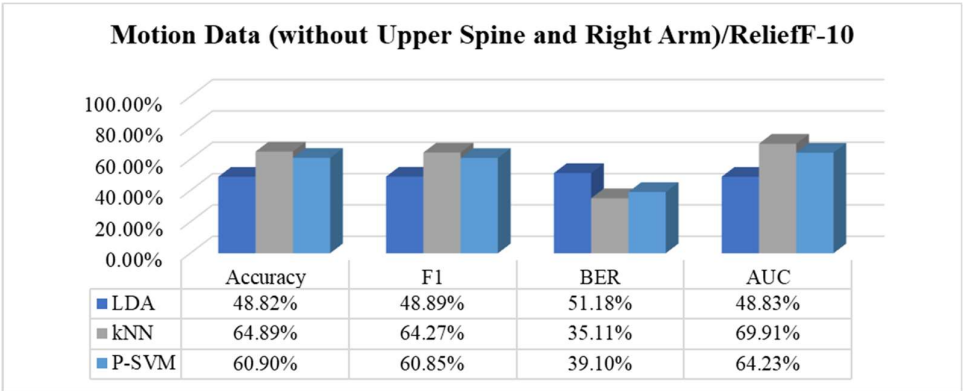
	Accuracy	F1	BER	AUC
LDA	48.19%	48.35%	51.81%	47.65%
kNN	69.74%	69.08%	30.26%	75.80%
P-SVM	65.58%	65.44%	34.42%	70.91%



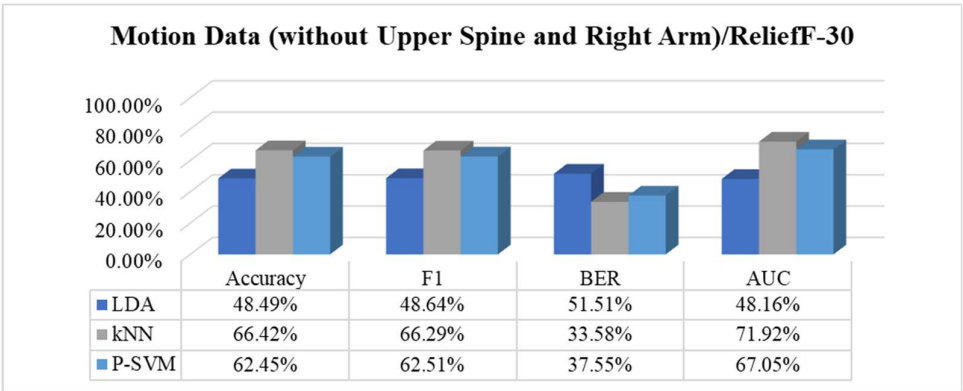
	Accuracy	F1	BER	AUC
LDA	48.85%	49.20%	51.15%	48.56%
kNN	67.09%	66.44%	32.91%	72.38%
P-SVM	62.12%	62.08%	37.88%	65.36%



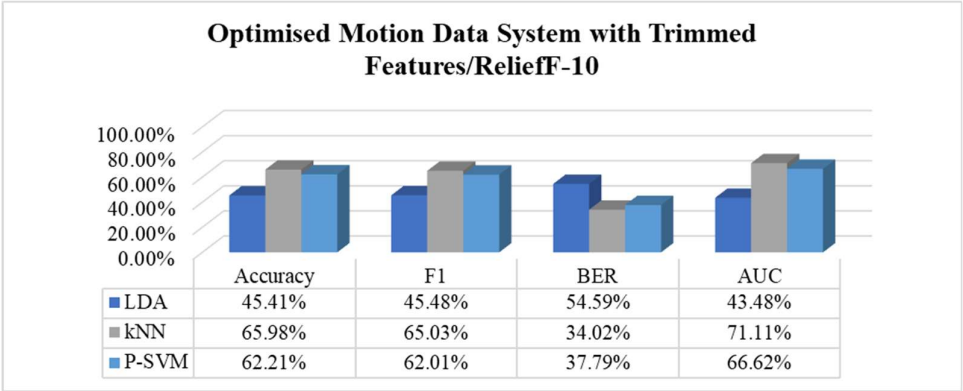
	Accuracy	F1	BER	AUC
LDA	48.75%	48.93%	51.25%	48.44%
kNN	66.59%	65.56%	33.41%	71.82%
P-SVM	62.45%	62.18%	37.55%	66.94%



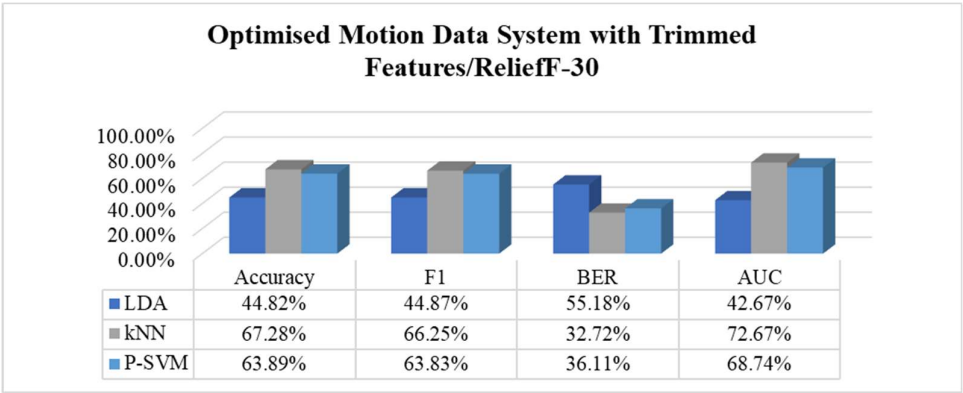
	Accuracy	F1	BER	AUC
LDA	48.82%	48.89%	51.18%	48.83%
kNN	64.89%	64.27%	35.11%	69.91%
P-SVM	60.90%	60.85%	39.10%	64.23%



	Accuracy	F1	BER	AUC
LDA	48.49%	48.64%	51.51%	48.16%
kNN	66.42%	66.29%	33.58%	71.92%
P-SVM	62.45%	62.51%	37.55%	67.05%



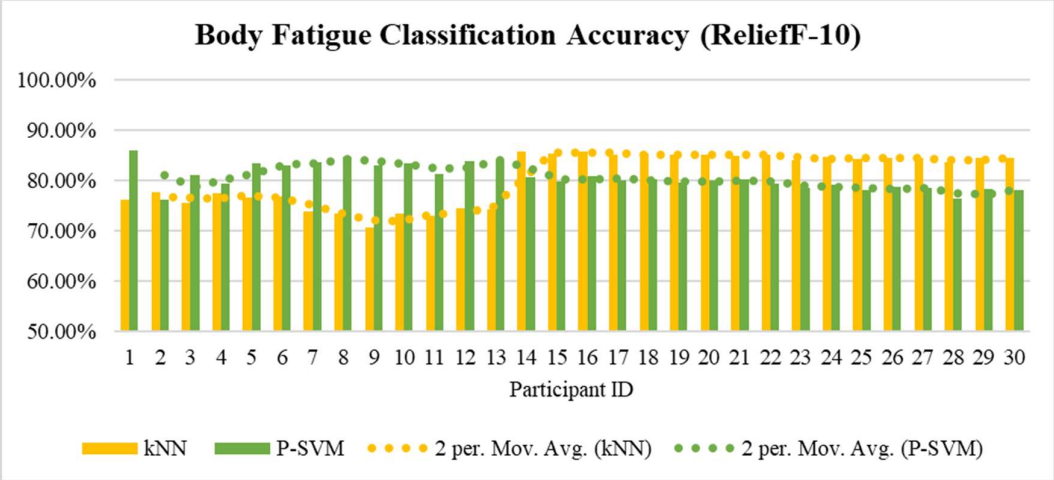
	Accuracy	F1	BER	AUC
LDA	45.41%	45.48%	54.59%	43.48%
kNN	65.98%	65.03%	34.02%	71.11%
P-SVM	62.21%	62.01%	37.79%	66.62%

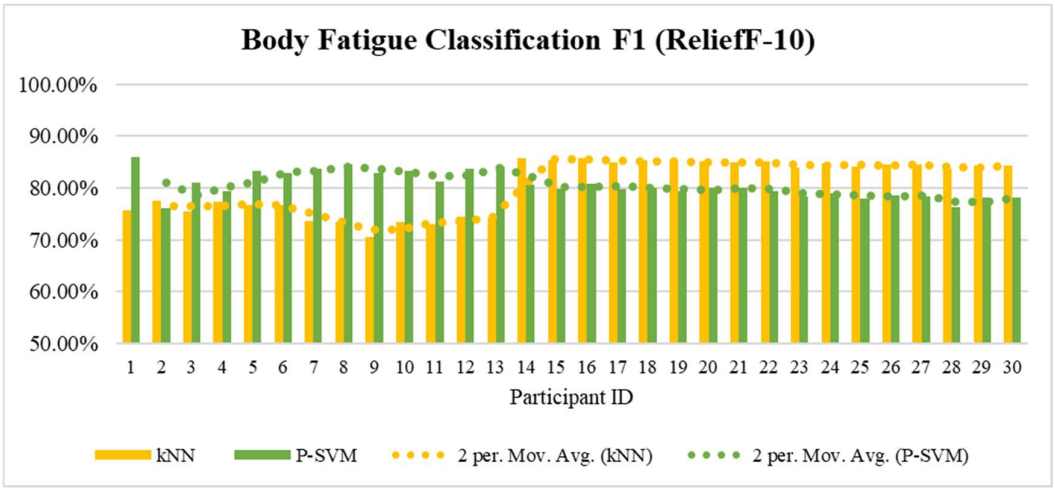


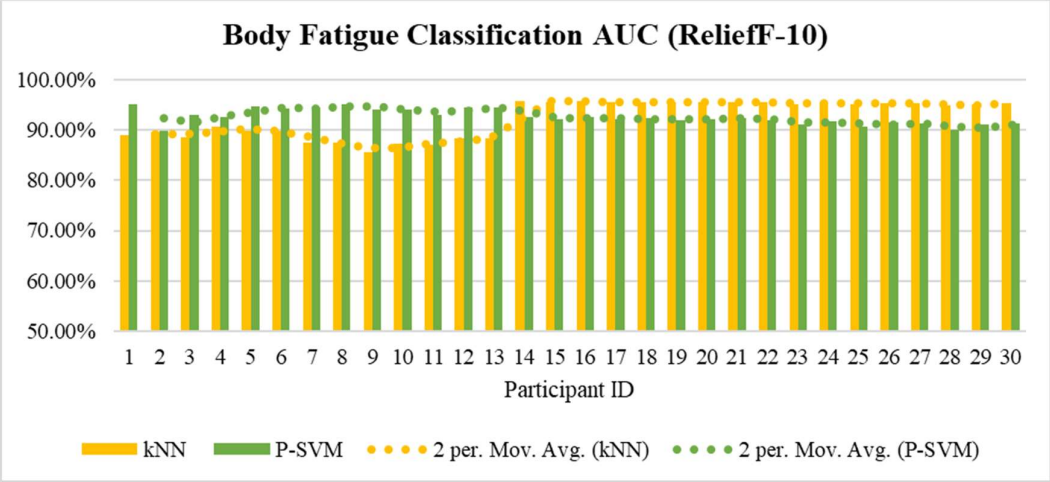
	Accuracy	F1	BER	AUC
LDA	44.82%	44.87%	55.18%	42.67%
kNN	67.28%	66.25%	32.72%	72.67%
P-SVM	63.89%	63.83%	36.11%	68.74%

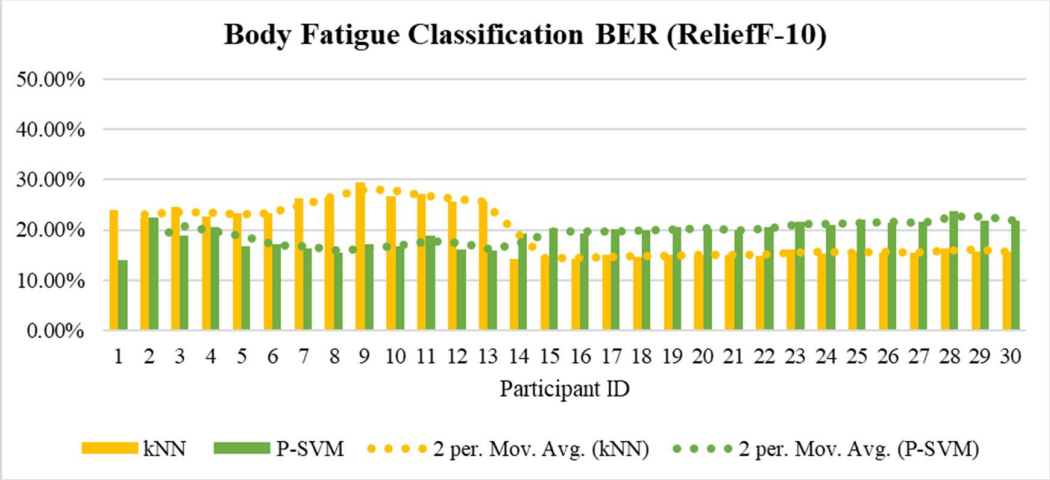
Appendix 10

Study Two Machine Learning Model Training Results (ReliefF-10)






Body Fatigue Classification AUC (ReliefF-10)
100.00%
90.00%
80.00%
70.00%
60.00%
50.00%
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Participant ID
kNN
P-SVM
2 per. Mov. Avg. (kNN)
2 per. Mov. Avg. (P-SVM)


Body Fatigue Classification BER (ReliefF-10)
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Participant ID
kNN
P-SVM
2 per. Mov. Avg. (kNN)
2 per. Mov. Avg. (P-SVM)